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Data Collection Nonresponse Bias Analysis
for Wave 7

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**PATH**Population Assessment
of Tobacco and Health*A collaboration between the NIH and FDA*

PATH Study

Wave 7 Data Collection

Nonresponse Bias Analysis Report

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1. Introduction

Westat prepared this report as a reference document for researchers using the Population Assessment of Tobacco and Health (PATH) Study interview and biomarker data. The PATH Study is a nationally representative, longitudinal cohort study of adults and youth in the United States, collecting data on those ages 12 years and older at Waves 1 to 4, Wave 5, and Wave 7, on those ages 12 to 17 at the Wave 4.5 special data collection, on those ages 13 to 19 at the Wave 5.5 special data collection, and on those ages 14 and older at Wave 6. A subsample of adults ages 20 and older also was surveyed in the latter part of 2020 via the PATH Adult Telephone Study (PATH-ATS). This report focuses on the Wave 7 response rates and potential nonresponse biases for select demographic, socio-economic, and tobacco-use measures.

Similar reports are also available for previous PATH Study data collections and can be found at the following websites for the National Addiction & HIV Data Archive Program (NAHDAP). The reports for the primary waves (i.e., the waves that included all available sample youth and adults) are located at one website and the reports for the special data collections, where subgroups of participants were surveyed, can be found at a separate website.

- Primary waves: Waves 1 to 4, 5, and 6:
<https://www.icpsr.umich.edu/web/NAHDAP/studies/36231/datadocumentation/>
- Special data collections: Waves 4.5, 5.5, and PATH-ATS:
<https://www.icpsr.umich.edu/web/NAHDAP/studies/37519/datadocumentation/>

In this document, the term “participants” is used to indicate people who agreed to be a part of the PATH Study, whether or not they completed an interview at a particular wave. The term “respondents” is used to indicate the subset of participants who completed an interview or, in the case of shadow youth (i.e., participants younger than 12 years old), verified their information with the study at a particular wave. Where needed for clarity, this term may be qualified to indicate the type of respondent (e.g., shadow youth respondent, adult interview respondent).

The PATH Study data support analyses of three cohorts: the Wave 1 Cohort, the Wave 4 Cohort, and the Wave 7 Cohort. The Wave 1 Cohort includes all Wave 1 respondents. These study participants were selected to represent adults ages 18 and older and youth ages 9 to 17 in the U.S. civilian, noninstitutionalized population (CNP) at the time of Wave 1 (September 12, 2013, to

December 14, 2014). The Wave 1 Cohort is to be used for longitudinal analysis at Wave 7. All Wave 1 respondents were eligible for Wave 7 data collection unless they were incarcerated, deceased, or resided outside of the United States at the time of Wave 7.

The Wave 4 Cohort consists of two components. The first component is the Wave 4 adult and youth respondents from the Wave 1 Cohort who were in the CNP and ages 12 and older at the time of Wave 4 (December 1, 2016, to January 3, 2018). The second component is the respondents to a replenishment sample designed to supplement the Wave 1 sample. The Wave 4 Cohort also serves the purpose of longitudinal analyses at Wave 7. All Wave 4 Cohort members were eligible for Wave 7 data collection unless they were incarcerated, deceased, or resided outside of the United States at the time of Wave 7.

The newly established Wave 7 Cohort also consists of two components. The first component is the Wave 7 adult and youth respondents from the Wave 4 Cohort who were in the CNP and at least 15 years old at the time of Wave 7 (January 6, 2022, to April 2, 2023). The second component is the respondents to a replenishment sample designed to supplement the Wave 4 Cohort. The Wave 7 Cohort can only be used for cross-sectional analysis at Wave 7.

This report summarizes the Wave 7 nonresponse bias analyses conducted separately for the three cohorts. Analyses for the Wave 1 and Wave 4 Cohorts use an all-waves perspective rather than a single-wave approach (see Section 3.2.1 for an explanation and why this better serves the research community). The analyses for the Wave 7 Cohort use a cross-sectional perspective and are similar to those conducted for the Wave 4 Cohort at Wave 4. The potential for nonresponse bias in Wave 7 estimates based on the Wave 1 Cohort is evaluated conditioning on Wave 1 response, whereas potential nonresponse bias in estimates based on the Wave 4 Cohort is assessed conditioning on Wave 4 response. With this in mind, a brief summary of the findings from the Wave 1 nonresponse bias analyses and from the Wave 4 nonresponse bias analyses for the Wave 4 Cohort is provided below.

Wave 1 Cohort Nonresponse Bias Analyses at Wave 1

- For the Wave 1 household screener and adult interview, most demographic and socioeconomic characteristics of the respondents at Wave 1 aligned with estimates from the 2013 American Community Survey 1-year Public Use Microdata Sample (2013 ACS) when using the inverse-of-probability-of-selection (IPS) weight for the PATH Study.

Exceptions were found for single-person households, sex, education, and ethnicity when comparing Wave 1 estimates using the IPS weight to the 2013 ACS estimates. Estimates of cigarette-smoking rates among adults at Wave 1 were within the range of estimates found in other national health studies. When the estimates were adjusted for nonresponse using the Wave 1 cross-sectional weight, they more closely approximated the 2013 ACS estimates for demographic and socio-economic characteristics and adult cigarette-smoking rates remained essentially the same.

- For the Wave 1 youth interview, most demographic characteristics of respondents were consistent with the estimates from the 2013 ACS, with the exception of ethnicity, when using the IPS weight. When the estimates were adjusted for nonresponse among youth, they more closely approximated the 2013 ACS estimates, but the ever-tried cigarette-smoking rates for youth at Wave 1 remained 2 to 12 percentage points lower than those found in other national studies.

Wave 4 Cohort Nonresponse Bias Analyses at Wave 4

- When using the Wave 4 cross-sectional weight, demographic and socio-economic estimates based on Wave 4 adults were nearly identical to corresponding estimates based on the 2016 American Community Survey 1-year Public Use Microdata Sample (2016 ACS). However, adults with health insurance were slightly underrepresented by the Wave 4 Cohort estimate. Estimates of current cigarette-smoking rates among adults at Wave 4 were within the range of estimates found in other national studies.
- When using the Wave 4 cross-sectional weight, demographic estimates based on Wave 4 youth were nearly identical to corresponding estimates based on the 2016 ACS. The ever-tried cigarette-smoking rates for Wave 4 youth were 1 to 6 percentage points lower than those found in other national studies.

This report is organized as follows: Chapter 2 provides an overview of the PATH Study sample design. Chapter 3 describes the methodology used for the analyses in this report. Chapters 4 and 5 present the results of nonresponse bias analyses for the Wave 1 Cohort interview data and Wave 4 Cohort interview data, respectively. Chapter 6 provides the household screener, adult interview, and youth interview response rates for the Wave 7 replenishment sample and the results of nonresponse bias analyses for the Wave 7 Cohort interview data. Chapter 7 summarizes the findings and discusses their implications.

2. Overview of Sample Design and Data Collection

This chapter provides an overview of the sample design for the PATH Study. The study's design allows for the longitudinal assessment of within-person and between-person change in the use of tobacco products, including initiation, cessation, relapse, and transitions between products, as well as in related health effects associated with use patterns.

At Wave 1, a four-stage, stratified probability sample design was used to select adults ages 18 and older and youth ages 12 to 17 from the CNP; an additional “shadow sample” of youth ages 9 to 11 was selected to be interviewed after reaching age 12 at later waves. All Wave 1 sample respondents together form the Wave 1 Cohort.

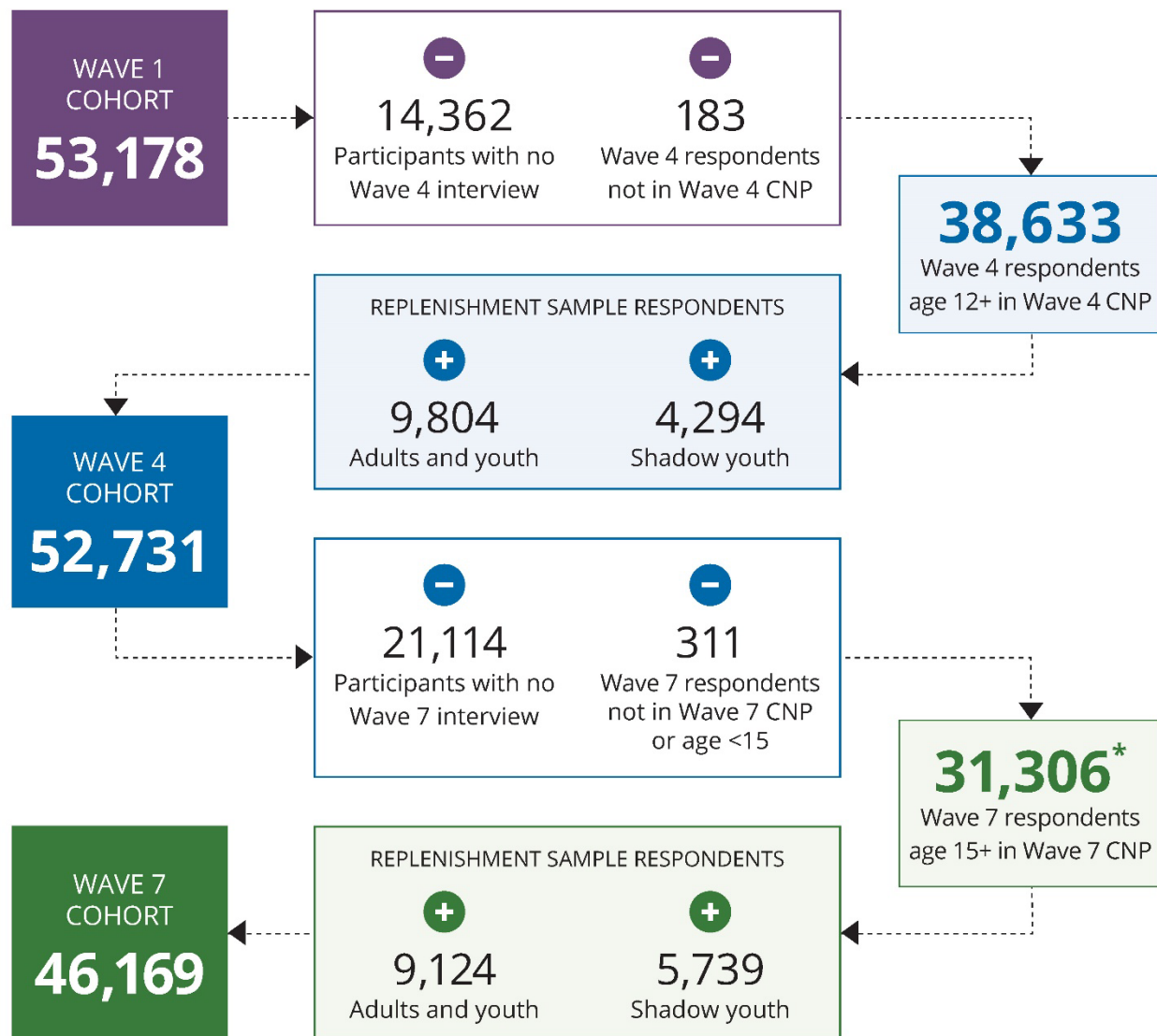
At Wave 4, the original set of Wave 1 sample respondents was replenished with a probability sample of adults, youth, and shadow youth ages 10 to 11¹ selected from the CNP at the time of Wave 4. The Wave 4 replenishment effort had two objectives: supplementation of the Wave 1 sample to address attrition and the changing composition of the study's adult respondents as Wave 1 youth (who were not oversampled with respect to tobacco use) turn 18 and age up to the adult interview, and selection of a shadow sample for interviewing at future waves. The Wave 4 replenishment effort also gave persons who had been overseas, in the military, or in an institutional setting at the time of Wave 1 (but were no longer at Wave 4) a chance of selection for the study. Because this “replenishment sample” was designed to supplement the Wave 1 sample, it is not intended to be used for estimation purposes on its own; rather, it was intended to be combined for estimation and analysis purposes with Wave 4 adult and youth respondents from the Wave 1 sample who were in the CNP at the time of Wave 4. This combined set of Wave 4 respondents forms the Wave 4 Cohort.

At Wave 7, the sample was replenished again with a probability sample of adults, youth, and shadow youth ages 9 to 11 selected from the CNP at the time of Wave 7. The objectives of this replenishment were the same as for Wave 4, particularly to account for the aging of the sample by

¹ The difference in the shadow youth age ranges between the Wave 1 sample and the Wave 7 replenishment sample (ages 9 to 11) compared to the Wave 4 replenishment sample (ages 10 to 11) reflects the timings of future data collections and periodic efforts to replenish the PATH Study sample that were planned at the time of Wave 4 development.

selecting new samples of youth ages 12 to 14 and shadow youth ages 9 to 11 because the youngest participants in the Wave 4 Cohort at the start of Wave 7 were expected to be 15 years old. This new sample was pooled with Wave 7 respondents from the Wave 4 Cohort who were in the CNP and ages 15 and older at the time of Wave 7 to form a new PATH Study cohort, the Wave 7 Cohort. Figure 1 illustrates the sample sizes for the Wave 1 Cohort, Wave 4 Cohort, and Wave 7 Cohort.

Figure 1. Illustration of the sample sizes for the Wave 1 Cohort, Wave 4 Cohort, and Wave 7 Cohort



*77.5% of Wave 4 Cohort Wave 7 respondents ages 15 and older in the Wave 7 CNP were sampled at Wave 1.

Key features of the Wave 1 sample design, as well as the Wave 4 and Wave 7 replenishment sample designs are summarized below. Further details can be found in Chapter 2 of the PATH Study Restricted Use Files (RUFs) User Guide

(<https://www.icpsr.umich.edu/web/NAHDAP/studies/36231/datadocumentation/>). At the end of Chapter 2 of this report, Table 2-1 provides data collection start and end dates for each wave of the PATH Study, as well as a summary of the results of the data and biospecimen collection efforts by wave, cohort, and interview type.

2.1 Wave 1

The target population of the PATH Study at Wave 1 is the CNP ages 9 and older at the time of Wave 1 (2013-2014). College students were sampled through their permanent residence rather than at their dormitory. Active-duty members of the military (Army, Navy, Marines, Air Force, and Coast Guard) were ineligible, as were all persons living in institutional and noninstitutional group quarters other than college dormitories. Spouses and children of active-duty military living in the United States were eligible for the study. For Wave 1, a four-stage, stratified area probability sample design was used with a two-phase design for sampling adults at the final stage. The sampling rates for adults varied by age, race, and tobacco-use status. At the first stage, a stratified sample of geographical primary sampling units (PSUs) was selected, in which a PSU was a county or group of counties. For the second stage, within each selected PSU, smaller geographical segments (consisting of one or more census blocks) were formed and then a systematic sample of these segments was drawn. At the third stage, a systematic sample of addresses within sampled segments was drawn from listings of addresses; the main source of these addresses was the U.S. Postal Service (USPS) Computerized Delivery Sequence Files (CDSFs). The CDSFs provide very high coverage of the residential addresses in the United States. The fourth stage was the random selection of persons within sampled households.

For within-household selection, a roster of all the members of the sampled household was constructed using the household screener. An adult household member, the household screener respondent, was asked to list members of the household and provide demographic, and for adults,

tobacco-use information about each person.² This information was used for sampling three main groups of interest:

- Adults ages 18 and older (up to two adults per household);
- Children ages 12 to 17 (i.e., “youth,” generally up to two per household);³ and
- Children ages 9 to 11 (i.e., “shadow youth,” generally up to two per household) to be interviewed at later waves after reaching age 12.

Since two-phase sampling was used to select adults from the replenishment sample, there were Phase 1 domains and Phase 2 domains. Phase 1 domains were defined based on the household screener respondent’s report of the adult’s age, race, and tobacco-use status. Phase 2 domains were defined based on the adult’s self-report of these characteristics and whether or not the tobacco-use statuses reported by the household screener respondent and adult agreed. Sampling rates for the two phases were designed to oversample adults ages 18 to 24 and adults of all ages who use tobacco, thereby increasing precision for estimates relating to these subgroups.⁴

All adult interview respondents were asked to provide urine and blood specimens. A subsample of the adults who provided urine forms the Wave 1 Biomarker Core, and these participants were flagged for ongoing urine collection after Wave 1. Information about the subsequent waves of the PATH Study is provided in the following sections.

2.2 Wave 2 and Wave 3

There was no additional sampling for Wave 2 or Wave 3 of the PATH Study. Wave 1 respondents were eligible for Wave 2 if they were still residents of the United States and not incarcerated.

² The household screener collected information on adult household members’ use of several different types of tobacco products. For example, it collected information on current use of products with relatively high prevalence or well-established use, such as cigarettes, cigars, and pipes; and on ever use of products with relatively low prevalence or newly emerging use, such as electronic cigarettes or e-cigarettes.

³ Given a special analytic interest in multiple-birth youth (e.g., twins), the shadow youth and youth sampling procedures were modified when households containing multiple-birth youths were encountered so that the multiple-birth youths would have relatively higher probabilities of selection. This resulted in some households with more than two children selected for the youth and/or shadow youth samples.

⁴ In addition, Black or African American adults were slightly oversampled because of their higher rate of use of menthol cigarettes compared to other racial/ethnic groups. For this purpose, Black or African American was defined as Black or African American alone or in combination with other race(s), whether the adult was Hispanic or non-Hispanic.

During the Wave 2 data collection period, attempts were made to contact the Wave 1 adult and youth respondents as well as members of the shadow youth sample established at Wave 1. Shadow youth who reached age 12 by Wave 2 and were permitted by a parent or guardian to participate in the study were asked for assent to be interviewed for the first time at Wave 2. Similarly, Wave 1 youth respondents who reached age 18 by Wave 2 were asked to complete the adult interview.

In addition to the members of the Wave 1 Biomarker Core, the study flagged a subset of other Wave 1 adult respondents who had provided urine at Wave 1 for urine collection at Wave 2. Those who completed the PATH Study adult interview for the first time (referred to as “first-time adult interview respondents”) were asked to provide urine and blood specimens at Wave 2.

Wave 1 respondents were eligible for Wave 3 if they were still residents of the United States and not incarcerated. Wave 2 nonrespondents were asked to participate at Wave 3 unless they were deceased, had moved out of the United States permanently, were incarcerated long term, had specifically requested withdrawal from the study, gave a firm or hostile refusal at Wave 2, had a language barrier,⁵ or had a physical or mental disability or chronic illness that prevented participation in the study. Shadow youth from previous waves who reached age 12 by Wave 3 and were permitted by a parent or guardian to participate in the study were asked for assent to be interviewed for the first time at Wave 3. Similarly, youth from previous waves who reached age 18 by Wave 3 were asked to complete the adult interview.

In addition to the members of the Wave 1 Biomarker Core, the study flagged a subset of other adults who had provided urine at a previous wave for urine collection at Wave 3. Those who completed the PATH Study adult interview for the first time at Wave 3 were asked to provide urine and blood specimens.

⁵ The PATH Study interviews can be conducted in English or Spanish. At Wave 2, the cases assigned a final nonresponse status due to language problems included two groups: (a) Wave 1 respondents who mentioned difficulty completing the Wave 1 interview in English or Spanish as their main reason for not completing an interview at Wave 2, and (b) Wave 1 shadow youth who did not feel comfortable completing an interview in English or Spanish when asked to do so by the study for the first time.

2.3 Wave 4

Wave 1 respondents were eligible for Wave 4 of the PATH Study if they were still residents of the United States and not incarcerated. Nonrespondents from previous waves were asked to participate at Wave 4 unless they were deceased, had moved out of the United States permanently, were incarcerated long term, had specifically requested withdrawal from the study, gave a firm or hostile refusal at a previous wave, had a language barrier, or had a physical or mental disability or chronic illness that prevented participation in the study. In addition, those who had refused participation at both Wave 2 and Wave 3 were not asked to participate.⁶ (Note that refusal is a specific type of nonresponse, so some nonrespondents to Wave 2 and Wave 3 who did not refuse at both those waves were asked to participate at Wave 4.) Shadow youth from previous waves who reached age 12 by Wave 4 and were permitted by a parent or guardian to participate in the study were asked for assent to be interviewed for the first time at Wave 4. Similarly, youth from previous waves who reached age 18 by Wave 4 were asked to complete the adult interview and to provide urine and blood specimens. In addition to the members of the Wave 1 Biomarker Core, the study flagged a subset of other adults who had provided urine at a previous wave for urine collection at Wave 4. Unlike at previous waves, youth interview respondents were asked to provide a urine specimen.

At Wave 4, a probability replenishment sample was selected from the CNP to account for the aging of Wave 1 respondents and sample attrition among the Wave 1 Cohort. The Wave 4 replenishment sample also provided the opportunity for a panel conditioning assessment to monitor the potential effects of participating in the PATH Study over time.⁷ As part of the Wave 4 replenishment effort, adults, youth, and shadow youth (ages 10 and 11) were sampled within the existing PATH Study sample segments from among the addresses not selected for Wave 1.

The Wave 4 within-household sampling procedures mirrored those used at Wave 1. However, the sampling rates were designed to bring the Wave 4 adult and youth sample sizes up to Wave 1 levels by sampling domain. This required consideration for the expected combined effect of the aging of

⁶ For youth, refusal could have come from the youth or their parent/guardian.

⁷ Panel conditioning is a term used to describe the situation where the reported attitudes and/or behaviors of survey respondents who have been retained in a study for an extended period of time begin to be influenced by study participation itself. After the Wave 4 replenishment, such an assessment was conducted comparing PATH Study estimates between two portions of the Wave 4 Cohort: those sampled at Wave 4, and those sampled at Wave 1. The results of this analysis are presented in Opsomer et al. (2023).

Wave 1 respondents and the loss of sample size due to attrition on the PATH Study sample by the time of Wave 4. In particular, it was necessary to oversample adults who use tobacco given that the Wave 1 shadow youth and youth respondents who became adults after Wave 1 (and those who will continue to do so) were not oversampled with respect to tobacco-use status. All adult interview respondents from the replenishment sample were asked to provide urine and blood specimens, and all youth interview respondents from this sample were asked to provide a urine specimen.

2.4 Wave 4.5

Wave 4.5 was a special data collection of the PATH Study; its focus was on continuing the annual collection of data among youth ages 12 to 17 (and their parents). There was no additional sampling for Wave 4.5 and no biospecimens were collected. All study participants were eligible for Wave 4.5 if they were ages 12 to 17, still residents of the United States, and not incarcerated. Nonrespondents from previous waves were asked to participate at Wave 4.5 unless they were deceased, had moved out of the United States permanently, were incarcerated long term, had specifically requested withdrawal from the study, gave a firm or hostile refusal at a previous wave, had a language barrier, or had a physical or mental disability or chronic illness that prevented participation in the study. In addition, those who had refused participation at any two consecutive waves were not asked to participate. Shadow youth from previous waves who reached age 12 by Wave 4.5 and were permitted by a parent or guardian to participate in the study were asked for assent to be interviewed for the first time at Wave 4.5.

2.5 Wave 5

There was no additional sampling for Wave 5, and all study members were eligible if they were still residents of the United States and not incarcerated. Nonrespondents from previous waves were asked to participate at Wave 5 unless they were deceased, had moved out of the United States permanently, were incarcerated long term, had specifically requested withdrawal from the study, gave a firm or hostile refusal at a previous wave, had a language barrier, or had a physical or mental disability or chronic illness that prevented participation in the study. In addition, those who had

refused participation at any two consecutive waves were not asked to participate.⁸ Shadow youth from previous waves who reached age 12 by Wave 5 and were permitted by a parent or guardian to participate in the study were asked for assent to be interviewed for the first time at Wave 5 and to provide a urine specimen. Urine specimens were also requested from Wave 5 youth who completed their first youth interview at the Wave 4.5 special data collection (when no biospecimens were requested) or who consented to urine specimen collection at Wave 4.⁹ Similarly, participants who completed their first adult interview at Wave 5 were asked to provide urine and blood specimens. Urine specimens were also requested from all adults belonging to the Wave 1 Biomarker Core.

2.6 Wave 5.5

Wave 5.5 was a special data collection for members of the Wave 4 Cohort, and only youth ages 13 to 17 and young adults ages 18 and 19 at the time of Wave 5.5 were interviewed. By design (see footnote 1), all PATH Study participants were expected to be at least age 13 at Wave 5.5. There was no additional sampling for Wave 5.5 and no biospecimens were collected. All Wave 4 Cohort members ages 13 to 19 at Wave 5.5 were eligible if they were still residents of the United States and not incarcerated. Nonrespondents from previous waves were asked to participate at Wave 5.5 unless they were deceased, had moved out of the United States permanently, were incarcerated long term, had specifically requested withdrawal from the study, gave a firm or hostile refusal at a previous wave, were unable to complete a previous interview in English or Spanish, or had a physical or mental disability or chronic illness that prevented participation in the study. In addition, those who had refused participation at Wave 4.5 and Wave 5 were not asked to participate.

Wave 5.5 began on December 1, 2019, with the same in-person data collection methods used at prior waves of the PATH Study and was originally scheduled to continue until November 30, 2020. However, in-person data collection was suspended on March 17, 2020, due to the Coronavirus

⁸ A refusal at Wave 4.5 was counted in the calculation of “two consecutive waves” of refusal for participants who were part of that special data collection.

⁹ In addition to the main groups of Wave 5 youth respondents described above, urine specimens were also requested from approximately 60 youth respondents who did not complete the Wave 4 youth interview or did not consent to urine specimen collection at Wave 4, where the reason for nonresponse or lack of consent was something other than a refusal.

Disease (COVID-19) pandemic. Data collection resumed via telephone on July 3, 2020, and continued until December 31, 2020.

If the data collected in person, before COVID-19 and related restrictions were widespread in the United States, had been simply pooled with the data obtained in the latter portion of 2020 via telephone, the resulting Wave 5.5 estimates would have the potential for bias. This is because even though the combined data would represent tobacco-use behavior over most of the year, it would not include the 15-week period during which U.S. COVID-19 cases were rising and most state populations were under stay-at-home orders; such factors may have influenced tobacco-use behaviors. For this reason, it was decided that only data collected via telephone would be used to produce Wave 5.5 estimates, and the target population would be redefined to cover the latter portion of 2020 (specifically, July 3, 2020, to December 31, 2020). However, the PATH Study participants eligible for Wave 5.5 were scheduled for interviews approximately 1 year after their Wave 5 interview, and not in random subsamples. This meant that the participants scheduled for interviews in the latter portion of the year were not a random subsample of those eligible for Wave 5.5, so analyses of data collected from only these participants could also result in biased estimates. To overcome this concern, all participants who completed a Wave 5.5 interview in person before data collection was suspended and were still age eligible for Wave 5.5 (i.e., ages 13 to 19) were re-contacted for an interview by telephone starting August 3, 2020.¹⁰

During the data collection hiatus, some of the 19-year-olds interviewed in person reached age 20, and others would reach age 20 after the resumption of data collection. To ensure that the in-person respondents re-contacted by telephone would still be age eligible for Wave 5.5, August 31, 2020, was used as a date of demarcation to allow some older 19-year-olds (e.g., those with 20th birthdays in September) to be part of Wave 5.5. Those participants ages 13 to 19 on August 31, 2020, who completed the Wave 5.5 interview in person were eligible for the Wave 5.5 telephone re-interview.

¹⁰ Most participants whose in-person interview status had been finalized as “nonrespondent” prior to the resumption of data collection on July 3, 2020, were participants who refused to respond to the interview. Because it was presumed that these participants would also refuse a request to complete a telephone interview, it was decided that no participants finalized as nonrespondent prior to the resumption of data collection would be re-contacted for a Wave 5.5 telephone interview. These cases were accounted for in the weighting process (see Appendix section B.7 of the RUFs User Guide for details).

More generally, participants from the Wave 4 Cohort who were ages 19 and younger¹¹ on August 31, 2020, were eligible for Wave 5.5 data collection in the latter portion of 2020.

Only data collected via telephone interview are included in the Wave 5.5 data files. The target population for the Wave 4 Cohort at Wave 5.5 is the resident (not incarcerated) U.S. population who were ages 13 to 19 in the latter portion of 2020 and part of the CNP at the time of Wave 4.

2.7 PATH-ATS

The PATH-ATS was a special data collection for which a subsample of adults ages 20 and older were interviewed. No biospecimens were collected for the PATH-ATS. The survey was developed in 2020 in light of new Federal tobacco policies, such as the nationwide Tobacco 21 policy and flavor restrictions, and the COVID-19 pandemic and its potential impact on the way people used and purchased tobacco. The objectives of the PATH-ATS were to (1) provide data that can be used to understand the impact of the Federal policy and regulatory changes in 2020; (2) provide data that can be used to assess the relationship between tobacco use and the COVID-19 pandemic; and (3) fill an unanticipated gap¹² in nationally representative data on adult tobacco use, prevalence, cessation, initiation, and relapse.

The PATH-ATS was conducted from September 10, 2020, to December 20, 2020. Because the planned data collection period was shorter than most PATH Study data collection efforts, only a subsample of participants were interviewed. Participants eligible for the PATH-ATS subsample were ages 20 and older on August 31, 2020, part of the Wave 4 Cohort, and respondents to the Wave 5 adult interview. A stratified random sample of 18,601 PATH Study participants was selected for the PATH-ATS from a total of 31,343 eligible participants. Those who at Wave 5 were ages 20 to 24, used electronic nicotine delivery systems (ENDS), smoked cigarettes, and more recently used tobacco were oversampled.

¹¹ This includes some participants who were age 12 on August 31, 2020; these participants were contacted for an interview on or after their 13th birthday.

¹² Although the PATH Study did not originally plan to collect data on adults ages 20 and older in 2020, the COVID-19 pandemic also interrupted planned data collections for other national surveys used to understand tobacco use.

The target population for PATH-ATS estimates based on the Wave 4 Cohort is the resident (not incarcerated) U.S. population ages 20 and older in the latter portion of 2020 who were in the CNP at the time of Wave 4 and the resident (not incarcerated) U.S. adult population at Wave 5.

In light of the pandemic, it was of analytic interest to learn how the public health crisis affected the full target population(s) of the PATH Study, adults as well as youth. As noted above, the PATH-ATS was designed to complement the Wave 5.5 sample consisting of those ages 13 to 19 who were interviewed by telephone in the latter portion of 2020. Because the same date of demarcation was applied to determine age eligibility for the Wave 5.5 telephone interview and the PATH-ATS, the two samples are mutually exclusive in terms of their coverage of adults and jointly exhaustive for the adult population. Thus, the Wave 5.5 and the PATH-ATS data may be combined for longitudinal and cross-sectional analyses for the Wave 4 Cohort.

2.8 Wave 6

There was no additional sampling for Wave 6 and no biospecimens were collected; all study members were eligible if they were still residents of the United States and not incarcerated. Nonrespondents from previous waves were asked to participate at Wave 6 unless they were deceased, had moved out of the United States permanently, were incarcerated long term, had specifically requested withdrawal from the study, gave a firm or hostile refusal at a previous wave, had a language barrier, or had a physical or mental disability or chronic illness that prevented participation in the study. In addition, those who had refused participation at any two consecutive waves or otherwise had not responded at three consecutive waves were not asked to participate.¹³

Both in-person data collection and telephone data collection were conducted for Wave 6. Due to the COVID-19 pandemic, and for the continued safety of PATH Study participants and field interviewers, Wave 6 data collection began with telephone interviewing. As conditions improved in certain parts of the country, in-person interviews began on May 7, 2021. Where and when in-person visits were permitted by Westat, the National Institute on Drug Abuse (NIDA) and the Food and

¹³ Refusals and other nonresponse at Wave 4.5 or Wave 5.5 for participants who were part of those special data collections were counted in the calculations of “two consecutive waves” of refusal or “three consecutive waves” of nonresponse. However, for youth and adults ages 18 to 19 approached for re-interview by telephone at Wave 5.5 and for adults selected for the PATH-ATS, refusing or nonresponding to the interview was not counted in the calculation of “two consecutive waves” of refusal nor “three consecutive waves” of nonresponse.

Drug Administration (FDA), and local jurisdictions, in-person data collection was given priority, though each participant made the final decision regarding their interview mode. All in-person contacts with participants were conducted in compliance with local and state restrictions for COVID-19 mitigation. Of the 36,168 Wave 6 interviews, 69 percent (24,792) were conducted by telephone. Among all telephone interviews at Wave 6, 54 percent (13,297) were conducted before May 7, 2021.

2.9 Wave 7

The sample was replenished again at Wave 7, and a new cohort was formed. The main objectives of the Wave 7 sample design were to replenish the sample to account for attrition and the aging of participants. Particularly, at the start of Wave 7, the youngest participants in the Wave 4 Cohort were primarily 15 years old, so the replenishment effort provided a means of selecting a new sample of youth ages 12 to 14, in addition to a new shadow sample of youth ages 9 to 11 to be interviewed after they reach age 12 at later waves. The replenishment also provided the opportunity for a second panel conditioning assessment.

Participants from the Wave 1 Cohort and/or Wave 4 Cohort were eligible for Wave 7 if they were still U.S. residents and not incarcerated. Nonrespondents from previous waves were asked to participate at Wave 7 unless they were deceased, had moved out of the United States permanently, were incarcerated long term, had specifically requested withdrawal from the study, gave a firm or hostile refusal at a previous wave, were unable to complete a previous interview in English or Spanish, or had a physical or mental disability or chronic illness that prevented participation in the study. In addition, participants with two consecutive refusals or three consecutive waves of any type of nonresponse were not asked to participate.¹⁴ Among the Wave 4 Cohort members at Wave 7, 28,165 completed the adult interview, 3,452 completed the youth interview, and 311 completed a Wave 7 interview but were not included in the Wave 7 Cohort because they were younger than age 15 and/or not in the Wave 7 CNP at the time of the interview (see Figure 1).

A probability replenishment sample of adults, youth, and shadow youth (ages 9 to 11) was selected from the CNP at Wave 7, including persons who were not in the CNP at the time of Wave 4 (such

¹⁴ Starting at Wave 6, participants whose three previous consecutive waves were any type of nonresponse were not asked to participate at the current wave or at future waves.

as recent immigrants or those returning from the military). Only members of the Wave 4 Cohort who responded, were at least age 15, and were in the CNP at the time of Wave 7 were combined with the Wave 7 replenishment sample to form the new Wave 7 Cohort. As a result, the target population for the Wave 7 Cohort is the CNP who were ages 9 and older at the time of Wave 7.

As part of the Wave 7 replenishment effort, adults, youth, and shadow youth were sampled within the existing PATH Study sample segments from among the addresses not selected for Wave 1 or Wave 4. A total of 244,338 mailing addresses were selected and randomly divided into three subsamples. A subset of 111,423 addresses (or close to 45 percent) were screened solely to identify young youth ages 9 to 14, and another subset of 96,752 addresses (or close to 40 percent) were screened to identify youth ages 9 to 17. The remaining 36,163 addresses were screened for adults, youth, and shadow youth ages 9 to 11. These subsamples are referred to as the “YYO” (young youth only), “YO” (youth only), and “AYS” (adults, youth, and shadow youth) replenishment subsamples, respectively.

The Wave 7 within-household sampling procedures were similar to those used at Wave 1 and Wave 4, with sampling rates varying for adults by age, race, and tobacco-use status.¹⁵ The within-household sampling rates were designed to bring the Wave 7 adult and youth sample sizes up to a level commensurate with Wave 4 sample sizes. This required consideration for the expected combined effect of the aging of continuing participants and the loss of sample size due to attrition by the time of Wave 7. The number of responding adults from the continuing sample at Wave 7 was projected to be very close to Wave 4 levels without replenishment. However, small samples of adults were selected from each of the sampling domains to allow representation of immigrants and other adults who had joined the CNP since Wave 4, and to support a panel conditioning assessment to monitor the potential effects of participating in the PATH Study over time. The numbers of targeted adults made it feasible to sample adults at Phase 1 based only on age and race, and to subsample at Phase 2 based on self-reported tobacco use, in addition to self-reported age and race.

In person was the preferred data collection mode for Wave 7, with the option to conduct interviews by telephone as needed due to any public health considerations. In addition, a small number of

¹⁵ The main difference between the definition of tobacco use for sampling adults at Waves 1, 4, and 7 is that “e-cigarettes” were replaced by “electronic nicotine products” starting at Wave 4, and ever use of IQOS, nicotine pouches, and other types of oral nicotine products were added at Wave 7.

participants recruited at earlier waves completed a Web version of the Wave 7 interview as part of a Web pilot test (see Section 3.10.4 of the PATH Study RUFs User Guide). Biospecimen collection was resumed at Wave 7 with procedures mirroring those used at Wave 1. Subsamples of adults recruited at earlier waves and at Wave 7 were asked for urine, blood, or both specimens until blood collection was discontinued on March 4, 2022. A subsample of youth recruited at earlier waves were asked for urine specimens. Youth from the Wave 7 replenishment sample were not asked for a urine specimen at Wave 7.

Table 2-1. PATH Study data and biospecimen collection summary by wave, cohort, and interview type

Wave	Data collection start date	Data collection end date	Cohort ^a	Interview type ^b	Interviews conducted	Urine specimens collected ^c	Blood specimens collected ^d
1	September 12, 2013	December 14, 2014	Wave 1 Cohort	Adults	32,320	21,801	14,520
				Youth	13,651	N/A	N/A
2	October 23, 2014	October 30, 2015	Wave 1 Cohort	Adults	28,362	13,696	908
				Youth	12,172	N/A	N/A
3	October 19, 2015	October 23, 2016	Wave 1 Cohort	Adults	28,148	14,979	835
				Youth	11,814	N/A	N/A
4	December 1, 2016	January 3, 2018	Wave 1 Cohort	Adults	27,757	16,131	890
				Youth	11,059	9,892	N/A
			Wave 4 Cohort	Adults	33,644	20,928	3,602
				Youth	14,793	13,095	N/A
4.5	December 1, 2017	December 1, 2018	Wave 1 Cohort	Youth	8,761	N/A	N/A
			Wave 4 Cohort	Youth	12,918	N/A	N/A
5	December 1, 2018	November 30, 2019	Wave 1 Cohort	Adults	28,970	11,270	1,640
				Youth	6,760	5,948	N/A
			Wave 4 Cohort	Adults	32,687	11,429	1,929
				Youth	11,976	10,490	N/A
5.5	July 3, 2020 ^e	December 31, 2020	Wave 4 Cohort	Adults	3,628	N/A	N/A
				Youth	7,129	N/A	N/A
PATH-ATS	September 10, 2020	December 20, 2020	Wave 4 Cohort	Adults	8,874	N/A	N/A
6	March 1, 2021	November 30, 2021	Wave 1 Cohort	Adults	25,604	N/A	N/A
				Youth	2,377	N/A	N/A
			Wave 4 Cohort	Adults	29,516	N/A	N/A
				Youth	5,585	N/A	N/A
7	January 6, 2022	April 2, 2023	Wave 1 Cohort	Adults	24,408	10,839	486
				Youth	899	122	N/A
			Wave 4 Cohort	Adults	28,165	11,951	540
				Youth	3,452	497	N/A
			Wave 7 Cohort	Adults	29,780	12,485	546
				Youth	10,650	483	N/A

^a As illustrated in Figure 1, there is substantial overlap between the members of the Wave 1 Cohort, the Wave 4 Cohort, and the Wave 7 Cohort, and therefore in the counts of interviews conducted by cohort (for relevant waves).

^b Only Wave 4 Cohort adults ages 18 and 19, and Wave 4 Cohort youth ages 13 to 17 completed interviews at Wave 5.5. Only a subsample of Wave 4 Cohort adults ages 20 and older were interviewed for the PATH-ATS. At Wave 6, all interview respondents were at least age 14. For other waves, there was no cohort restriction or subsampling, and the age range was 18 and older for adults and 12 to 17 for youth.

^c The PATH Study requested urine specimens from: all first-time adult interview respondents at Waves 1 to 4 and Wave 5; subsets of other adult respondents at Waves 2 to 4 and Wave 5; a subset of all adult interview respondents (including first-time adult interview respondents) at Wave 7; all youth at Wave 4; a subset of youth respondents at Wave 5; and a subset of Wave 7 replenishment sample youth at Wave 7.

^d The PATH Study requested blood specimens from first-time adult interview respondents at Waves 1-4 and Wave 5. At Wave 7, the PATH Study requested blood specimens from a subset of all adult interview respondents until blood collection was discontinued on March 4, 2022.

^e Data collection for Wave 5.5 originally began in person on December 1, 2019, but was suspended due to the COVID-19 pandemic. The released data for this wave were collected via telephone starting on July 3, 2020.

3. Methodology

This chapter first discusses the factors affecting nonresponse bias, and then describes the methods used for evaluating potential nonresponse bias in this report.

3.1 Factors Affecting Nonresponse Bias

Bias is the difference between a survey estimate and the actual population value. Although nonresponse bias can be a major concern in multi-stage household surveys, nonresponse does not necessarily lead to nonresponse bias in survey estimates. Assuming nonresponse to be a fixed property of an individual in the population, the nonresponse bias of an estimate can be expressed mathematically to show the relationship between the bias and two factors—the amount of nonresponse and the difference between respondents and nonrespondents (Groves, 2006):

$$Bias(\bar{y}_r) = (1 - r) \times (\bar{Y}_r - \bar{Y}_m)$$

where $\bar{y}_r$ is the estimated population mean based on the sample respondents only, r is the response rate in the target population, $\bar{Y}_r$ is the mean for respondents in the target population, and $\bar{Y}_m$ is the mean for nonrespondents in the target population.

In practice, nonresponse is not necessarily a fixed property of individuals in the population and the magnitude of nonresponse bias depends on the correlation between response propensity and the measure of interest. Within the same survey, different estimates can be subject to different levels of nonresponse bias. Some measures, unrelated to the propensity to respond, can be immune from the biasing effect of nonresponse; others, in the same survey, can be subject to large biases (Groves, 2006). For example, if people who smoke cigarettes were less inclined to respond to the PATH Study than those who do not smoke cigarettes, then the estimated prevalence of cigarette smoking would be subject to bias, prior to any weighting adjustments for nonresponse. However, if people who use and do not use smokeless tobacco were equally likely to respond to the study, then the estimated prevalence of smokeless tobacco use would not be biased due to nonresponse. If propensity to respond is completely random, then nonresponse reduces sample sizes for analysis but does not lead to bias, even if the overall response rate is lower than desired.

Survey practitioners often attempt to decrease potential nonresponse bias by increasing the overall response rate and improving cooperation from “difficult to reach” subgroups. Effective statistical adjustments can also help reduce potential nonresponse biases in survey estimates.

3.2 Analyses for Evaluating Wave 7 Potential Nonresponse Bias

It is not always possible to measure the actual bias due to nonresponse; however, various approaches can be used to help identify potential sources of nonresponse bias. At Wave 7, the Wave 1 Cohort data and Wave 4 Cohort data serve the purpose of longitudinal analysis, whereas the Wave 7 Cohort data support cross-sectional analyses. Separate sets of analyses were conducted for the three cohorts to assess potential nonresponse bias in estimates using interview data from Wave 7 of the PATH Study.

3.2.1 Weights and Corresponding Analysis Perspective Used by Cohort

As described in Appendix section B.10 of the PATH Study RUFs User Guide, the study provides four different types of weights for researchers analyzing Wave 7 data: a cross-sectional weight, a single-wave weight, an all-waves weight, and a special-collection all-waves weight. A cross-sectional weight was created for all Wave 7 Cohort members. Separately for the Wave 1 and Wave 4 Cohorts, a single-wave weight was created for all cohort members who responded at Wave 7. Similarly, a separate all-waves weight for each of the two cohorts was created for Wave 7 respondents who also responded at each primary wave relevant to the cohort. A special collection all-waves weight was created for each of the first two cohorts and applies to Wave 7 respondents who also responded at all previous waves relevant to the cohort and at all previous special data collections as appropriate. Weighting adjustments are often used to account for differential response propensities across population subgroups. Among numerous sources, the United Nations handbook on household surveys (2005, Chapter 6) and Särndal and Lundström (2005) discuss the methods and theory of using weight adjustments for nonresponse.

In terms of conducting longitudinal analyses using either the Wave 1 Cohort data or the Wave 4 Cohort data, the single-wave weight has less utility than the all-waves weight. The single-wave weight allows researchers to relate a behavior or outcome at a later wave (Wave *n*) to characteristics of the same person at the first wave of a cohort. An example of an analysis that would use the single-wave

weight for Wave 7 data from the Wave 1 Cohort would be estimating the proportion of people who used e-cigarettes at Wave 1 among those who used e-cigarettes at Wave 7. This is a longitudinal analysis using the participants who have data at both Wave 1 and Wave 7. However, if it is desired to relate Wave *w* outcomes to characteristics from an earlier wave that is not the first wave of a cohort then the all-waves weight is needed. An example of an all-waves analysis would be using logistic regression to estimate a person's e-cigarette use at Wave 7 from the person's e-cigarette use and demographic characteristics at Waves 1 through 6.

As a result, and as waves of data collection accrue, the preponderance of analyses for the Wave 1 and Wave 4 Cohorts are likely to use all-waves respondents and their corresponding weights. To address the majority of analyses the research community will likely conduct using Wave 7 data from the first two cohorts, the evaluation of potential nonresponse bias in estimates based on the Wave 7 interview data for either the Wave 1 Cohort or the Wave 4 Cohort will be from an all-waves perspective.

The Wave 7 Cohort will be used for cross-sectional analyses at Wave 7 (like those presented in Sections 6.2.3, 6.2.4, 6.3.3, and 6.3.4).

3.2.2 Comparison of Response Rates Across Subgroups

Although a response rate does not yield direct estimates of potential nonresponse biases on key measures, examining response rates by subgroups (for example, males versus females) may reveal sources of potential nonresponse bias. Wave 7 interview response rates were calculated for the Wave 1 Cohort, the Wave 4 Cohort, and the Wave 7 replenishment sample separately, as described below, using the RR3 response rate formula provided by the American Association for Public Opinion Research (AAPOR, 2023).

Wave 7 all-waves response rates for the Wave 1 and Wave 4 Cohorts

For the Wave 1 Cohort, the Wave 7 interview all-waves response rates are conditioning on Wave 1 response, so the subgroups are based on Wave 1 information. Analogously, the Wave 7 interview all-waves response rates for the Wave 4 Cohort are conditioning on Wave 4 response, so most of the subgroups are based on Wave 4 information—the exceptions are the recruitment wave subgroups,

which distinguish cohort members recruited at Wave 1 from those recruited at Wave 4. This should be kept in mind when reviewing the response rates in Sections 4.1.1, 4.2.1, 5.1.1, and 5.2.1.

We introduce the adjective “continuing” when referring to study participants who were recruited at Wave 1 or Wave 4. For the continuing sample participants at Wave 7, those ages 18 and older were asked to complete an adult interview and those younger than age 18 were asked to complete a youth interview. Wave 7 continuing sample nonrespondents do not have a Wave 7 interview date, so their Wave 7 age and interview eligibility (i.e., adult interview versus youth interview) were determined using the latest available date of birth or age information. Prior to Wave 7 data collection, each continuing sample participant was assigned an “anniversary month” for Wave 7, based primarily on the most recent interview date of the study member(s) in their household from prior waves. For most previous waves, the age classification date for nonrespondents was 1 month after the last day of their anniversary month or the final date of the data collection, whichever was earlier. However, because the final date of Wave 7 data collection was extended from December 31, 2022, to April 2, 2023, to meet completed interview targets, a different approach was required. Wave 7 continuing sample nonrespondents were assigned an age classification date based on whether they were asked to participate or not (see Section 2.9). The age classification date for nonrespondents not asked to participate was 1 month after the last day of their anniversary month, as described above. For Wave 7 continuing sample nonrespondents asked to participate, the date of assigning a final disposition of nonresponse was used as the age classification date. Nonrespondents were included in the Wave 7 youth analyses if their age was determined to be younger than 18 on the age classification date or included in the Wave 7 adult analyses if their age was determined to be 18 or older on the age classification date.

Both weighted and unweighted response rates are reported for the adult and youth interviews. The unweighted response rate measures the success of survey operations in obtaining responses from the sampled persons. The weighted response rate estimates the proportion of the population represented by the sampled persons that would have responded if they all had been asked to participate in the study, and thus provides a measure of the potential impact of nonresponse on the study estimates.

The Wave 1 IPS weight was used for calculating the Wave 1 Cohort weighted all-waves response rates. A Wave 4 IPS weight does not exist for the Wave 4 Cohort. As discussed in Chapter 2, the

Wave 4 Cohort consists of the Wave 4 adult and youth respondents from the Wave 1 sample (except those not in the CNP at Wave 4) and the respondents from the replenishment sample. The Wave 1 sample and the Wave 4 replenishment sample were selected at different times from two different sampling frames, so the probabilities of selection (and IPS weights) were computed for the two samples separately. An adjustment to the IPS weights computed for the separate samples is needed for persons with multiple chances of selection for the Wave 4 Cohort. The information needed to make this adjustment (e.g., Wave 1 age, race, and tobacco-use status from which the probability of an adult being selected at Wave 1 could be determined) is known only for some Wave 4 replenishment sample respondents and is unknown for persons from the replenishment sample who did not respond at Wave 4. Because the Wave 4 cross-sectional weight is the closest available approximation to an IPS weight for the Wave 4 Cohort,¹⁶ the Wave 4 cross-sectional weight was used for calculating the Wave 4 Cohort weighted response rates.

Wave 7 replenishment sample response rates

As described in Chapter 2, the Wave 7 Cohort is comprised of participants recruited into the PATH Study at different times (Wave 1, Wave 4, or Wave 7). Members of the Wave 7 replenishment sample were asked to participate in the study for the first time at Wave 7; whereas those sampled at Wave 1 or Wave 4 had received multiple follow-up requests. As demonstrated by results for the Wave 4 Cohort presented in Sections 5.1.1, 5.2.1, and Appendix C, differences in response status prior to Wave 7 led to large differences in Wave 7 response rates. Furthermore, follow-up wave response rates are conditioning on Wave 4 response for those in the Wave 4 Cohort, whereas response rates for the adults and youth from the Wave 7 replenishment sample are conditioning on completion of the Wave 7 household screener. It is clear that the Wave 7 response rates for the different groups of study members comprising the Wave 7 Cohort are conceptually different, so no attempt was made to compute “blended” response rates for the Wave 7 Cohort.¹⁷

¹⁶ The sum of the Wave 4 cross-sectional weights provides the closest approximation to the Wave 4 Cohort target population at Wave 4, i.e., the CNP ages 10 and older as of Wave 4. Details on the creation of the Wave 4 cross-sectional weight can be found in Appendix sections B.4.3 and B.4.4 of the PATH Study RUFs User Guide.

¹⁷ The response propensities of the component of the Wave 7 Cohort that comes from the Wave 4 Cohort are reflected in the Wave 4 Cohort single-wave response rates provided in Appendix C, and separate response rates are provided for the Wave 7 replenishment sample (as described below).

The Wave 7 household screener response rates in the main body of this report were computed for the entire Wave 7 replenishment sample, as well as the Wave 7 AYS replenishment sample (from which all Wave 7 replenishment sample adults were selected). Wave 4 Cohort members who were younger than age 15 and/or not in the CNP at the time of Wave 7 contribute to the Wave 4 Cohort response rates but do not belong to the Wave 7 Cohort. Because this group includes a relatively small number of cases (see Figure 2-1), their inclusion or exclusion has no meaningful impact on Wave 7 response rates.

In addition to the overall household screener response rates for the entire Wave 7 replenishment sample and the Wave 7 AYS replenishment address subsample, response rates are reported for subgroups defined by geography or neighborhood characteristics. For the Wave 7 replenishment sample, adult and youth interview response rates are reported for subgroups based on information collected through the household screener (e.g., age of sampled person) or the location of the selected housing unit.

3.2.3 Comparisons Using Data from Cohort's First Wave

In a longitudinal study, the information collected for respondents at the cohort's first wave is available regardless of whether they are respondents or nonrespondents at future waves, and thus can be used for evaluating potential nonresponse bias (Bose & West, 2002; Javitz & Wagner, 2005; Brownstein et al., 2009). For the PATH Study Wave 1 Cohort, estimates of Wave 1 demographic, socio-economic, and tobacco-use characteristics based on Wave 7 all-waves respondents were compared to estimates of the same Wave 1 characteristics based on cohort members eligible to be Wave 7 all-waves respondents. Analogously, for the Wave 4 Cohort, estimates of Wave 4 demographic, socio-economic, and tobacco-use characteristics based on Wave 7 all-waves respondents were compared to estimates of the same Wave 4 characteristics based on cohort members eligible to be Wave 7 all-waves respondents.

For both cohorts, this comparison helps identify characteristics that might be associated with bias due to nonresponse at any primary wave since the cohort's first wave, after compensating for possible undercoverage of the target population (due to sampling frame deficiencies) and nonresponse at the cohort's first wave. Participants known to have been ineligible at any primary wave after the cohort's first wave were excluded from the all-waves response rates and nonresponse

bias analyses in this report. The focus of both analyses is on assessing bias due to nonresponse by participants known to be eligible or whose eligibility status was unknown.

Wave 1 Cohort

One set of Wave 1 Cohort estimates was computed based on the cohort members who were eligible to be Wave 7 all-waves respondents. These estimates were computed using the Wave 1 cross-sectional weight, which was designed to reduce potential nonresponse bias at Wave 1.

Two sets of Wave 1 Cohort estimates were computed based on the Wave 7 all-waves respondents. The first set of estimates used the Wave 1 cross-sectional weight. Comparing this set of estimates to the estimates based on cohort members eligible to be Wave 7 all-waves respondents (which also used the Wave 1 cross-sectional weight) shows the representativeness of the Wave 7 all-waves respondents *before* statistical adjustment for nonresponse at any primary wave since Wave 1.

However, the study provides a Wave 7 all-waves weight for the Wave 1 Cohort that was specifically designed to reduce potential bias due to nonresponse at any primary wave since Wave 1. The Wave 6 nonresponse-adjusted all-waves weights for the Wave 1 Cohort (prior to raking and trimming) served as the initial weights for developing the Wave 7 all-waves weights. The initial weights were adjusted to account for nonresponse to the Wave 7 interview, and the resulting weights were raked to control totals. Some of the control totals came from the 2013 ACS; others involving tobacco use were sample-based rather than population-based and reflected estimated Wave 1 population characteristics. Raking to sample-based control totals, often employed in longitudinal studies (see, for example, Brick, Lê, & West (2003)), can limit “drifting” in some important baseline characteristics that might arise through the applications of nonresponse adjustments over time. Lundström and Särndal (1999) provide theoretical discussions about sample-based calibration together with empirical evidence that such calibration can help reduce both variance and nonresponse bias. General discussions about the calibration method can be found in Särndal and Lundström (2005) and Särndal (2007). Details about the construction of the Wave 7 all-waves weight for the Wave 1 Cohort appear in Appendix section B.9.1 of the PATH Study RUFs User Guide.

Therefore, the second set of estimates based on the Wave 7 all-waves respondents used the Wave 7 all-waves weight for the Wave 1 Cohort. Comparing this set of estimates to the estimates based on

Wave 1 Cohort members eligible to be Wave 7 all-waves respondents shows the representativeness of the Wave 7 all-waves respondents *after* statistical adjustment for nonresponse at any primary wave. That is, the results of this comparison show the extent to which the statistical weighting adjustments might have reduced potential nonresponse bias in Wave 1 Cohort estimates based on the Wave 7 all-waves respondents.

For ease of reference, the terms “before Wave 7 weighting adjustment” and “after Wave 7 weighting adjustment” are used to refer to the two sets of estimates based on Wave 7 all-waves respondents and associated analyses in Sections 4.1.2 and 4.2.2.

Wave 4 Cohort

For the Wave 4 Cohort, the analyses are nearly identical to the ones described above for the Wave 1 Cohort, except that they involve Wave 4 estimates, the Wave 4 cross-sectional weight, and the Wave 7 all-waves weight for the Wave 4 Cohort. In addition, the population-based control totals for the Wave 7 all-waves weight for the Wave 4 Cohort came from the 2016 ACS. Details about the construction of the Wave 7 all-waves weight for the Wave 4 Cohort appear in Appendix section B.10.4 of the PATH Study RUFs User Guide.

Similar to the Wave 1 Cohort analyses, the terms “before Wave 7 weighting adjustment” and “after Wave 7 weighting adjustment” are used to refer to the two sets of estimates based on Wave 7 all-waves respondents and associated analyses in Sections 5.1.2 and 5.2.2.

3.2.4 Wave 7 Cohort: Comparison to External Data Sources

For the first three waves of the PATH Study, the Wave 1 Cohort can be used to produce cross-sectional estimates (which are approximate for Waves 2 and 3).¹⁸ Similarly, for Wave 4 and subsequent waves up to and including Wave 6, the Wave 4 Cohort serves the purpose of cross-sectional analysis (which is approximate after Wave 4). For Wave 7, the Wave 7 Cohort should be used to produce cross-sectional estimates. A widely used approach for assessing potential nonresponse bias in cross-sectional estimates is benchmarking to external data sources. A strength

¹⁸ See Section 6.2 of the PATH Study RUFs User Guide for further information about how the slight changes in the target population between Wave 1, Wave 2, and Wave 3 lead to cross-sectional estimates that are approximate for Waves 2 and 3 when using the Wave 1 Cohort.

of this method is that estimates from external sources are independent of the PATH Study. On the other hand, key outcomes from the PATH Study are not necessarily available from external sources; the measurements may not be exactly the same across studies, and the external sources may be subject to coverage and nonresponse errors as well (Groves, 2006).

Two types of estimates were obtained using the Wave 7 Cohort adult interview and youth interview data. First, weighted distributions of demographic and socio-economic variables were compared to those from the 2022 American Community Survey 1-year Public Use Microdata Sample (2022 ACS). Second, several PATH Study cigarette-smoking and ENDS-use¹⁹ measures were compared to those from other national studies of tobacco and health. The PATH Study measures a range of tobacco-use behaviors; many of these variables are not available in other studies. However, responses to the Wave 7 questions on cigarette smoking and ENDS use can be compared with estimates from other national studies of tobacco and health (for the most similar time frames for which data were available). Wave 7 estimates for adults and youth were obtained using the Wave 7 cross-sectional weight for the Wave 7 Cohort.

The external data sources used for analyses of cigarette smoking were the National Health and Nutrition Examination Survey 2017-prepandemic 2020 (2017-2020 NHANES), the National Youth Risk Behavior Survey 2021 (2021 NYRBS), Monitoring the Future 2022 (2022 MTF), the National Health Interview Survey 2022 (2022 NHIS), the National Survey on Drug Use and Health 2022 (2022 NSDUH), the National Youth Tobacco Survey 2022 (2022 NYTS), and the Tobacco Use Supplement to the Current Population Survey 2022 (2022 TUS-CPS). The external data sources used for comparison of ENDS use were 2021 NYRBS, 2022 MTF, 2022 NSDUH, 2022 NYTS, and 2022 TUS-CPS. Estimates from these surveys were obtained using public use files, with the exception of 2022 MTF. The variance stratum and cluster variables, which are needed to account appropriately for MTF's complex sample design when estimating the variance of estimates and corresponding confidence intervals, were available only on restricted use files. When computing estimates of youth cigarette smoking or youth ENDS use, the data from the other national studies (i.e., 2017-2020

¹⁹ Although the term “electronic nicotine products” is used in the PATH Study instrument to represent all electronic nicotine devices such as e-cigarettes, e-cigars, e-pipes, and e-hookahs, this report uses the more technical term “ENDS” because it is more commonly used in the tobacco-use literature.

NHANES, 2021 NYRBS, 2022 MTF, 2022 NSDUH, and 2022 NYTS) were restricted to ages 12 to 17 to match the age range for PATH Study youth at Wave 7.

With data collection for Wave 7 of the PATH Study occurring primarily in 2022 and given the possible effects of the COVID-19 pandemic on tobacco-use behaviors, each of the external data sources ideally would include data from after the start of the pandemic in mid-March 2020. For example, over the past 5 years, estimates of young adult ENDS use have steadily increased and estimates of adult cigarette smoking have decreased (Patrick, Miech, Johnston, & O'Malley, 2023). However, NHANES is not conducted every year and new data are not scheduled to be released in time to be included in this report.

Appendixes A and B describe the questions used to define cigarette smoking and ENDS use by these surveys, as well as the PATH Study, and outline differences in their target populations. The PATH Study's reports on nonresponse bias assessments for waves prior to Wave 5.5 included two estimates of adult current cigarette smoking based on NSDUH data—one using NSDUH's "original definition" and one using a "modified definition." (See Ryan, Trosclair, & Gfroerer, 2012, for the reasoning for creating a modified definition.) The latter adds the restriction that the adult has smoked at least 100 cigarettes in their lifetime, and is the only definition considered in this report. Although not labeled as such, the 2022 NSDUH estimates of adult current cigarette smoking therefore correspond to NSDUH estimates using the "modified definition" in PATH Study reports about nonresponse bias prior to Wave 5.5.

For both the PATH Study and the external data sources for cigarette smoking and ENDS use, item nonresponse was handled by excluding respondents with missing values for an item from the counts and estimates regarding that item. The proportions of item-missing values were generally very low: no greater than 3.1 percent for all adult estimates at the overall level, and below 2.4 percent for youth estimates at the overall level for all studies except 2017-2020 NHANES and 2021 NYRBS.²⁰ The missing data rates for youth estimates for 2017-2020 NHANES were 5.3 percent for ever tried cigarette smoking and 10.2 percent for past 30-day cigarette smoking. For 2021 NYRBS, the highest

²⁰ However, note that the 2022 NSDUH public use file included only "usable" cases that met specified minimum item response requirements.

missing data rates were 21.0 percent for ever tried cigarette smoking and 6.5 percent for past 30-day ENDS use among high school students.

3.3 Estimation Method and Software Packages

SAS/STAT software version 15.3 was used to calculate all the estimates presented in this report. Fay's balanced repeated replication (BRR) method with a factor of 0.3 was used for variance estimation with the Wave 7 data. This method accounts for the impact of the stratification and clustering involved in the study's sample design and the weighting adjustments made to the IPS weights. For estimates involving data from one of the external sources, different variance estimation methods were used according to each study's sample design. Confidence intervals were estimated using the modified Wilson approach (Wilson, 1927) with the default SAS options (SAS Institute, 2022). No adjustments were made for multiple comparisons; however, all the statistical tests were based on pre-planned (i.e., not post-hoc) comparisons and are presented in this report.

The preferred approach for testing whether two point estimates differ is to examine the confidence interval for the difference between the two point estimates. If the 95 percent confidence interval does not include zero (i.e., the lower and upper confidence interval endpoints are both negative or both positive), it can be concluded that the difference between the two estimates is statistically significant at the 0.05 significance level.²¹ This approach was used for comparisons of the type described in Section 3.2.3, which involve only PATH Study estimates. Due to the nature of these analyses, some of the point estimates and confidence interval endpoints for the difference between two estimates are very close to zero. Although it is usually of no practical importance, a negative sign appears in front of a point estimate or confidence interval endpoint when the exact value is less than zero but the displayed value rounds to zero (e.g., “-0.0%”).

Another approach is to examine whether the confidence intervals for the two point estimates overlap; if the confidence intervals for two proportions do not overlap, then the difference between the two proportions is considered statistically significant. This approach was used to compare estimates between the PATH Study and the external data sources (as described in Section 3.2.4) due

²¹ Testing whether or not a $100(1-\alpha)$ percent confidence interval for a difference between two population proportions includes zero is equivalent to a two-sided test of the null hypothesis that the difference is zero at the α significance level. See, for example, Hanushek and Jackson (1977) and Heeringa, West, and Berglund (2017, Section 5.6.1).

to the different complex sample designs used by each study. However, Schenker and Gentleman (2001) show that using this approach results in a conservative test.

4. Results of Wave 1 Cohort Nonresponse Bias Analyses at Wave 7

This chapter presents results of nonresponse bias analyses for the Wave 1 Cohort interview data at Wave 7. The nonresponse bias analyses for the Wave 1 Cohort at Wave 7 are all conditioning on Wave 1 response. Sections 4.1.1 and 4.2.1 describe how the all-waves response rates vary across select subgroups. Sections 4.1.2 and 4.2.2 compare demographic and socio-economic characteristics as well as tobacco-use status between those eligible to be Wave 7 all-waves respondents and the Wave 7 all-waves respondents. The results are presented separately for adults and youth.

4.1 Adult Interview

4.1.1 All-Waves Response Rates Conditioning on Wave 1 Response

Among those in the Wave 1 Cohort, unweighted and weighted all-waves response rates were calculated for the Wave 7 adult interview using the following formulas:

$$RR_{A_AW} = C_{A_AW} / (C_{A_AW} + N_{A_AW} + e_{A_AW} \times U_{A_AW})$$

$$e_{A_AW} = (C_{A_AW} + N_{A_AW}) / (C_{A_AW} + N_{A_AW} + I_{A_AW})$$

where

RR_{A_AW} = Wave 7 adult interview all-waves response rate;

C_{A_AW} = number of Wave 7 adults who were a respondent at each of primary Waves 2 through 7;

N_{A_AW} = number of Wave 7 adults known to be eligible at each wave, but a nonrespondent at least once for primary Waves 2 through 7;

U_{A_AW} = number of Wave 7 adults with unknown eligibility status at least once and never ineligible for primary Waves 2 through 7;

e_{A_AW} = estimated proportion of eligible adults among the Wave 7 adults with unknown eligibility status (from an all-waves perspective); and

I_{A_AW} = number of Wave 7 adults who were ineligible at least once for primary Waves 2 through 7.

Unweighted counts and weighted counts based on the Wave 1 IPS weight were obtained for the all-waves response status categories C_{A_AW} , N_{A_AW} , U_{A_AW} , and I_{A_AW} for the unweighted all-waves response rates (RR_{A_AW}) and weighted all-waves response rates ($RR_{A_W_AW}$), respectively.

Table 4-1 provides all-waves response rates for the Wave 7 adult interview. In addition to the overall row, the table includes rows for subgroups based on Wave 1 sex, age, race/ethnicity, census region, highest education level attained, health insurance status, and tobacco-use status. Participants for whom the Wave 1 interview (or Wave 1 household screener in the case of shadow youth) had missing information for the demographic, socio-economic, or tobacco-use characteristic were excluded from the response rate calculation for that subgroup. The education level and health insurance status subgroups exclude the Wave 7 adults who were youth or shadow youth at Wave 1 because they were not asked about their highest level of education attained or health insurance status at that time.

The unweighted and weighted all-waves response rates for the Wave 7 adult interview were 44.6 percent and 48.6 percent, respectively. For many subgroups the weighted all-waves response rate was at least one percentage point higher than the unweighted all-waves response rate, with Non-Hispanic White alone adults having the greatest difference between weighted (48.9 percent) and unweighted (43.4 percent) response rates. The unweighted and weighted single-wave response rates for the Wave 7 adult interview were higher than the all-waves response rates, at 50.0 percent and 52.4 percent, respectively.

Table 4-1 shows some differences in the all-waves response rates across the various subgroups (e.g., male versus female). The largest variations in weighted response rates for Wave 7 were associated with age group and education level. It is generally considered harder to reach younger adults, and the Wave 7 weighted all-waves response rates were lower for participants younger than age 18 at Wave 1 (42.2 percent) and ages 18 to 24 at Wave 1 (40.8 percent) than for those in the older Wave 1 age groups (who had response rates ranging from 47.3 percent to 55.3 percent). Moreover, adults with a bachelor's degree or advanced degree at Wave 1 had a higher propensity to be all-waves respondents (56.7 percent) than adults with less education (who had response rates ranging from 45.5 percent to 48.5 percent).

Notable variations in response rate occurred for the other Wave 1 characteristics as well. The weighted all-waves response rate was lower among males (45.9 percent) than among females (50.9 percent); this pattern is consistent with most household surveys (Groves & Couper, 1998; Stoop, 2005). Non-Hispanic Black alone adults had a slightly higher weighted all-waves response rate (50.6 percent) than the other Wave 1 race/ethnicity groups (47.8 percent for Hispanic, 48.9 percent for Non-Hispanic White alone, and 46.6 percent for Non-Hispanic other race or multiple races). Across the census regions, those residing in the Midwest at Wave 1 had the highest weighted all-waves response rate (50.5 percent) whereas those in the Northeast had the lowest (45.8 percent). Adults with health insurance at Wave 1 were more likely to be all-waves respondents (51.3 percent) than adults without health insurance (43.2 percent). Finally, the weighted all-waves response rates were 8.3 percentage points lower for Wave 1 adults with current established use of tobacco at Wave 1 than for Wave 1 adults with no current established use of tobacco, and 9.3 percentage points lower for Wave 1 youth who had ever used tobacco at Wave 1 than for Wave 1 youth who had never used tobacco.

Variation in response rates by subgroups is to be expected in large-scale data collection efforts, especially at later waves of a longitudinal study. In particular, Cunradi, Moore, Killoran, and Ames (2005) and Young, Powers, and Bell (2006) found that people who smoke were less likely to be retained in subsequent waves of surveys than those who do not smoke. Therefore, given the differences in response propensity shown in Table 4-1, it is pertinent to examine the degree to which the Wave 7 all-waves weighting adjustments addressed differences in response propensity. The results of these analyses are discussed in the next section.

Table 4-1. Wave 7 adult interview all-waves response rates conditioning on Wave 1 response

Wave 1 characteristic ^a	$C_{A_{AW}}$ Completed Wave 7 adult interview and responded at each primary wave (n)	$I_{A_{AW}}$ Ineligible for at least one primary wave (n)	$N_{A_{AW}}$ Nonresponse for at least one primary wave and known to be eligible for each primary wave (n)	$U_{A_{AW}}$ Nonresponse with unknown eligibility status for at least one primary wave and never ineligible for a primary wave (n)	$RR_{A_{AW}}$ Unweighted all-waves response rate ^b (%)	$RR_{A_{W_{AW}}}$ Weighted all-waves response rate ^b (%)
Overall	20,675	2,918	5,474	22,443	44.6	48.6
Sex						
Male	9,674	1,776	2,847	11,853	42.2	45.9
Female	10,977	1,136	2,617	10,553	47.0	50.9
Age						
Under 18	7,769	431	2,456	8,533	42.2	42.2
18-24	3,279	398	875	4,557	39.4	40.8
25-44	4,637	521	1,041	5,068	44.9	47.3
45-64	3,977	844	821	3,175	53.0	55.3
65 or older	1,010	724	280	1,096	50.7	50.9
Race/ethnicity ^c						
Hispanic	4,582	528	1,158	4,719	45.5	47.8
Non-Hispanic White alone	11,099	1,589	3,105	12,655	43.4	48.9
Non-Hispanic Black alone	3,056	467	681	2,867	48.6	50.6
Non-Hispanic other race or multiple races	1,651	244	437	1,840	44.2	46.6
Census region						
Northeast	2,947	383	898	3,666	41.1	45.8
Midwest	4,944	719	1,285	4,905	46.5	50.5
South	7,577	1,190	1,966	8,767	43.7	47.5
West	5,207	626	1,325	5,105	46.5	50.3
Education (Wave 1 adults only)						
Less than high school or GED	2,265	822	579	2,784	45.3	46.5
High school graduate	2,739	580	748	3,481	42.3	45.5
Some college but no degree, or associate's degree	4,537	686	1,098	4,978	45.0	48.5
Bachelor's degree or advanced degree	3,315	375	573	2,546	53.4	56.7

Table 4-1. Wave 7 adult interview all-waves response rates conditioning on Wave 1 response (continued)

Wave 1 characteristic ^a	$C_{A_{AW}}$ Completed Wave 7 adult interview and responded at each primary wave (n)	$I_{A_{AW}}$ Ineligible for at least one primary wave (n)	$N_{A_{AW}}$ Nonresponse for at least one primary wave and known to be eligible for each primary wave (n)	$U_{A_{AW}}$ Nonresponse with unknown eligibility status for at least one primary wave and never ineligible for a primary wave (n)	$RR_{A_{AW}}$ Unweighted all-waves response rate ^b (%)	$RR_{A_{W_{AW}}}$ Weighted all-waves response rate ^b (%)
Health insurance (Wave 1 adults only)						
Yes	10,612	1,985	2,426	10,731	47.5	51.3
No	2,189	464	559	2,944	41.6	43.2
Wave 1 current established tobacco use ^d (Wave 1 adults only)						
Yes	5,085	1,373	1,354	6,557	42.9	43.6
No	7,490	989	1,581	6,949	48.8	51.9
Wave 1 ever tobacco use ^e (Wave 1 youth only)						
Yes	915	133	382	1,438	35.2	35.1
No	4,391	212	1,334	4,306	44.5	44.4

^a For each characteristic, the sum of the counts across the categories may not be equal to the count in the overall row due to missing values.

^b The adult interview unweighted all-waves response rate formula is provided earlier in this section. The weighted response rates were calculated using a similar approach except that the Wave 1 IPS weight was used to obtain weighted counts.

^c Hispanic refers to Hispanic or Latino; Black refers to Black or African American. Alone means that the participant reported only one race.

^d Current established use of a tobacco product means the person currently uses the product every day or some days and—for cigarettes—has smoked at least 100 cigarettes in their lifetime and—for any other tobacco product—has reported they ever used that product fairly regularly. This characteristic considers the following products covered by the Wave 1 adult interview: cigarettes, traditional cigars, cigarillos, filtered cigars, pipes, smokeless tobacco, snus, hookah, dissolvable tobacco, and e-cigarettes.

^e Ever use of a tobacco product means the person has ever used the product, even one or two times. This characteristic considers the following products covered by the Wave 1 youth interview: cigarettes, traditional cigars, cigarillos, filtered cigars, pipes, smokeless tobacco, snus, hookah, dissolvable tobacco, e-cigarettes, bidis, and kreteks.

4.1.2 Comparisons Between Those Eligible to be Wave 7 All-Waves Respondents and Wave 7 All-Waves Respondents (Among the Wave 1 Cohort)

This section compares weighted estimates between the Wave 1 respondents who were eligible to be Wave 7 adult all-waves respondents and the Wave 7 adult all-waves respondents in the Wave 1 Cohort. Section 3.2.3 describes the methods used to obtain these estimates. The weighted estimates cover Wave 1 demographic and socio-economic characteristics (shown in Table 4-2) as well as Wave 1 tobacco-use status (shown in Tables 4-3 and 4-4). Each table shows two sets of comparisons. The first set is for the “before Wave 7 weighting adjustment” comparison, and the second set is for the “after Wave 7 weighting adjustment” comparison.

In Table 4-2, the percentages sum to 100 percent²² over the categories associated with each characteristic. For example, for both the Wave 7 all-waves respondents and those eligible to be Wave 7 all-waves respondents, the estimated percentage of males and the estimated percentage of females add to 100 percent. Table 4-2 has two columns labeled “Unweighted count” and Tables 4-3 and 4-4 each have two columns labeled “Sample size.” These columns provide the unweighted number of adults in the subgroup (e.g., males) associated with the table row. They are labeled differently in each table because the (weighted) number constitutes the numerator of the corresponding percentage in Table 4-2 (e.g., the estimated percentage of adults who were male), whereas it is the denominator of the corresponding percentage in Tables 4-3 and 4-4 (e.g., for Table 4-3, the estimated percentage of current established use of tobacco among adult males at Wave 1).²³

²² The sum may be not exactly 100 percent due to rounding.

²³ These definitions generalize to all tables in this report that include a column heading of “Unweighted count” or “Sample size.”

Table 4-2. Comparison of Wave 1 demographic and socio-economic characteristics between Wave 1 respondents eligible to be Wave 7 adult all-waves respondents and Wave 1 respondents who were Wave 7 adult all-waves respondents

Wave 1 characteristic	Wave 1 respondents who were eligible to be Wave 7 adult all-waves respondents		Wave 1 respondents who were Wave 7 adult all-waves respondents				
	Unweighted count	A: Weighted percentage, using Wave 1 cross-sectional weight [95% confidence interval]	Unweighted count	Before Wave 7 weighting adjustment		After Wave 7 weighting adjustment	
				B: Weighted percentage, using Wave 1 cross-sectional weight [95% confidence interval]	Difference in weighted percentages [B – A] [95% confidence interval]	C: Weighted percentage, using Wave 7 all-waves weight [95% confidence interval]	Difference in weighted percentages [C – A] [95% confidence interval]
Sex							
Male	24,374	47.9% [47.5%, 48.4%]	9,674	44.8% [44.0%, 45.6%]	-3.1% [-3.9%, -2.3%]	47.9% [47.2%, 48.6%]	0.0% [-0.2%, 0.2%]
Female	24,147	52.1% [51.6%, 52.5%]	10,977	55.2% [54.4%, 56.0%]	3.1% [2.3%, 3.9%]	52.1% [51.4%, 52.8%]	-0.0% [-0.2%, 0.2%]
Age							
Under 18	18,758	13.3% [13.0%, 13.6%]	7,769	12.1% [11.6%, 12.5%]	-1.3% [-1.5%, -1.0%]	13.3% [12.9%, 13.8%]	0.0% [-0.1%, 0.1%]
18-24	8,711	11.7% [11.5%, 12.0%]	3,279	10.1% [9.7%, 10.5%]	-1.6% [-1.9%, -1.3%]	11.9% [11.4%, 12.3%]	0.1% [0.1%, 0.2%]
25-44	10,746	31.1% [30.7%, 31.5%]	4,637	30.5% [29.7%, 31.4%]	-0.5% [-1.3%, 0.3%]	31.4% [30.8%, 32.0%]	0.3% [0.2%, 0.5%]
45-64	7,973	30.2% [29.8%, 30.7%]	3,977	34.6% [33.8%, 35.5%]	4.4% [3.6%, 5.2%]	30.5% [29.9%, 31.1%]	0.2% [0.1%, 0.4%]
65 or older	2,386	13.6% [13.3%, 13.9%]	1,010	12.6% [12.0%, 13.3%]	-1.0% [-1.6%, -0.4%]	12.9% [12.5%, 13.4%]	-0.7% [-0.9%, -0.4%]
Race/ethnicity ^a							
Hispanic	10,459	16.3% [16.0%, 16.7%]	4,582	16.0% [15.4%, 16.7%]	-0.3% [-1.0%, 0.4%]	16.4% [15.9%, 16.9%]	0.1% [-0.1%, 0.2%]
Non-Hispanic White alone	26,859	64.4% [63.9%, 64.8%]	11,099	64.5% [63.7%, 65.4%]	0.2% [-0.7%, 1.0%]	64.3% [63.6%, 64.9%]	-0.1% [-0.3%, 0.1%]
Non-Hispanic Black alone	6,604	11.5% [11.2%, 11.7%]	3,056	11.9% [11.4%, 12.3%]	0.4% [-0.1%, 0.9%]	11.4% [10.9%, 11.8%]	-0.1% [-0.2%, 0.1%]
Non-Hispanic other race or multiple races	3,928	7.8% [7.6%, 8.1%]	1,651	7.6% [7.1%, 8.1%]	-0.3% [-0.7%, 0.2%]	8.0% [7.6%, 8.3%]	0.1% [-0.1%, 0.3%]

Table 4-2. Comparison of Wave 1 demographic and socio-economic characteristics between Wave 1 respondents eligible to be Wave 7 adult all-waves respondents and Wave 1 respondents who were Wave 7 adult all-waves respondents (continued)

Wave 1 characteristic	Wave 1 respondents who were eligible to be Wave 7 adult all-waves respondents		Wave 1 respondents who were Wave 7 adult all-waves respondents				
	Unweighted count	A: Weighted percentage, using Wave 1 cross-sectional weight [95% confidence interval]	Unweighted count	Before Wave 7 weighting adjustment		After Wave 7 weighting adjustment	
				B: Weighted percentage, using Wave 1 cross-sectional weight [95% confidence interval]	Difference in weighted percentages [B – A] [95% confidence interval]	C: Weighted percentage, using Wave 7 all-waves weight [95% confidence interval]	Difference in weighted percentages [C – A] [95% confidence interval]
Census region							
Northeast	7,511	18.1% [17.8%, 18.5%]	2,947	17.3% [16.8%, 17.9%]	-0.8% [-1.4%, -0.3%]	18.1% [17.6%, 18.7%]	-0.0% [-0.2%, 0.1%]
Midwest	11,134	21.4% [21.0%, 21.7%]	4,944	22.3% [21.5%, 23.1%]	0.9% [0.1%, 1.7%]	21.3% [20.8%, 21.9%]	-0.0% [-0.2%, 0.1%]
South	18,310	37.0% [36.6%, 37.4%]	7,577	35.8% [34.7%, 36.8%]	-1.2% [-2.3%, -0.2%]	37.0% [36.3%, 37.6%]	-0.0% [-0.2%, 0.2%]
West	11,637	23.5% [23.1%, 23.9%]	5,207	24.6% [23.8%, 25.5%]	1.1% [0.3%, 1.9%]	23.6% [23.0%, 24.1%]	0.1% [-0.1%, 0.2%]
Education (Wave 1 adults only)							
Less than high school or GED	5,628	15.7% [15.3%, 16.1%]	2,265	13.7% [13.1%, 14.4%]	-2.0% [-2.6%, -1.4%]	15.4% [14.8%, 16.0%]	-0.3% [-0.6%, -0.0%]
High school	6,968	23.9% [23.4%, 24.4%]	2,739	21.5% [20.7%, 22.3%]	-2.5% [-3.2%, -1.7%]	23.7% [23.0%, 24.5%]	-0.2% [-0.5%, 0.1%]
Some college but no degree, or associate's degree	10,613	31.6% [31.0%, 32.1%]	4,537	31.1% [30.3%, 31.9%]	-0.5% [-1.2%, 0.2%]	31.5% [30.7%, 32.3%]	-0.0% [-0.3%, 0.2%]
Bachelor's degree or advanced degree	6,434	28.8% [28.3%, 29.3%]	3,315	33.7% [32.9%, 34.5%]	4.9% [4.2%, 5.7%]	29.3% [28.5%, 30.1%]	0.5% [0.3%, 0.7%]
Health insurance (Wave 1 adults only)							
Yes	23,769	85.3% [84.7%, 85.9%]	10,612	87.4% [86.6%, 88.2%]	2.1% [1.6%, 2.6%]	85.7% [84.9%, 86.4%]	0.4% [-0.1%, 0.8%]
No	5,692	14.7% [14.1%, 15.3%]	2,189	12.6% [11.8%, 13.4%]	-2.1% [-2.6%, -1.6%]	14.3% [13.6%, 15.1%]	-0.4% [-0.8%, 0.1%]

^a Hispanic refers to Hispanic or Latino; Black refers to Black or African American. Alone means that the participant reported only one race.

Table 4-3. Comparison of Wave 1 current established tobacco-use estimates between Wave 1 adults eligible to be Wave 7 all-waves respondents and Wave 1 adults who were Wave 7 all-waves respondents*

Wave 1 characteristic ^a	Wave 1 adults who were eligible to be Wave 7 all-waves respondents		Wave 1 adults who were Wave 7 all-waves respondents				
	Sample size	A: Weighted estimate, using Wave 1 cross-sectional weight [95% confidence interval]	Sample size	Before Wave 7 weighting adjustment	After Wave 7 weighting adjustment		
				B: Weighted estimate, using Wave 1 cross-sectional weight [95% confidence interval]	Difference in weighted estimates [B – A] [95% confidence interval]	C: Weighted estimate, using Wave 7 all-waves weight [95% confidence interval]	Difference in weighted estimates [C – A] [95% confidence interval]
Overall	29,016	23.4% [22.8%, 24.0%]	12,575	19.6% [18.9%, 20.3%]	-3.8% [-4.3%, -3.3%]	23.0% [22.3%, 23.8%]	-0.3% [-0.5%, -0.2%]
Sex							
Male	14,440	29.2% [28.3%, 30.0%]	5,729	23.6% [22.5%, 24.8%]	-5.5% [-6.4%, -4.7%]	28.6% [27.5%, 29.8%]	-0.5% [-0.8%, -0.2%]
Female	14,552	18.1% [17.4%, 18.8%]	6,835	16.4% [15.5%, 17.3%]	-1.7% [-2.3%, -1.2%]	18.0% [17.1%, 18.9%]	-0.1% [-0.3%, 0.0%]
Age							
18-24	8,577	28.8% [27.4%, 30.2%]	3,235	23.8% [22.0%, 25.8%]	-5.0% [-6.1%, -3.8%]	28.5% [27.0%, 30.1%]	-0.3% [-0.5%, -0.0%]
25-44	10,535	28.4% [27.5%, 29.4%]	4,556	23.6% [22.4%, 24.8%]	-4.8% [-5.8%, -3.9%]	28.1% [26.8%, 29.4%]	-0.4% [-1.2%, 0.4%]
45-64	7,676	21.9% [21.0%, 22.8%]	3,832	18.7% [17.5%, 20.0%]	-3.2% [-4.0%, -2.4%]	21.1% [19.8%, 22.4%]	-0.8% [-1.5%, -0.0%]
65 or older	2,217	9.6% [8.4%, 10.9%]	949	8.3% [6.7%, 10.2%]	-1.3% [-2.3%, -0.3%]	9.7% [8.0%, 11.7%]	0.1% [-1.2%, 1.4%]
Race/ethnicity ^b							
Hispanic	4,977	17.2% [16.1%, 18.2%]	2,176	15.6% [14.1%, 17.1%]	-1.6% [-2.6%, -0.6%]	17.0% [15.5%, 18.7%]	-0.1% [-0.5%, 0.2%]
Non-Hispanic White alone	17,461	24.9% [24.0%, 25.8%]	7,486	19.9% [18.9%, 20.8%]	-5.0% [-5.7%, -4.4%]	24.5% [23.5%, 25.5%]	-0.4% [-0.6%, -0.2%]
Non-Hispanic Black alone	3,991	26.0% [24.6%, 27.4%]	1,862	26.3% [24.1%, 28.7%]	0.3% [-1.2%, 1.9%]	26.7% [24.7%, 28.7%]	0.7% [0.1%, 1.2%]
Non-Hispanic other race or multiple races	2,170	19.1% [17.5%, 20.9%]	888	15.0% [12.8%, 17.5%]	-4.1% [-5.9%, -2.3%]	18.4% [16.0%, 21.1%]	-0.8% [-1.5%, 0.0%]

Table 4-3. Comparison of Wave 1 current established tobacco-use estimates between Wave 1 adults eligible to be Wave 7 all-waves respondents and Wave 1 adults who were Wave 7 all-waves respondents* (continued)

Wave 1 characteristic ^a	Wave 1 adults who were eligible to be Wave 7 all-waves respondents		Wave 1 adults who were Wave 7 all-waves respondents				
	Sample size	A: Weighted estimate, using Wave 1 cross-sectional weight [95% confidence interval]	Sample size	Before Wave 7 weighting adjustment		After Wave 7 weighting adjustment	
				B: Weighted estimate, using Wave 1 cross-sectional weight [95% confidence interval]	Difference in weighted estimates [B – A] [95% confidence interval]	C: Weighted estimate, using Wave 7 all-waves weight [95% confidence interval]	Difference in weighted estimates [C – A] [95% confidence interval]
Census region							
Northeast	4,587	21.8% [20.3%, 23.3%]	1,857	17.4% [15.6%, 19.4%]	-4.4% [-5.5%, -3.2%]	21.5% [19.7%, 23.4%]	-0.3% [-0.6%, 0.0%]
Midwest	6,889	26.2% [24.5%, 27.9%]	3,172	22.7% [20.7%, 24.9%]	-3.4% [-4.7%, -2.2%]	25.8% [24.2%, 27.6%]	-0.3% [-0.7%, 0.0%]
South	10,897	25.1% [23.8%, 26.5%]	4,556	21.1% [19.7%, 22.6%]	-4.0% [-4.8%, -3.1%]	24.7% [23.3%, 26.1%]	-0.5% [-0.7%, -0.2%]
West	6,643	19.3% [17.7%, 21.0%]	2,990	16.0% [14.5%, 17.6%]	-3.3% [-4.3%, -2.4%]	19.2% [17.7%, 20.8%]	-0.1% [-0.4%, 0.2%]
Education (Wave 1 adults only)							
Less than high school or GED	5,447	35.0% [33.3%, 36.7%]	2,197	34.6% [32.2%, 37.0%]	-0.4% [-2.2%, 1.4%]	35.0% [33.0%, 37.0%]	0.0% [-0.8%, 0.9%]
High school	6,742	27.6% [26.2%, 29.1%]	2,658	24.4% [22.6%, 26.3%]	-3.3% [-4.6%, -1.9%]	27.3% [25.7%, 29.1%]	-0.3% [-0.8%, 0.2%]
Some college but no degree, or associate's degree	10,374	26.0% [25.0%, 27.1%]	4,430	22.1% [20.8%, 23.4%]	-4.0% [-4.9%, -3.1%]	25.7% [24.4%, 27.0%]	-0.4% [-0.7%, -0.1%]
Bachelor's degree or advanced degree	6,284	10.6% [9.9%, 11.4%]	3,247	8.3% [7.4%, 9.3%]	-2.3% [-2.9%, -1.8%]	10.6% [9.6%, 11.7%]	-0.0% [-0.2%, 0.2%]
Health insurance (Wave 1 adults only)							
Yes	23,140	20.8% [20.2%, 21.4%]	10,344	17.3% [16.6%, 18.1%]	-3.5% [-4.0%, -3.0%]	20.4% [19.7%, 21.2%]	-0.4% [-0.7%, -0.1%]
No	5,557	38.1% [36.4%, 39.8%]	2,138	35.0% [32.3%, 37.7%]	-3.1% [-5.1%, -1.2%]	38.5% [35.8%, 41.1%]	0.4% [-1.5%, 2.2%]

* Current established use of a tobacco product means the person currently uses the product every day or some days and—for cigarettes—has smoked at least 100 cigarettes in their lifetime and—for any other tobacco product—has reported they ever used that product fairly regularly. This characteristic considers the following products covered by the Wave 1 adult interview: cigarettes, traditional cigars, cigarillos, filtered cigars, pipes, smokeless tobacco, snus, hookah, dissolvable tobacco, and e-cigarettes.

^a For each characteristic, the sum of the counts across the categories may not be equal to the count in the overall row due to missing values.

^b Hispanic refers to Hispanic or Latino; Black refers to Black or African American. Alone means that the participant reported only one race.

Table 4-4. Comparison of Wave 1 ever tobacco-use estimates between Wave 1 youth eligible to be Wave 7 all-waves respondents and Wave 1 youth who were Wave 7 all-waves respondents (among Wave 7 adults)*

Wave 1 characteristic ^a	Wave 1 youth who were eligible to be Wave 7 adult all-waves respondents		Wave 1 youth who were Wave 7 adult all-waves respondents				
	Sample size	A: Weighted estimate, using Wave 1 cross-sectional weight [95% confidence interval]	Sample size	Before Wave 7 weighting adjustment	After Wave 7 weighting adjustment		
				B: Weighted estimate, using Wave 1 cross-sectional weight [95% confidence interval]	Difference in weighted estimates [B – A] [95% confidence interval]	C: Weighted estimate, using Wave 7 all-waves weight [95% confidence interval]	Difference in weighted estimates [C – A] [95% confidence interval]
Overall	12,766	21.4% [20.4%, 22.4%]	5,306	17.0% [16.0%, 18.0%]	-4.4% [-5.3%, -3.4%]	21.2% [20.2%, 22.4%]	-0.1% [-0.5%, 0.3%]
Sex							
Male	6,415	22.6% [21.5%, 23.9%]	2,560	18.4% [16.9%, 19.9%]	-4.3% [-5.7%, -2.8%]	22.4% [20.8%, 24.0%]	-0.3% [-1.2%, 0.6%]
Female	6,321	20.1% [19.0%, 21.3%]	2,736	15.7% [14.3%, 17.3%]	-4.4% [-5.6%, -3.2%]	20.2% [18.7%, 21.8%]	0.0% [-0.9%, 1.0%]
Race/ethnicity ^b							
Hispanic	3,610	21.0% [19.4%, 22.7%]	1,590	17.4% [15.2%, 19.8%]	-3.6% [-5.7%, -1.6%]	20.9% [18.8%, 23.2%]	-0.1% [-1.7%, 1.5%]
Non-Hispanic White alone	6,122	23.0% [21.6%, 24.6%]	2,350	17.8% [16.2%, 19.5%]	-5.2% [-6.6%, -3.9%]	22.5% [20.8%, 24.2%]	-0.6% [-1.4%, 0.3%]
Non-Hispanic Black alone	1,670	18.3% [16.4%, 20.4%]	768	17.2% [14.5%, 20.3%]	-1.1% [-3.2%, 0.9%]	20.8% [17.8%, 24.0%]	2.5% [0.3%, 4.6%]
Non-Hispanic other race or multiple races	1,145	18.5% [16.3%, 20.8%]	510	13.4% [10.7%, 16.6%]	-5.1% [-7.3%, -3.0%]	17.2% [14.2%, 20.7%]	-1.3% [-3.8%, 1.3%]

Table 4-4. Comparison of Wave 1 ever tobacco-use estimates between Wave 1 youth eligible to be Wave 7 all-waves respondents and Wave 1 youth who were Wave 7 all-waves respondents (among Wave 7 adults)* (continued)

Wave 1 characteristic ^a	Wave 1 youth who were eligible to be Wave 7 adult all-waves respondents		Wave 1 youth who were Wave 7 adult all-waves respondents				
	Sample size	A: Weighted estimate, using Wave 1 cross-sectional weight [95% confidence interval]	Sample size	Before Wave 7 weighting adjustment	After Wave 7 weighting adjustment		
				B: Weighted estimate, using Wave 1 cross-sectional weight [95% confidence interval]	Difference in weighted estimates [B – A] [95% confidence interval]	C: Weighted estimate, using Wave 7 all-waves weight [95% confidence interval]	Difference in weighted estimates [C – A] [95% confidence interval]
Census region							
Northeast	1,925	20.9% [18.7%, 23.2%]	709	16.2% [13.7%, 19.1%]	-4.7% [-7.5%, -1.8%]	20.0% [17.3%, 23.1%]	-0.8% [-2.2%, 0.6%]
Midwest	2,786	21.8% [19.7%, 24.0%]	1,156	18.1% [16.0%, 20.5%]	-3.7% [-5.3%, -2.0%]	22.6% [20.1%, 25.4%]	0.8% [-0.5%, 2.1%]
South	4,806	21.9% [20.4%, 23.5%]	1,980	17.1% [15.5%, 18.8%]	-4.9% [-6.5%, -3.3%]	21.5% [19.8%, 23.4%]	-0.4% [-1.5%, 0.7%]
West	3,249	20.5% [18.4%, 22.8%]	1,461	16.4% [14.2%, 18.9%]	-4.1% [-6.0%, -2.2%]	20.4% [18.1%, 22.8%]	-0.2% [-1.8%, 1.4%]

*Ever use of a tobacco product means the person has ever used the product, even one or two times. This characteristic considers the following products covered by the Wave 1 youth interview: cigarettes, traditional cigars, cigarillos, filtered cigars, pipes, smokeless tobacco, snus, hookah, dissolvable tobacco, e-cigarettes, bidis, and kreteks.

^a For each characteristic, the sum of the counts across the categories may not be equal to the count in the overall row due to missing values.

^b Hispanic refers to Hispanic or Latino; Black refers to Black or African American. Alone means that the participant reported only one race.

Apart from the confidence intervals for the Wave 1 race/ethnicity characteristic, the majority of the 95 percent confidence intervals for the estimated differences in the first set of comparisons in Table 4-2 did not include zero. For the Wave 1 race/ethnicity characteristic, the confidence interval for the estimated difference included zero for each subgroup. Prior to Wave 7 all-waves weighting adjustments, the most noticeable underrepresentation among the Wave 7 adult all-waves respondents was for males, who tend to have a lower response propensity than females in most household surveys (Groves & Couper, 1998; Stoop, 2005). People younger than age 25 or older than age 64 at Wave 1, those who resided in the Northeast or South census regions at Wave 1, and adults who had never attended college at Wave 1 were also underrepresented among the Wave 7 adult all-waves respondents; whereas adults ages 45 to 64 at Wave 1, those residing in the Midwest or West census regions at Wave 1, adults with a bachelor's degree or advanced degree at Wave 1, and adults with health insurance at Wave 1 were overrepresented.

As shown in the last column of Table 4-2, the Wave 7 all-waves weighting adjustments for the Wave 1 Cohort reduced all the differences between the Wave 1 respondents eligible to be Wave 7 adult all-waves respondents and the Wave 1 respondents who were Wave 7 all-waves respondents. The 95 percent confidence intervals for the estimated differences either included or were in the proximity of zero, with three exceptions. Among the Wave 7 adult all-waves respondents, adults ages 65 and older at Wave 1 were underrepresented, whereas adults ages 18 to 64 at Wave 1 and adults with a bachelor's degree or advanced degree at Wave 1 were slightly overrepresented.

Tables 4-3 and 4-4 show the comparisons for the Wave 1 tobacco-use measures between those eligible to be Wave 7 adult all-waves respondents and the Wave 7 adult all-waves respondents, among the Wave 1 Cohort. The Wave 1 respondents were divided into two groups based on their Wave 1 interview type (i.e., adult versus youth) and analyzed separately. Table 4-3 covers current established tobacco use for the Wave 1 adult respondents and Table 4-4 covers ever use of tobacco for the Wave 1 youth respondents.

For the “before Wave 7 weighting adjustment” estimates, most of the 95 percent confidence intervals did not include zero, with the differences consistently indicating an underrepresentation of Wave 1 adult and youth tobacco use among the Wave 7 adult all-waves respondents. The Wave 7 all-waves weighting adjustments for the Wave 1 Cohort greatly reduced almost all the differences and the majority of the corresponding confidence intervals either included or were in the proximity

of zero. However, using the Wave 7 all-waves weight, the Wave 7 adult all-waves respondents tended to underrepresent Wave 1 adult current established tobacco use overall, and among males, non-Hispanic White alone adults, adults who resided in the South census region at Wave 1, adults with some college but no degree or with an associate's degree at Wave 1, and adults with health insurance at Wave 1; whereas Wave 1 adult current established tobacco use was slightly overestimated among non-Hispanic Black alone adults (see Table 4-3). After Wave 7 weighting adjustments, the Wave 7 adult all-waves respondents tended to overrepresent Wave 1 youth ever use of tobacco among non-Hispanic Black alone persons (see Table 4-4).

Overall, these results reflect positively on the use of demographic, socio-economic, and tobacco-use variables from Wave 1 for calibrating the Wave 7 all-waves weight for the Wave 1 Cohort. However, they reflect the challenges faced by panel studies as waves of nonresponse accrue. If the Wave 1 characteristics in Tables 4-2 through 4-4 are correlated with key tobacco- and health-related outcome measures at Wave 7 of the PATH Study, the results suggest a small amount of potential bias in the adult estimates for the Wave 1 Cohort due to nonresponse at any primary wave since recruitment. Researchers are encouraged to think carefully about the choice of covariates in their analyses and to consider conducting sensitivity analyses to support their findings.

4.2 Youth Interview

4.2.1 All-Waves Response Rates Conditioning on Wave 1 Response

Among those in the Wave 1 Cohort, unweighted and weighted all-waves response rates were calculated for the Wave 7 youth interview using the following formulas:

$$RR_{Y_AW} = C_{Y_AW} / (C_{Y_AW} + N_{Y_AW} + e_{Y_AW} \times U_{Y_AW})$$

$$e_{Y_AW} = (C_{Y_AW} + N_{Y_AW}) / (C_{Y_AW} + N_{Y_AW} + I_{Y_AW})$$

where

RR_{Y_AW} = Wave 7 youth interview all-waves response rate;

C_{Y_AW} = number of Wave 7 youth who were respondents at each of primary Waves 2 through 7;

- N_{Y_AW} = number of Wave 7 youth known to be eligible at each wave, but a nonrespondent at least once for primary Waves 2 through 7;
- U_{Y_AW} = number of Wave 7 youth with unknown eligibility status at least once and never ineligible for primary Waves 2 through 7;
- e_{Y_AW} = estimated proportion of eligible youth among the Wave 7 youth with unknown eligibility status (from an all-waves perspective); and
- I_{Y_AW} = number of Wave 7 youth who were ineligible at least once for primary Waves 2 through 7.

Unweighted counts and weighted counts based on the Wave 1 IPS weight were obtained for the all-waves response status categories C_{Y_AW} , N_{Y_AW} , U_{Y_AW} , and I_{Y_AW} for the unweighted all-waves response rates (RR_{Y_AW}) and weighted all-waves response rates ($RR_{Y_W_AW}$), respectively.

Table 4-5 provides all-waves response rates for the Wave 7 youth interview. In addition to the overall row, the table includes rows for subgroups based on Wave 1 sex, race/ethnicity, and census region.²⁴ Almost all the Wave 7 youth in the Wave 1 Cohort were shadow youth who were age 9 at Wave 1. The Wave 1 demographic information for Wave 1 shadow youth comes from information provided by the household screener respondent in the Wave 1 household screener. Participants for whom the Wave 1 household screener (or Wave 1 interview in the case of Wave 1 youth) had missing information for the demographic characteristic were excluded from the response rate calculation for that subgroup.

²⁴ Response rates by Wave 1 tobacco-use status are not included in Table 4-5 because almost all the Wave 7 youth in the Wave 1 Cohort were shadow youth at Wave 1 and so were not asked about their tobacco use at that time.

Table 4-5. Wave 7 youth interview all-waves response rates conditioning on Wave 1 response

Wave 1 characteristic ^a	C_{Y_AW} Completed Wave 7 youth interview and responded at each primary wave (n)	I_{Y_AW} Ineligible for at least one primary wave (n)	N_{Y_AW} Nonresponse for at least one primary wave and known to be eligible for each primary wave (n)	U_{Y_AW} Nonresponse with unknown eligibility status for at least one primary wave and never ineligible for a primary wave (n)	RR_{Y_AW} Unweighted all- waves response rate ^b (%)	$RR_{Y_W_AW}$ Weighted all- waves response rate ^b (%)
Overall	807	16	183	662	49.2	49.3
Sex						
Male	422	10	102	339	49.3	49.3
Female	385	6	81	323	49.1	49.2
Race/ethnicity ^c						
Hispanic	239	7	62	183	49.8	50.0
Non-Hispanic White alone	404	1	82	329	49.6	49.9
Non-Hispanic Black alone	104	4	17	88	50.4	49.6
Non-Hispanic other race or multiple races	59	4	22	61	42.4	42.7
Census region						
Northeast	113	6	26	102	47.7	48.2
Midwest	195	3	46	160	48.9	49.3
South	271	4	63	241	47.4	47.0
West	228	3	48	159	52.6	52.8

^a For each characteristic, the sum of the counts across the categories may not be equal to the count in the overall row due to missing values.

^b The youth interview unweighted all-waves response rate formula is provided earlier in this section. The weighted response rates were calculated using a similar approach except that the Wave 1 IPS weight was used to obtain weighted counts.

^c Hispanic refers to Hispanic or Latino; Black refers to Black or African American. Alone means that the participant reported only one race.

The unweighted and weighted all-waves response rates for the Wave 7 youth interview were 49.2 percent and 49.3 percent, respectively. The unweighted and weighted all-waves response rates were also similar to each other by subgroup, with non-Hispanic Black alone youth having the greatest difference between unweighted (50.4 percent) and weighted (49.6 percent) response rates. The unweighted and weighted single-wave response rates for the Wave 7 youth interview were higher than the all-waves response rates; both were 54.3 percent.

The Wave 7 weighted all-waves response rates varied most by Wave 1 race/ethnicity and census region, and only slightly by sex. For the Wave 1 race/ethnicity groups, the weighted all-waves response rate was highest among Hispanic youth (50.0 percent) and lowest among non-Hispanic youth of other race or multiple races (42.7 percent). Weighted all-waves response rates also differed across the census regions; Wave 7 youth residing in the West at Wave 1 had the highest response rate (52.8 percent) whereas those in the South had the lowest (47.0 percent).

As mentioned in Section 4.1.1, variation in response rates by subgroups is to be expected in large-scale data collection efforts. However, it is pertinent to examine the degree to which the Wave 7 all-waves weighting adjustments addressed differences in response propensity. The results of these analyses are discussed in the next section.

4.2.2 Comparisons Between Those Eligible to be Wave 7 All-Waves Respondents and Wave 7 All-Waves Respondents (Among the Wave 1 Cohort)

This section compares weighted demographic estimates between the Wave 1 respondents who were eligible to be Wave 7 youth all-waves respondents and the Wave 7 youth all-waves respondents in the Wave 1 Cohort. Note that the Wave 1 respondents who were eligible to be Wave 7 youth all-waves respondents were almost all age 9 at Wave 1—this reduces the number of cases for analysis and also means that estimates of Wave 1 tobacco use are not available. The weighted estimates in Table 4-6 cover Wave 1 demographic characteristics. The structure of this table is similar to Table 4-2 in Section 4.1.2 for the Wave 7 adults. For example, the percentages sum to 100 percent²⁵ over the categories associated with each characteristic.

²⁵ The sum may be not exactly 100 percent due to rounding.

The estimated 95 percent confidence intervals for the “before Wave 7 weighting adjustment” differences either included or were in the proximity of zero for all the Wave 1 demographics examined. However, using the Wave 7 all-waves weight, the Wave 7 youth all-waves respondents slightly underrepresented youth who resided in the Midwest census region at Wave 1. Overall, these results reflect positively on the use of demographic variables from Wave 1 for calibrating the Wave 7 youth all-waves weight for the Wave 1 Cohort. However, the analysis is constrained by the fact that almost all the Wave 7 youth in the Wave 1 Cohort were age 9 at Wave 1 and limited data—particularly no data on Wave 1 tobacco use—are available for assessing nonresponse bias for this group. It is therefore challenging to make statements about the possible extent of bias in Wave 7 youth estimates for the Wave 1 Cohort due to nonresponse at any primary wave since recruitment.

Table 4-6. Comparison of Wave 1 demographic characteristics between Wave 1 respondents eligible to be Wave 7 youth all-waves respondents and Wave 1 respondents who were Wave 7 youth all-waves respondents

Wave 1 characteristic	Wave 1 respondents who were eligible to be Wave 7 youth all-waves respondents		Wave 1 respondents who were Wave 7 youth all-waves respondents				
	Unweighted count	A: Weighted percentage, using Wave 1 cross-sectional weight [95% confidence interval]	Unweighted count	Before Wave 7 weighting adjustment		After Wave 7 weighting adjustment	
				B: Weighted percentage, using Wave 1 cross-sectional weight [95% confidence interval]	Difference in weighted percentages [B – A] [95% confidence interval]	C: Weighted percentage, using Wave 7 all-waves weight [95% confidence interval]	Difference in weighted percentages [C – A] [95% confidence interval]
Sex							
Male	863	51.5% [49.1%, 53.9%]	422	51.1% [47.7%, 54.6%]	-0.4% [-2.6%, 1.9%]	51.4% [48.0%, 54.8%]	-0.1% [-1.9%, 1.8%]
Female	789	48.5% [46.1%, 50.9%]	385	48.9% [45.4%, 52.3%]	0.4% [-1.9%, 2.6%]	48.6% [45.2%, 52.0%]	0.1% [-1.8%, 1.9%]
Race/ethnicity ^a							
Hispanic	484	24.5% [22.5%, 26.7%]	239	24.4% [21.6%, 27.5%]	-0.1% [-2.6%, 2.3%]	24.1% [21.2%, 27.1%]	-0.5% [-1.9%, 0.9%]
Non-Hispanic White alone	815	53.2% [50.8%, 55.6%]	404	54.3% [50.8%, 57.7%]	1.1% [-1.5%, 3.7%]	54.5% [51.0%, 57.9%]	1.3% [-0.7%, 3.3%]
Non-Hispanic Black alone	209	12.1% [10.6%, 13.7%]	104	12.1% [10.1%, 14.6%]	0.1% [-1.6%, 1.7%]	12.2% [10.2%, 14.7%]	0.2% [-1.0%, 1.4%]
Non-Hispanic other race or multiple races	142	10.2% [8.8%, 11.7%]	59	9.2% [7.2%, 11.5%]	-1.0% [-2.8%, 0.7%]	9.2% [7.4%, 11.4%]	-1.0% [-2.2%, 0.2%]
Census region							
Northeast	241	15.9% [14.2%, 17.7%]	113	15.8% [13.4%, 18.4%]	-0.1% [-2.2%, 2.0%]	16.3% [13.9%, 19.0%]	0.4% [-1.0%, 1.9%]
Midwest	401	22.7% [20.7%, 24.8%]	195	22.5% [19.7%, 25.5%]	-0.2% [-2.5%, 2.0%]	21.2% [18.5%, 24.1%]	-1.5% [-2.7%, -0.3%]
South	575	36.5% [34.2%, 38.9%]	271	34.7% [31.5%, 38.1%]	-1.8% [-4.4%, 0.8%]	37.0% [33.8%, 40.4%]	0.5% [-1.2%, 2.3%]
West	435	24.9% [22.9%, 27.1%]	228	27.1% [24.1%, 30.2%]	2.1% [0.0%, 4.2%]	25.5% [22.6%, 28.6%]	0.6% [-0.7%, 1.8%]

^a Hispanic refers to Hispanic or Latino; Black refers to Black or African American. Alone means that the participant reported only one race.

5. Results of Wave 4 Cohort Nonresponse Bias Analyses at Wave 7

This chapter presents results of nonresponse bias analyses for the Wave 4 Cohort interview data at Wave 7. The nonresponse bias analyses for the Wave 4 Cohort at Wave 7 are all conditioning on Wave 4 response. Sections 5.1.1 and 5.2.1 describe how the all-waves response rates vary across select subgroups, including recruitment wave. Sections 5.1.2 and 5.2.2 compare demographic and socio-economic characteristics as well as tobacco-use status between those eligible to be Wave 7 all-waves respondents and the Wave 7 all-waves respondents. The results are presented separately for adults and youth.

5.1 Adult Interview

5.1.1 All-Waves Response Rates Conditioning on Wave 4 Response

This section summarizes the Wave 4 Cohort all-waves response rates for the Wave 7 adult interview, conditioning on Wave 4 response. The unweighted and weighted all-waves response rates were calculated for the Wave 7 adult interview among those in the Wave 4 Cohort using the following formulas:

$$RR_{A_AW} = C_{A_AW} / (C_{A_AW} + N_{A_AW} + e_{A_AW} \times U_{A_AW})$$

$$e_{A_AW} = (C_{A_AW} + N_{A_AW}) / (C_{A_AW} + N_{A_AW} + I_{A_AW})$$

where

RR_{A_AW} = Wave 7 adult interview all-waves response rate;

C_{A_AW} = number of Wave 7 adults who were a respondent at each of primary Waves 5 through 7;

N_{A_AW} = number of Wave 7 adults known to be eligible at each wave, but a nonrespondent at least once for primary Waves 5 through 7;

U_{A_AW} = number of Wave 7 adults with unknown eligibility status at least once and never ineligible for primary Waves 5 through 7;

e_{A_AW} = estimated proportion of eligible adults among the Wave 7 adults with unknown eligibility status (from an all-waves perspective); and

I_{A_AW} = number of Wave 7 adults who were ineligible at least once for primary Waves 5 through 7

Unweighted counts and weighted counts based on the Wave 4 cross-sectional weight were obtained for the all-waves response status categories C_{A_AW} , N_{A_AW} , U_{A_AW} , and I_{A_AW} for the unweighted all-waves response rates (RR_{A_AW}) and weighted all-waves response rates ($RR_{A_W_AW}$), respectively.

Table 5-1 provides all-waves response rates for the Wave 7 adult interview. In addition to the overall row, the table includes rows for subgroups based on recruitment wave as well as the following Wave 4 characteristics: sex, age, race/ethnicity, census region, highest education level attained, health insurance status, and tobacco-use status. Participants for whom the Wave 4 interview had missing information for the demographic, socio-economic, or tobacco-use characteristic were excluded from the response rate calculation for that subgroup. The education level and health insurance status subgroups exclude the Wave 7 adults who were youth at Wave 4 because they were not asked about their highest level of education attained or health insurance status at that time.

The unweighted and weighted all-waves response rates for the Wave 7 adult interview were 57.6 percent and 63.1 percent, respectively. For many subgroups the weighted all-waves response rate was at least one percentage point higher than the unweighted all-waves response rate, with Non-Hispanic White alone adults having the greatest difference between weighted (64.0 percent) and unweighted (57.3 percent) response rates. The unweighted and weighted single-wave response rates for the Wave 7 adult interview were higher than the all-waves response rates, at 62.8 percent and 66.9 percent, respectively.

The weighted response rates varied moderately by Wave 4 sex, race/ethnicity, and census region, and more so by Wave 4 health insurance status, current established tobacco use (among those who were adults at Wave 4), and ever tobacco use (among those who were youth at Wave 4). The response rates varied most by Wave 4 age and educational level, as well as recruitment wave. Adults ages 45 to 64 at Wave 4 had the highest weighted all-waves response rate (69.4 percent) among the age groups, whereas those ages 18 to 24 at Wave 4 had the lowest (52.3 percent). Adults with a bachelor's degree or advanced degree at Wave 4 had a higher propensity to respond at Waves 5, 6, and 7 (71.9 percent) than adults with less education (who had weighted all-waves response rates

ranging from 58.6 percent to 63.0 percent). The weighted all-waves response rate for Wave 4 Cohort adults recruited at Wave 1 was 14.6 percentage points higher than for those recruited as part of the Wave 4 replenishment sample. A possible explanation for the large differences in subgroup response rates by recruitment wave is that Wave 4 Cohort members who have responded to multiple previous waves may be more cooperative than other cohort members. The Wave 1 sample members of the Wave 4 Cohort have persisted with the PATH Study over a number of years, and all completed an interview at Wave 4.

Variation in response rates by subgroups is to be expected in large-scale data collection efforts. However, it is pertinent to examine the effects of the Wave 7 all-waves weighting adjustments for the Wave 4 Cohort, given the differences in response propensity noted above. The results of these analyses are discussed in the next section.

Table 5-1. Wave 7 adult interview all-waves response rates conditioning on Wave 4 response (Wave 4 Cohort)

Characteristic ^a	<i>C_{A,AW}</i> : Completed Wave 7 adult interview and responded at each primary wave since Wave 4 (n)	<i>I_{A,AW}</i> : Ineligible for at least one primary wave since Wave 4 (n)	<i>N_{A,AW}</i> : Nonresponse for at least one primary wave and known to be eligible for each primary wave since Wave 4 (n)	<i>U_{A,AW}</i> : Nonresponse with unknown eligibility status for at least one primary wave and never ineligible for a primary wave since Wave 4 (n)	<i>RR_{A,AW}</i> : Unweighted all-waves response rate ^b (%)	<i>RR_{A,W,AW}</i> : Weighted all- waves response rate ^b (%)
Overall	25,366	1,864	6,280	13,154	57.6	63.1
Wave 4 sex						
Male	11,932	1,109	3,305	6,863	55.2	60.8
Female	13,396	750	2,962	6,270	59.9	65.3
Wave 4 age						
Under 18	6,776	210	2,000	4,034	53.3	52.8
18-24	5,578	268	1,559	3,808	51.6	52.3
25-44	6,439	325	1,354	3,217	59.2	62.0
45-64	4,943	524	950	1,568	67.4	69.4
65 or older	1,629	537	417	524	66.2	65.4
Wave 4 race/ethnicity ^c						
Hispanic	5,681	352	1,369	3,103	56.8	60.8
Non-Hispanic White alone	13,381	964	3,491	6,830	57.3	64.0
Non-Hispanic Black alone	3,724	323	769	1,853	59.9	63.2
Non-Hispanic other race or multiple races	2,076	155	499	1,108	57.3	62.1
Wave 4 census region						
Northeast	3,566	219	1,051	1,916	55.3	61.6
Midwest	5,978	462	1,458	2,973	58.4	64.3
South	9,425	800	2,280	5,232	56.8	61.5
West	6,397	383	1,491	3,031	59.3	65.7
Wave 4 education (Wave 4 adults only)						
Less than high school or GED	3,102	527	772	1,935	55.6	58.6
High school graduate	4,157	416	1,078	2,463	55.3	59.4
Some college but no degree, or associate's degree	6,612	440	1,650	3,195	58.5	63.0
Bachelor's degree or advanced degree	4,665	257	755	1,470	68.4	71.9

Table 5-1. Wave 7 adult interview all-waves response rates conditioning on Wave 4 response (Wave 4 Cohort) (continued)

Characteristic ^a	<i>C_{A,AW}</i> : Completed Wave 7 adult interview and responded at each primary wave since Wave 4 (n)	<i>I_{A,AW}</i> : Ineligible for at least one primary wave since Wave 4 (n)	<i>N_{A,AW}</i> : Nonresponse for at least one primary wave and known to be eligible for each primary wave since Wave 4 (n)	<i>U_{A,AW}</i> : Nonresponse with unknown eligibility status for at least one primary wave and never ineligible for a primary wave since Wave 4 (n)	<i>RR_{A,AW}</i> : Unweighted all-waves response rate ^b (%)	<i>RR_{A,W,AW}</i> : Weighted all- waves response rate ^b (%)
Wave 4 health insurance (Wave 4 adults only)						
Yes	15,775	1,360	3,510	7,055	61.0	65.3
No	2,645	270	726	1,942	51.2	55.5
Wave 4 current established tobacco use ^d (Wave 4 adults only)						
Yes	6,345	854	1,615	3,770	55.8	56.6
No	12,190	788	2,646	5,317	61.3	66.3
Wave 4 ever tobacco use ^e (Wave 4 youth only)						
Yes	1,332	72	489	1,055	47.0	46.1
No	5,099	119	1,387	2,703	55.8	55.4
Recruitment wave						
Wave 1	21,341	1,506	4,920	9,662	60.3	65.3
Wave 4	4,025	358	1,360	3,492	46.5	50.7

^a For each characteristic, the sum of the counts across the categories may not be equal to the count in the overall row due to missing values.

^b The adult interview unweighted all-waves response rate formula is provided earlier in this section. The weighted response rates were calculated using a similar approach except that the Wave 4 cross-sectional weight was used to obtain weighted counts.

^c Hispanic refers to Hispanic or Latino; Black refers to Black or African American. Alone means that the participant reported only one race.

^d Current established use of a tobacco product means the person currently uses the product every day or some days and—for cigarettes—has smoked at least 100 cigarettes in their lifetime and—for any other tobacco product—has reported they ever used that product fairly regularly. This characteristic considers the following products covered by the Wave 4 adult interview: cigarettes, traditional cigars, cigarillos, filtered cigars, pipes, smokeless tobacco, snus, hookah, and electronic nicotine delivery systems or ENDS (such as e-cigarettes, vape pens, personal vaporizers and mods, e-cigars, e-pipes, e-hookahs, and hookah pens).

^e Ever use of a tobacco product means the person has ever used the product, even one or two times. This characteristic considers the following products covered by the Wave 4 youth interview: cigarettes, traditional cigars, cigarillos, filtered cigars, pipes, smokeless tobacco, snus, hookah, electronic nicotine delivery systems or ENDS (such as e-cigarettes, vape pens, personal vaporizers and mods, e-cigars, e-pipes, e-hookahs, and hookah pens), dissolvable tobacco, bidis, and kreteks.

5.1.2 Comparisons Between Those Eligible to be Wave 7 All-Waves Respondents and Wave 7 All-Waves Respondents (Among the Wave 4 Cohort)

This section compares weighted estimates between the members of the Wave 4 Cohort who were eligible to be Wave 7 adult all-waves respondents and the Wave 7 adult all-waves respondents in the Wave 4 Cohort. The weighted estimates cover Wave 4 demographic and socio-economic characteristics (shown in Table 5-2) as well as Wave 4 tobacco-use status (shown in Tables 5-3 and 5-4). The structure of these tables is similar to the tables in Section 4.1.2.

For the “before Wave 7 weighting adjustments” in Table 5-2, the largest overrepresentation among Wave 7 adult all-waves respondents was of those with a bachelor’s degree or advanced degree at Wave 4. Females, adults ages 45 to 64 at Wave 4, non-Hispanic White alone individuals, people residing in the West census region at Wave 4, and adults with health insurance at Wave 4 were also overrepresented among the Wave 7 adult all-waves respondents. In contrast, those younger than age 25 at Wave 4, people who identify as Hispanic, people residing in the South census region at Wave 4, and adults with at most a high school diploma at Wave 4 were underrepresented.

For the “after Wave 7 weighting adjustment” differences, the estimated 95 percent confidence intervals (in the last column of Table 5-2) either included or were in the proximity of zero for most of the Wave 4 characteristics examined. However, among the Wave 7 adult all-waves respondents from the Wave 4 Cohort, adults with health insurance at Wave 4 were slightly overrepresented and those younger than age 18 at Wave 4 were slightly underrepresented.

Table 5-2. Comparison of Wave 4 demographic and socio-economic characteristics between Wave 4 respondents eligible to be Wave 7 adult all-waves respondents and Wave 4 respondents who were Wave 7 adult all-waves respondents (Wave 4 Cohort)

Wave 4 characteristic	Wave 4 respondents who were eligible to be Wave 7 adult all-waves respondents		Wave 4 respondents who were Wave 7 adult all-waves respondents				
	Unweighted count	A: Weighted percentage, using Wave 4 cross-sectional weight [95% confidence interval]	Unweighted count	Before Wave 7 weighting adjustment		After Wave 7 weighting adjustment	
				B: Weighted percentage, using Wave 4 cross-sectional weight [95% confidence interval]	Difference in weighted percentages [B – A] [95% confidence interval]	C: Weighted percentage, using Wave 7 all-waves weight [95% confidence interval]	Difference in weighted percentages [C – A] [95% confidence interval]
Sex							
Male	22,100	47.9% [47.5%, 48.4%]	11,932	45.9% [45.3%, 46.5%]	-2.0% [-2.6%, -1.4%]	47.9% [47.3%, 48.5%]	0.0% [-0.1%, 0.1%]
Female	22,628	52.1% [51.6%, 52.5%]	13,396	54.1% [53.5%, 54.7%]	2.0% [1.4%, 2.6%]	52.1% [51.5%, 52.7%]	-0.0% [-0.1%, 0.1%]
Age							
Under 18	12,810	8.5% [8.3%, 8.8%]	6,776	7.2% [6.9%, 7.5%]	-1.3% [-1.5%, -1.1%]	8.4% [8.1%, 8.8%]	-0.1% [-0.1%, -0.1%]
18-24	10,945	11.7% [11.4%, 12.0%]	5,578	9.8% [9.4%, 10.1%]	-2.0% [-2.2%, -1.7%]	11.7% [11.4%, 12.1%]	0.0% [-0.0%, 0.1%]
25-44	11,010	32.2% [31.8%, 32.7%]	6,439	32.0% [31.4%, 32.5%]	-0.3% [-0.9%, 0.3%]	32.3% [31.7%, 32.9%]	0.1% [-0.0%, 0.2%]
45-64	7,461	31.5% [31.0%, 31.9%]	4,943	34.8% [34.2%, 35.4%]	3.3% [2.7%, 4.0%]	31.6% [31.0%, 32.1%]	0.1% [0.0%, 0.2%]
65 or older	2,570	16.1% [15.7%, 16.4%]	1,629	16.3% [15.7%, 16.9%]	0.2% [-0.3%, 0.8%]	16.0% [15.5%, 16.4%]	-0.1% [-0.3%, 0.1%]

Table 5-2. Comparison of Wave 4 demographic and socio-economic characteristics between Wave 4 respondents eligible to be Wave 7 adult all-waves respondents and Wave 4 respondents who were Wave 7 adult all-waves respondents (Wave 4 Cohort) (continued)

Wave 4 characteristic	Wave 4 respondents who were eligible to be Wave 7 adult all-waves respondents		Wave 4 respondents who were Wave 7 adult all-waves respondents				
	Unweighted count	A: Weighted percentage, using Wave 4 cross-sectional weight [95% confidence interval]	Unweighted count	Before Wave 7 weighting adjustment		After Wave 7 weighting adjustment	
				B: Weighted percentage, using Wave 4 cross-sectional weight [95% confidence interval]	Difference in weighted percentages [B – A] [95% confidence interval]	C: Weighted percentage, using Wave 7 all-waves weight [95% confidence interval]	Difference in weighted percentages [C – A] [95% confidence interval]
Race/ethnicity ^a							
Hispanic	10,153	16.6% [16.3%, 17.0%]	5,681	15.9% [15.4%, 16.5%]	-0.7% [-1.2%, -0.1%]	16.6% [16.2%, 17.1%]	0.0% [-0.1%, 0.1%]
Non-Hispanic White alone	23,702	63.6% [63.2%, 64.1%]	13,381	64.5% [63.7%, 65.2%]	0.8% [0.2%, 1.5%]	63.6% [63.0%, 64.2%]	0.0% [-0.1%, 0.2%]
Non-Hispanic Black alone	6,346	11.4% [11.1%, 11.7%]	3,724	11.4% [11.0%, 11.8%]	-0.0% [-0.4%, 0.3%]	11.4% [11.0%, 11.8%]	-0.1% [-0.2%, 0.0%]
Non-Hispanic other race or multiple races	3,683	8.3% [8.1%, 8.6%]	2,076	8.2% [7.8%, 8.6%]	-0.1% [-0.5%, 0.3%]	8.4% [8.0%, 8.7%]	0.0% [-0.1%, 0.2%]
Census region							
Northeast	6,533	17.9% [17.5%, 18.3%]	3,566	17.5% [17.0%, 18.1%]	-0.4% [-0.9%, 0.2%]	17.8% [17.4%, 18.3%]	-0.1% [-0.1%, 0.0%]
Midwest	10,409	21.0% [20.6%, 21.4%]	5,978	21.4% [20.9%, 22.0%]	0.4% [-0.1%, 1.0%]	21.0% [20.5%, 21.5%]	-0.0% [-0.1%, 0.1%]
South	16,937	37.4% [36.9%, 37.8%]	9,425	36.3% [35.6%, 37.0%]	-1.1% [-1.8%, -0.4%]	37.4% [36.8%, 38.0%]	0.0% [-0.1%, 0.1%]
West	10,919	23.7% [23.3%, 24.1%]	6,397	24.8% [24.2%, 25.4%]	1.0% [0.4%, 1.7%]	23.8% [23.3%, 24.3%]	0.1% [-0.0%, 0.2%]

Table 5-2. Comparison of Wave 4 demographic and socio-economic characteristics between Wave 4 respondents eligible to be Wave 7 adult all-waves respondents and Wave 4 respondents who were Wave 7 adult all-waves respondents (Wave 4 Cohort) (continued)

Wave 4 characteristic	Wave 4 respondents who were eligible to be Wave 7 adult all-waves respondents		Wave 4 respondents who were Wave 7 adult all-waves respondents				
	Unweighted count	A: Weighted percentage, using Wave 4 cross-sectional weight [95% confidence interval]	Unweighted count	Before Wave 7 weighting adjustment		After Wave 7 weighting adjustment	
				B: Weighted percentage, using Wave 4 cross-sectional weight [95% confidence interval]	Difference in weighted percentages [B – A] [95% confidence interval]	C: Weighted percentage, using Wave 7 all-waves weight [95% confidence interval]	Difference in weighted percentages [C – A] [95% confidence interval]
Education (Wave 4 adults only)							
Less than high school or GED	5,809	15.1% [14.7%, 15.5%]	3,102	13.4% [12.9%, 14.0%]	-1.7% [-2.1%, -1.2%]	15.0% [14.5%, 15.5%]	-0.1% [-0.3%, 0.1%]
High school	7,698	23.3% [22.9%, 23.8%]	4,157	21.4% [20.8%, 22.0%]	-1.9% [-2.4%, -1.4%]	23.3% [22.7%, 23.9%]	-0.0% [-0.2%, 0.2%]
Some college but no degree, or associate's degree	11,457	31.6% [31.1%, 32.1%]	6,612	31.2% [30.6%, 31.9%]	-0.4% [-0.9%, 0.1%]	31.6% [30.9%, 32.3%]	-0.0% [-0.2%, 0.1%]
Bachelor's degree or advanced degree	6,890	29.9% [29.4%, 30.5%]	4,665	33.9% [33.3%, 34.6%]	4.0% [3.5%, 4.5%]	30.1% [29.5%, 30.8%]	0.2% [0.0%, 0.3%]
Health insurance (Wave 4 adults only)							
Yes	26,340	88.1% [87.5%, 88.6%]	15,775	89.8% [89.1%, 90.4%]	1.7% [1.2%, 2.2%]	88.7% [87.9%, 89.4%]	0.6% [0.1%, 1.1%]
No	5,313	11.9% [11.4%, 12.5%]	2,645	10.2% [9.6%, 10.9%]	-1.7% [-2.2%, -1.2%]	11.3% [10.6%, 12.1%]	-0.6% [-1.1%, -0.1%]

^a Hispanic refers to Hispanic or Latino; Black refers to Black or African American. Alone means that the participant reported only one race.

Table 5-3. Comparison of Wave 4 current established tobacco-use estimates between Wave 4 adults eligible to be Wave 7 all-waves respondents and Wave 4 adults who were Wave 7 all-waves respondents (Wave 4 Cohort)*

Wave 4 characteristic ^a	Wave 4 adults who were eligible to be Wave 7 all-waves respondents		Wave 4 adults who were Wave 7 all-waves respondents				
	Sample size	A: Weighted estimate, using Wave 4 cross-sectional weight [95% confidence interval]	Sample size	Before Wave 7 weighting adjustment		After Wave 7 weighting adjustment	
				B: Weighted estimate, using Wave 4 cross-sectional weight [95% confidence interval]	Difference in weighted estimates [B – A] [95% confidence interval]	C: Weighted estimate, using Wave 7 all-waves weight [95% confidence interval]	Difference in weighted estimates [C – A] [95% confidence interval]
Overall	31,883	22.5% [21.9%, 23.0%]	18,535	19.6% [18.9%, 20.2%]	-2.9% [-3.3%, -2.4%]	22.5% [21.9%, 23.1%]	0.0% [-0.3%, 0.3%]
Sex							
Male	15,476	27.9% [27.0%, 28.8%]	8,565	24.0% [22.9%, 25.2%]	-3.9% [-4.7%, -3.1%]	27.8% [26.9%, 28.8%]	-0.1% [-0.6%, 0.4%]
Female	16,374	17.5% [16.9%, 18.2%]	9,947	15.8% [15.1%, 16.6%]	-1.7% [-2.1%, -1.3%]	17.6% [16.9%, 18.4%]	0.1% [-0.2%, 0.4%]
Age							
18-24	10,905	22.3% [21.1%, 23.5%]	5,557	18.9% [17.4%, 20.4%]	-3.4% [-4.4%, -2.5%]	21.8% [20.4%, 23.2%]	-0.5% [-1.4%, 0.3%]
25-44	10,979	28.2% [27.2%, 29.2%]	6,422	24.2% [22.9%, 25.5%]	-4.0% [-4.9%, -3.2%]	28.3% [27.0%, 29.5%]	0.1% [-0.7%, 0.8%]
45-64	7,444	22.9% [21.9%, 23.9%]	4,933	20.2% [19.1%, 21.3%]	-2.8% [-3.5%, -2.0%]	22.7% [21.5%, 23.9%]	-0.3% [-0.9%, 0.4%]
65 or older	2,553	10.1% [9.0%, 11.3%]	1,622	9.6% [8.3%, 11.1%]	-0.5% [-1.1%, 0.2%]	10.8% [9.3%, 12.4%]	0.7% [-0.1%, 1.4%]

Table 5-3. Comparison of Wave 4 current established tobacco-use estimates between Wave 4 adults eligible to be Wave 7 all-waves respondents and Wave 4 adults who were Wave 7 all-waves respondents (Wave 4 Cohort)* (continued)

Wave 4 characteristic ^a	Wave 4 adults who were eligible to be Wave 7 all-waves respondents		Wave 4 adults who were Wave 7 all-waves respondents				
	Sample size	A: Weighted estimate, using Wave 4 cross-sectional weight [95% confidence interval]	Sample size	Before Wave 7 weighting adjustment		After Wave 7 weighting adjustment	
				B: Weighted estimate, using Wave 4 cross-sectional weight [95% confidence interval]	Difference in weighted estimates [B – A] [95% confidence interval]	C: Weighted estimate, using Wave 7 all-waves weight [95% confidence interval]	Difference in weighted estimates [C – A] [95% confidence interval]
Race/ethnicity ^b							
Hispanic	6,337	17.1% [16.0%, 18.3%]	3,570	15.1% [13.9%, 16.4%]	-2.1% [-3.1%, -1.0%]	17.3% [16.1%, 18.6%]	0.2% [-0.6%, 1.0%]
Non-Hispanic White alone	17,971	24.2% [23.4%, 25.0%]	10,536	20.5% [19.6%, 21.4%]	-3.7% [-4.3%, -3.2%]	24.0% [23.2%, 24.9%]	-0.1% [-0.5%, 0.2%]
Non-Hispanic Black alone	4,643	25.2% [23.8%, 26.6%]	2,763	24.9% [23.0%, 26.9%]	-0.3% [-1.6%, 1.0%]	26.1% [24.5%, 27.8%]	0.9% [-0.1%, 1.9%]
Non-Hispanic other race or multiple races	2,469	15.7% [14.2%, 17.4%]	1,409	13.6% [11.9%, 15.5%]	-2.1% [-3.4%, -0.8%]	15.6% [13.8%, 17.6%]	-0.1% [-1.4%, 1.2%]
Census region							
Northeast	4,750	20.5% [18.9%, 22.3%]	2,695	18.1% [16.4%, 19.9%]	-2.4% [-3.4%, -1.5%]	20.6% [19.1%, 22.3%]	0.1% [-0.6%, 0.8%]
Midwest	7,597	25.7% [24.1%, 27.3%]	4,510	22.4% [20.7%, 24.2%]	-3.2% [-4.4%, -2.1%]	25.5% [24.0%, 27.0%]	-0.2% [-0.8%, 0.4%]
South	12,005	24.6% [23.4%, 25.9%]	6,814	21.6% [20.3%, 23.0%]	-3.0% [-3.6%, -2.4%]	24.5% [23.3%, 25.8%]	-0.1% [-0.6%, 0.4%]
West	7,531	17.7% [16.4%, 19.1%]	4,516	15.1% [13.6%, 16.8%]	-2.6% [-3.5%, -1.6%]	17.9% [16.5%, 19.4%]	0.2% [-0.3%, 0.8%]

Table 5-3. Comparison of Wave 4 current established tobacco-use estimates between Wave 4 adults eligible to be Wave 7 all-waves respondents and Wave 4 adults who were Wave 7 all-waves respondents (Wave 4 Cohort)* (continued)

Wave 4 characteristic ^a	Wave 4 adults who were eligible to be Wave 7 all-waves respondents		Wave 4 adults who were Wave 7 all-waves respondents				
	Sample size	A: Weighted estimate, using Wave 4 cross-sectional weight [95% confidence interval]	Sample size	Before Wave 7 weighting adjustment		After Wave 7 weighting adjustment	
				B: Weighted estimate, using Wave 4 cross-sectional weight [95% confidence interval]	Difference in weighted estimates [B – A] [95% confidence interval]	C: Weighted estimate, using Wave 7 all-waves weight [95% confidence interval]	Difference in weighted estimates [C – A] [95% confidence interval]
Education (Wave 4 adults only)							
Less than high school or GED	5,773	36.1% [34.2%, 38.1%]	3,078	34.9% [32.4%, 37.5%]	-1.2% [-2.7%, 0.3%]	36.5% [34.4%, 38.7%]	0.4% [-0.4%, 1.3%]
High school	7,665	28.0% [26.6%, 29.5%]	4,141	25.5% [23.9%, 27.3%]	-2.5% [-3.6%, -1.4%]	28.2% [26.7%, 29.7%]	0.2% [-0.5%, 0.8%]
Some college but no degree, or associate's degree	11,436	23.8% [22.8%, 24.8%]	6,603	20.7% [19.5%, 21.9%]	-3.1% [-3.8%, -2.4%]	23.6% [22.5%, 24.7%]	-0.2% [-0.7%, 0.3%]
Bachelor's degree or advanced degree	6,881	9.7% [9.0%, 10.5%]	4,661	8.6% [7.8%, 9.5%]	-1.1% [-1.5%, -0.7%]	9.8% [8.9%, 10.7%]	0.0% [-0.4%, 0.5%]
Health insurance (Wave 4 adults only)							
Yes	26,274	20.6% [20.0%, 21.1%]	15,743	17.9% [17.3%, 18.5%]	-2.6% [-3.1%, -2.2%]	20.5% [19.9%, 21.2%]	-0.0% [-0.4%, 0.3%]
No	5,279	36.2% [34.5%, 37.9%]	2,625	33.3% [31.0%, 35.7%]	-2.9% [-4.8%, -1.0%]	36.8% [34.5%, 39.2%]	0.6% [-1.2%, 2.5%]

* Current established use of a tobacco product means the person currently uses the product every day or some days and—for cigarettes—has smoked at least 100 cigarettes in their lifetime and—for any other tobacco product—has reported they ever used that product fairly regularly. This characteristic considers the following products covered by the Wave 4 adult interview: cigarettes, traditional cigars, cigarillos, filtered cigars, pipes, smokeless tobacco, snus, hookah, and electronic nicotine delivery systems or ENDS (such as e-cigarettes, vape pens, personal vaporizers and mods, e-cigars, e-pipes, e-hookahs, and hookah pens).

^a For each characteristic, the sum of the counts across the categories may not be equal to the count in the overall row due to missing values.

^b Hispanic refers to Hispanic or Latino; Black refers to Black or African American. Alone means that the participant reported only one race.

Table 5-4. Comparison of Wave 4 ever tobacco-use estimates between Wave 4 youth eligible to be Wave 7 all-waves respondents and Wave 4 youth who were Wave 7 all-waves respondents (among Wave 7 adults in the Wave 4 Cohort)*

Wave 4 characteristic ^a	Wave 4 youth who were eligible to be Wave 7 adult all-waves respondents		Wave 4 youth who were Wave 7 adult all-waves respondents				
	Sample size	A: Weighted estimate, using Wave 4 cross-sectional weight [95% confidence interval]	Sample size	Before Wave 7 weighting adjustment		After Wave 7 weighting adjustment	
				B: Weighted estimate, using Wave 4 cross-sectional weight [95% confidence interval]	Difference in weighted estimates [B – A] [95% confidence interval]	C: Weighted estimate, using Wave 7 all-waves weight [95% confidence interval]	Difference in weighted estimates [C – A] [95% confidence interval]
Overall	12,065	23.8% [22.7%, 24.8%]	6,431	20.5% [19.4%, 21.6%]	-3.3% [-4.0%, -2.6%]	22.9% [21.8%, 24.0%]	-0.9% [-1.3%, -0.4%]
Sex							
Male	6,145	25.6% [24.4%, 26.9%]	3,151	20.9% [19.4%, 22.5%]	-4.8% [-6.0%, -3.6%]	24.0% [22.5%, 25.5%]	-1.7% [-2.6%, -0.8%]
Female	5,892	21.8% [20.5%, 23.2%]	3,271	20.1% [18.5%, 21.8%]	-1.7% [-2.8%, -0.6%]	21.9% [20.3%, 23.5%]	0.0% [-0.8%, 0.8%]
Race/ethnicity ^b							
Hispanic	3,536	24.1% [22.4%, 25.9%]	1,977	21.7% [19.7%, 23.8%]	-2.4% [-4.0%, -0.8%]	24.0% [21.9%, 26.3%]	-0.1% [-1.5%, 1.4%]
Non-Hispanic White alone	5,428	25.7% [24.1%, 27.3%]	2,715	21.8% [20.2%, 23.5%]	-3.8% [-5.0%, -2.6%]	24.7% [23.1%, 26.4%]	-0.9% [-1.8%, -0.1%]
Non-Hispanic Black alone	1,572	19.3% [17.0%, 22.0%]	895	17.1% [14.1%, 20.5%]	-2.3% [-4.1%, -0.4%]	18.2% [15.1%, 21.9%]	-1.1% [-3.1%, 0.9%]
Non-Hispanic other race or multiple races	1,136	20.1% [17.5%, 23.0%]	629	17.9% [14.9%, 21.4%]	-2.2% [-5.1%, 0.6%]	19.6% [16.6%, 23.0%]	-0.5% [-3.0%, 1.9%]

Table 5-4. Comparison of Wave 4 ever tobacco-use estimates between Wave 4 youth eligible to be Wave 7 all-waves respondents and Wave 4 youth who were Wave 7 all-waves respondents (among Wave 7 adults in the Wave 4 Cohort)* (continued)

Wave 4 characteristic ^a	Wave 4 youth who were eligible to be Wave 7 adult all-waves respondents		Wave 4 youth who were Wave 7 adult all-waves respondents				
	Sample size	A: Weighted estimate, using Wave 4 cross-sectional weight [95% confidence interval]	Sample size	Before Wave 7 weighting adjustment		After Wave 7 weighting adjustment	
				B: Weighted estimate, using Wave 4 cross-sectional weight [95% confidence interval]	Difference in weighted estimates [B – A] [95% confidence interval]	C: Weighted estimate, using Wave 7 all-waves weight [95% confidence interval]	Difference in weighted estimates [C – A] [95% confidence interval]
Census region							
Northeast	1,664	25.4% [23.4%, 27.6%]	822	22.6% [19.8%, 25.6%]	-2.9% [-4.8%, -0.9%]	25.5% [22.6%, 28.5%]	0.0% [-1.4%, 1.5%]
Midwest	2,651	24.9% [23.0%, 27.0%]	1,379	21.4% [19.3%, 23.6%]	-3.6% [-4.8%, -2.4%]	24.5% [22.2%, 26.9%]	-0.5% [-1.7%, 0.8%]
South	4,593	22.8% [21.4%, 24.2%]	2,458	19.3% [17.7%, 21.0%]	-3.5% [-4.7%, -2.2%]	21.3% [19.6%, 23.1%]	-1.4% [-2.5%, -0.4%]
West	3,157	23.1% [20.3%, 26.3%]	1,772	20.2% [17.6%, 23.0%]	-3.0% [-4.4%, -1.5%]	22.2% [19.4%, 25.4%]	-0.9% [-1.9%, 0.1%]

* Ever use of a tobacco product means the person has ever used the product, even one or two times. This characteristic considers the following products covered by the Wave 4 youth interview: cigarettes, traditional cigars, cigarillos, filtered cigars, pipes, smokeless tobacco, snus, hookah, electronic nicotine delivery systems or ENDS (such as e-cigarettes, vape pens, personal vaporizers and mods, e-cigars, e-pipes, e-hookahs, and hookah pens), dissolvable tobacco, bidis, and kreteks.

^a For each characteristic, the sum of the counts across the categories may not be equal to the count in the overall row due to missing values.

^b Hispanic refers to Hispanic or Latino; Black refers to Black or African American. Alone means that the participant reported only one race.

Tables 5-3 and 5-4 show the comparisons for the Wave 4 tobacco-use measures between those eligible to be Wave 7 adult all-waves respondents and the Wave 7 adult all-waves respondents, among the Wave 4 Cohort.

For the “before Wave 7 weighting adjustment” estimates in Table 5-3, the largest difference between those eligible to be Wave 7 adult all-waves respondents and the Wave 7 adult all-waves respondents was 4.0 percentage points, for adults ages 25 to 44 at Wave 4. Nearly all the 95 percent confidence intervals for the estimated differences were below zero, indicating an underrepresentation of Wave 4 current established use of any tobacco among Wave 4 adults who were Wave 7 all-waves respondents. Similarly, all but one of the confidence intervals for the estimated differences in Table 5-4 were below zero, indicating an underrepresentation of Wave 4 ever use of any tobacco among Wave 4 youth who were Wave 7 adult all-waves respondents—the largest difference was 4.8 percentage points for males.

For the “after Wave 7 weighting adjustment” estimates in Table 5-3, the differences were mostly negligible and all the 95 percent confidence intervals included zero, indicating a relatively unbiased estimation of Wave 4 current established tobacco use based on the Wave 7 all-waves respondents who were adults at Wave 4. In Table 5-4, most of the confidence intervals for the differences in ever use of tobacco at Wave 4 between the Wave 4 youth who were eligible to be Wave 7 adult all-waves respondents and the Wave 4 youth who were Wave 7 adult all-waves respondents included zero. However, the confidence intervals were below zero overall, for males, for non-Hispanic White alone persons, and for those residing in the South census region at Wave 4, indicating an underrepresentation of Wave 4 ever use of tobacco among the Wave 7 adult all-waves respondents who were youth at Wave 4. The largest underrepresentation was again for males, but the magnitude of the difference (1.7 percentage points) was notably reduced after applying the Wave 7 weighting adjustments.

Overall, these results reflect positively on the use of demographic, socio-economic, and tobacco-use variables from Wave 4 for calibrating the Wave 7 all-waves weight for the Wave 4 Cohort. If the Wave 4 demographic, socio-economic, and tobacco-use characteristics in Tables 5-2 through 5-4 are correlated with key tobacco- and health-related outcome measures at Wave 7 of the PATH Study, the results suggest a very small amount of potential bias in the adult estimates for the Wave 4 Cohort due to nonresponse at any primary wave since Wave 4.

5.2 Youth Interview

5.2.1 All-Waves Response Rates Conditioning on Wave 4 Response

This section summarizes the Wave 4 Cohort all-waves response rates for the Wave 7 youth interview, conditioning on Wave 4 response. Unweighted and weighted all-waves response rates were calculated for the Wave 7 youth interview among those in the Wave 4 Cohort using the following formulas:

$$RR_{Y_AW} = C_{Y_AW} / (C_{Y_AW} + N_{Y_AW} + e_{Y_AW} \times U_{Y_AW})$$

$$e_{Y_AW} = (C_{Y_AW} + N_{Y_AW}) / (C_{Y_AW} + N_{Y_AW} + I_{Y_AW})$$

where

RR_{Y_AW} = Wave 7 youth interview all-waves response rate;

C_{Y_AW} = number of Wave 7 youth who were respondents at each of primary Waves 5 through 7;

N_{Y_AW} = number of Wave 7 youth known to be eligible at each wave, but a nonrespondent at least once for primary Waves 5 through 7;

U_{Y_AW} = number of Wave 7 youth with unknown eligibility status at least once and never ineligible for primary Waves 5 through 7;

e_{Y_AW} = estimated proportion of eligible youth among the Wave 7 youth with unknown eligibility status (from an all-waves perspective); and

I_{Y_AW} = number of Wave 7 youth who were ineligible at least once for primary Waves 5 through 7.

Unweighted counts and weighted counts based on the Wave 4 cross-sectional weight were obtained for the all-waves response status categories C_{Y_AW} , N_{Y_AW} , U_{Y_AW} , and I_{Y_AW} for the unweighted all-waves response rates (RR_{Y_AW}) and weighted all-waves response rates ($RR_{Y_W_AW}$), respectively.

Table 5-5 provides all-waves response rates for the Wave 7 youth interview. In addition to the overall row, the table includes rows for recruitment wave as well as the following Wave 4 characteristics: sex, age, race/ethnicity, census region, and tobacco-use status. Participants for whom the Wave 4 interview (or Wave 4 household screener in the case of shadow youth) had missing

information for the demographic or tobacco-use characteristic were excluded from the response rate calculation for that subgroup.

The unweighted and weighted all-waves response rates for the Wave 7 youth interview were 53.3 percent and 51.8 percent, respectively; the unweighted response rates were also generally slightly higher than the corresponding weighted response rates by subgroup. The unweighted and weighted single-wave response rates for the Wave 7 youth interview were higher than the all-waves response rates at 57.3 percent and 55.9 percent, respectively.

The weighted all-waves response rates varied moderately by Wave 4 sex and race/ethnicity. However, the response rates by Wave 4 age, census region, and ever use of tobacco, and by recruitment wave showed more variation. The weighted response rate for youth younger than age 12 at Wave 4 (47.6 percent) was noticeably lower than for youth ages 12 to 17 at that time (62.7 percent).²⁶ (In previous waves, a similar pattern has been observed for the youth interview response rates by age group among the Wave 1 and Wave 4 Cohorts.) For the Wave 4 Cohort, this is likely because the participants younger than age 12 were shadow youth at Wave 4 and became cohort members without having completed a Wave 4 interview. About half of the Wave 4 shadow youth were first asked to complete a PATH Study interview at the Wave 4.5 special data collection; the other half were first asked to complete an interview at Wave 5. Youth who resided in the West census region at Wave 4 had the highest all-waves response rate (57.9 percent) and youth who resided in the Midwest census region had the lowest (48.8 percent).

The weighted all-waves response rate was 10.7 percentage points lower for Wave 4 youth who had ever used tobacco at that time (52.3 percent) than for Wave 4 youth who had never used tobacco (63.0 percent). Lastly, the Wave 7 youth weighted all-waves response rate was 65.2 percent for those recruited at Wave 1 and 48.6 percent for those recruited at Wave 4; a possible explanation is that Wave 4 Cohort members who have responded to multiple previous waves, particularly those recruited at Wave 1, are likely to be more invested in the PATH Study and may be more cooperative than other cohort members. The Wave 1 sample members of the Wave 4 Cohort have persisted with the PATH Study over a number of years and all completed an interview at Wave 4, including

²⁶ In theory, all participants who were ages 13 to 17 at Wave 4 should have reached adulthood by Wave 7. However, some were still youth at Wave 7 as the result of age corrections since Wave 4. Due to the small sample size, these participants were combined into one category, youth ages 12 to 17 at Wave 4, for the purposes of this report.

the Wave 1 shadow youth. The Wave 4 replenishment sample members of the Wave 4 Cohort do not have the same length of history with the study and include Wave 4 shadow youth, who did not have to complete an interview at Wave 4 to become a study participant.

Variation in response rates by subgroups is to be expected in large-scale data collection efforts. However, it is pertinent to examine the effect of the Wave 7 all-waves weighting adjustments for the Wave 4 Cohort, given the differences in response propensity noted above. The results of these analyses are discussed in the next section.

Table 5-5. Wave 7 youth interview all-waves response rates conditioning on Wave 4 response (Wave 4 Cohort)

Characteristic ^a	<i>C_{Y,AW}</i> Completed Wave 7 youth interview and responded at each primary wave since Wave 4 (n)	<i>I_{Y,AW}</i> Ineligible for at least one primary wave since Wave 4 (n)	<i>N_{Y,AW}</i> Nonresponse for at least one primary wave and known to be eligible for each primary wave since Wave 4 (n)	<i>U_{Y,AW}</i> Nonresponse with unknown eligibility status for at least one primary wave and never ineligible for a primary wave since Wave 4 (n)	<i>RR_{Y,AW}</i> Unweighted all-waves response rate ^b (%)	<i>RR_{Y,W,AW}</i> Weighted all- waves response rate ^b (%)
Overall	3,198	42	766	2,060	53.3	51.8
Wave 4 sex						
Male	1,679	16	398	1,112	52.8	51.3
Female	1,513	26	367	943	53.8	52.3
Wave 4 age						
Under 12	2,089	31	541	1,624	49.3	47.6
12-17	1,109	11	225	435	62.8	62.7
Wave 4 race/ethnicity ^c						
Hispanic	910	16	234	586	52.8	51.9
Non-Hispanic White alone	1,544	11	370	985	53.4	51.7
Non-Hispanic Black alone	340	7	85	242	51.3	48.4
Non-Hispanic other race or multiple races	319	7	60	214	54.1	53.8
Wave 4 census region						
Northeast	450	6	118	307	51.6	50.3
Midwest	718	5	200	509	50.4	48.8
South	1,108	19	287	761	51.6	50.1
West	922	12	161	483	59.1	57.9
Wave 4 ever tobacco use ^d (Wave 4 youth only)						
Yes	44	0	13	23	55.0	52.3
No	1,043	10	208	407	63.0	63.0

Table 5-5. Wave 7 youth interview all-waves response rates conditioning on Wave 4 response (Wave 4 Cohort) (continued)

Characteristic ^a	<i>C_{Y,AW}</i> Completed Wave 7 youth interview and responded at each primary wave since Wave 4 (n)	<i>I_{Y,AW}</i> Ineligible for at least one primary wave since Wave 4 (n)	<i>N_{Y,AW}</i> Nonresponse for at least one primary wave and known to be eligible for each primary wave since Wave 4 (n)	<i>U_{Y,AW}</i> Nonresponse with unknown eligibility status for at least one primary wave and never ineligible for a primary wave since Wave 4 (n)	<i>RR_{Y,AW}</i> Unweighted all-waves response rate ^b (%)	<i>RR_{Y,W,AW}</i> Weighted all-waves response rate ^b (%)
Recruitment wave						
Wave 1	782	5	158	259	65.3	65.2
Wave 4	2,416	37	608	1,801	50.3	48.6

^a For each characteristic, the sum of the counts across the categories may not be equal to the count in the overall row due to missing values.

^b The youth interview unweighted all-waves response rate formula is provided earlier in this section. The weighted response rates were calculated using a similar approach except that the Wave 4 cross-sectional weight was used to obtain weighted counts.

^c Hispanic refers to Hispanic or Latino; Black refers to Black or African American. Alone means that the participant reported only one race.

^d Ever use of a tobacco product means the person has ever used the product, even one or two times. This characteristic considers the following products covered by the Wave 4 youth interview: cigarettes, traditional cigars, cigarillos, filtered cigars, pipes, smokeless tobacco, snus, hookah, electronic nicotine delivery systems or ENDS (such as e-cigarettes, vape pens, personal vaporizers and mods, e-cigars, e-pipes, e-hookahs, and hookah pens), dissolvable tobacco, bidis, and kreteks.

5.2.2 Comparisons Between Those Eligible to be Wave 7 All-Waves Respondents and Wave 7 All-Waves Respondents (Among the Wave 4 Cohort)

This section compares weighted estimates between the members of the Wave 4 Cohort who were eligible to be Wave 7 youth all-waves respondents and the Wave 7 youth all-waves respondents in the Wave 4 Cohort. The weighted estimates cover Wave 4 demographic characteristics (shown in Table 5-6) and Wave 4 tobacco-use status (shown in Table 5-7). The structure of Table 5-6 is similar to Table 4-6 in Section 4.2.2, and the structure of Table 5-7 is similar to Table 5-4 in Section 5.1.2.

Before applying the Wave 7 weighting adjustments, the distributions among those eligible to be Wave 7 youth all-waves respondents were similar to the distributions among the Wave 7 youth all-waves respondents for Wave 4 sex and race/ethnicity (see Table 5-6). The most noticeable underrepresentation among the Wave 7 youth all-waves respondents from the Wave 4 Cohort was for those younger than age 12 at Wave 4, with a corresponding overrepresentation of youth ages 12 to 17 at that time. Among the Wave 7 youth all-waves respondents, persons residing in the West census region at Wave 4 were overrepresented.

As shown in the last column of Table 5-6, the Wave 7 all-waves weighting adjustments reduced the differences between the Wave 4 Cohort members eligible to be Wave 7 youth all-waves respondents and those who were Wave 7 youth all-waves respondents. Furthermore, the estimated 95 percent confidence intervals for the “after Wave 7 weighting adjustment” differences included zero or were in the proximity of zero for all but two of the Wave 4 demographic subgroups examined. For Wave 4 age, the underrepresentation of those under age 12 (and consequent overrepresentation of ages 12 to 17) improved after Wave 7 weighting adjustments but persisted to some degree. Note that only those age 12 among the Wave 4 “12-17” age group would be expected still to be youth at Wave 7, but some participants ages 13 to 17 at Wave 4 also had not reached adulthood.²⁷ Moreover, some who were age 12 at Wave 4 were adults at Wave 7 and so were included in the adult tables in

²⁷ There were two ways Wave 7 youth all-waves respondents could have been ages 13 to 17 at Wave 4. First, those age 13 at Wave 4 could have completed the Wave 4 youth interview toward the end of the Wave 4 data collection period, just after they turned age 13, and completed the Wave 7 youth interview early in the Wave 7 data collection period before they turned age 18. Second, a small number of Wave 7 youth all-waves respondents had incorrect dates of birth recorded at Wave 4 and were younger than initially believed to be at Wave 4.

Section 5.1. Separate analyses, not shown here, confirmed that the Wave 4 age group distribution was preserved when the Wave 7 all-waves weight was applied to all Wave 7 all-waves respondents (i.e., youth and adults) together. That is, the results in Table 5-6 need to be interpreted in light of the fact that not all study participants were exactly 5 years older at the time of Wave 7 compared to their Wave 4 age. Those who resided in the Midwest census region at Wave 4 were slightly underrepresented among the Wave 7 youth all-waves respondents after Wave 7 weighting adjustments.

Table 5-6. Comparison of Wave 4 demographic characteristics between Wave 4 respondents eligible to be Wave 7 youth all-waves respondents and Wave 4 respondents who were Wave 7 youth all-waves respondents (Wave 4 Cohort)

Wave 4 characteristic	Wave 4 respondents who were eligible to be Wave 7 youth all-waves respondents		Wave 4 respondents who were Wave 7 youth all-waves respondents				
	Unweighted count	A: Weighted percentage, using Wave 4 cross-sectional weight [95% confidence interval]	Unweighted count	Before Wave 7 weighting adjustment		After Wave 7 weighting adjustment	
				B: Weighted percentage, using Wave 4 cross-sectional weight [95% confidence interval]	Difference in weighted percentages [B – A] [95% confidence interval]	C: Weighted percentage, using Wave 7 all-waves weight [95% confidence interval]	Difference in weighted percentages [C – A] [95% confidence interval]
Sex							
Male	3,189	51.2% [49.9%, 52.4%]	1,679	50.7% [49.0%, 52.5%]	-0.4% [-1.8%, 1.0%]	51.1% [49.3%, 52.8%]	-0.1% [-0.4%, 0.3%]
Female	2,823	48.8% [47.6%, 50.1%]	1,513	49.3% [47.5%, 51.0%]	0.4% [-1.0%, 1.8%]	48.9% [47.2%, 50.7%]	0.1% [-0.3%, 0.4%]
Age							
Under 12	4,254	72.1% [71.0%, 73.2%]	2,089	66.2% [64.6%, 67.8%]	-5.9% [-7.0%, -4.8%]	70.7% [69.1%, 72.2%]	-1.4% [-2.0%, -0.9%]
12-17	1,769	27.9% [26.8%, 29.0%]	1,109	33.8% [32.2%, 35.4%]	5.9% [4.8%, 7.0%]	29.3% [27.8%, 30.9%]	1.4% [0.9%, 2.0%]
Race/ethnicity ^a							
Hispanic	1,730	25.5% [24.4%, 26.7%]	910	25.7% [24.2%, 27.3%]	0.2% [-1.1%, 1.4%]	25.6% [24.1%, 27.2%]	0.0% [-0.3%, 0.4%]
Non-Hispanic White alone	2,899	51.5% [50.2%, 52.8%]	1,544	51.8% [50.0%, 53.5%]	0.2% [-1.3%, 1.7%]	51.5% [49.8%, 53.3%]	-0.0% [-0.5%, 0.5%]
Non-Hispanic Black alone	667	12.7% [11.8%, 13.5%]	340	11.8% [10.8%, 13.0%]	-0.8% [-1.8%, 0.1%]	12.8% [11.6%, 14.0%]	0.1% [-0.2%, 0.4%]
Non-Hispanic other race or multiple races	593	10.3% [9.5%, 11.1%]	319	10.7% [9.6%, 11.8%]	0.4% [-0.5%, 1.3%]	10.1% [9.1%, 11.2%]	-0.1% [-0.4%, 0.1%]

Table 5-6. Comparison of Wave 4 demographic characteristics between Wave 4 respondents eligible to be Wave 7 youth all-waves respondents and Wave 4 respondents who were Wave 7 youth all-waves respondents (Wave 4 Cohort) (continued)

Wave 4 characteristic	Wave 4 respondents who were eligible to be Wave 7 youth all-waves respondents		Wave 4 respondents who were Wave 7 youth all-waves respondents				
	Unweighted count	A: Weighted percentage, using Wave 4 cross-sectional weight [95% confidence interval]	Unweighted count	Before Wave 7 weighting adjustment		After Wave 7 weighting adjustment	
				B: Weighted percentage, using Wave 4 cross-sectional weight [95% confidence interval]	Difference in weighted percentages [B – A] [95% confidence interval]	C: Weighted percentage, using Wave 7 all-waves weight [95% confidence interval]	Difference in weighted percentages [C – A] [95% confidence interval]
Census region							
Northeast	875	16.1% [15.2%, 17.1%]	450	15.6% [14.4%, 16.9%]	-0.5% [-1.4%, 0.5%]	16.4% [15.1%, 17.7%]	0.3% [0.0%, 0.5%]
Midwest	1,427	21.2% [20.2%, 22.3%]	718	20.1% [18.7%, 21.5%]	-1.2% [-2.4%, 0.0%]	20.8% [19.5%, 22.3%]	-0.4% [-0.7%, -0.1%]
South	2,156	37.9% [36.7%, 39.2%]	1,108	36.7% [35.1%, 38.4%]	-1.2% [-2.7%, 0.2%]	38.2% [36.5%, 39.9%]	0.2% [-0.2%, 0.6%]
West	1,566	24.7% [23.6%, 25.8%]	922	27.6% [26.1%, 29.2%]	2.9% [1.7%, 4.2%]	24.6% [23.1%, 26.1%]	-0.1% [-0.4%, 0.2%]

^a Hispanic refers to Hispanic or Latino; Black refers to Black or African American. Alone means that the participant reported only one race.

Table 5-7. Comparison of Wave 4 ever tobacco-use estimates between Wave 4 youth eligible to be Wave 7 youth all-waves respondents and Wave 4 youth who were Wave 7 youth all-waves respondents (Wave 4 Cohort)*

Wave 4 characteristic ^a	Wave 4 youth who were eligible to be Wave 7 youth all-waves respondents		Wave 4 youth who were Wave 7 youth all-waves respondents				
	Sample size	A: Weighted estimate, using Wave 4 cross-sectional weight [95% confidence interval]	Sample size	Before Wave 7 weighting adjustment		After Wave 7 weighting adjustment	
				B: Weighted estimate, using Wave 4 cross-sectional weight [95% confidence interval]	Difference in weighted estimates [B – A] [95% confidence interval]	C: Weighted estimate, using Wave 7 all-waves weight [95% confidence interval]	Difference in weighted estimates [C – A] [95% confidence interval]
Overall	1,738	4.6% [3.6%, 5.9%]	1,087	3.9% [2.8%, 5.4%]	-0.7% [-1.7%, 0.2%]	4.6% [3.4%, 6.4%]	0.0% [-1.0%, 1.1%]
Sex							
Male	934	6.2% [4.7%, 8.3%]	585	4.9% [3.2%, 7.4%]	-1.3% [-2.9%, 0.2%]	6.1% [4.0%, 9.1%]	-0.2% [-2.0%, 1.7%]
Female	793	2.7% [1.7%, 4.2%]	496	2.8% [1.7%, 4.6%]	0.1% [-0.9%, 1.2%]	3.1% [1.9%, 5.1%]	0.5% [-0.7%, 1.6%]
Race/ethnicity ^b							
Hispanic	510	4.6% [2.9%, 7.4%]	320	2.9% [†] [1.5%, 5.4%]	-1.7% [-3.6%, 0.2%]	3.7% [†] [1.9%, 6.9%]	-1.0% [-3.1%, 1.2%]
Non-Hispanic White alone	776	5.0% [3.5%, 7.1%]	479	4.4% [2.7%, 7.0%]	-0.6% [-2.4%, 1.1%]	5.4% [3.3%, 8.6%]	0.4% [-1.6%, 2.4%]
Non-Hispanic Black alone	185	3.2% [†] [1.5%, 6.8%]	113	4.3% [†] [1.8%, 9.7%]	1.1% [-0.6%, 2.8%]	4.5% [†] [2.0%, 10.1%]	1.4% [-0.7%, 3.4%]
Non-Hispanic other race or multiple races	159	4.6% [†] [2.3%, 9.1%]	104	4.7% [†] [2.0%, 10.7%]	0.1% [-1.8%, 2.0%]	5.2% [†] [2.3%, 11.3%]	0.6% [-1.6%, 2.8%]

Table 5-7. Comparison of Wave 4 ever tobacco-use estimates between Wave 4 youth eligible to be Wave 7 youth all-waves respondents and Wave 4 youth who were Wave 7 youth all-waves respondents (Wave 4 Cohort)* (continued)

Wave 4 characteristic ^a	Wave 4 youth who were eligible to be Wave 7 youth all-waves respondents		Wave 4 youth who were Wave 7 youth all-waves respondents				
	Sample size	A: Weighted estimate, using Wave 4 cross-sectional weight [95% confidence interval]	Sample size	Before Wave 7 weighting adjustment		After Wave 7 weighting adjustment	
				B: Weighted estimate, using Wave 4 cross-sectional weight [95% confidence interval]	Difference in weighted estimates [B – A] [95% confidence interval]	C: Weighted estimate, using Wave 7 all-waves weight [95% confidence interval]	Difference in weighted estimates [C – A] [95% confidence interval]
Census region							
Northeast	260	5.1% [3.0%, 8.7%]	161	6.0% [†] [2.7%, 12.7%]	0.9% [-2.0%, 3.8%]	7.4% [†] [3.3%, 15.4%]	2.2% [-1.6%, 6.1%]
Midwest	407	4.2% [2.5%, 7.2%]	248	2.7% [1.3%, 5.6%]	-1.5% [-3.7%, 0.6%]	3.4% [†] [1.8%, 6.5%]	-0.8% [-3.2%, 1.6%]
South	606	5.5% [3.5%, 8.3%]	361	4.4% [2.7%, 7.1%]	-1.1% [-2.9%, 0.7%]	5.1% [3.3%, 7.9%]	-0.4% [-2.2%, 1.4%]
West	465	3.4% [2.1%, 5.4%]	317	2.7% [†] [1.3%, 5.7%]	-0.6% [-1.8%, 0.5%]	3.1% [†] [1.4%, 6.6%]	-0.3% [-1.7%, 1.2%]

* Ever use of a tobacco product means the person has ever used the product, even one or two times. This characteristic considers the following products covered by the Wave 4 youth interview: cigarettes, traditional cigars, cigarillos, filtered cigars, pipes, smokeless tobacco, snus, hookah, electronic nicotine delivery systems or ENDS (such as e-cigarettes, vape pens, personal vaporizers and mods, e-cigars, e-pipes, e-hookahs, and hookah pens), dissolvable tobacco, bidis, and kreteks.

^a For each characteristic, the sum of the counts across the categories may not be equal to the count in the overall row due to missing values.

^b Hispanic refers to Hispanic or Latino; Black refers to Black or African American. Alone means that the participant reported only one race.

[†] Estimate should be interpreted with caution because it has low precision. It is based on a denominator sample size of less than 50, or the coefficient of variation of the estimate or its complement is larger than 30 percent.

Table 5-7 shows the comparison of Wave 4 “ever tobacco-use” estimates between the Wave 4 youth respondents eligible to be Wave 7 youth all-waves respondents and the Wave 4 youth all-waves respondents (among the Wave 4 Cohort). The estimated 95 percent confidence intervals for both the “before Wave 7 weighting adjustment” differences and the “after Wave 7 weighting adjustment” differences included zero overall and for all the Wave 4 subgroups examined.

These results reflect positively on the use of both demographic and tobacco-use variables from Wave 4 for calibrating the Wave 7 all-waves weight for the Wave 4 Cohort. If the Wave 4 demographic and tobacco-use characteristics in Tables 5-6 and 5-7 are correlated with key tobacco- and health-related outcome measures at Wave 7 of the PATH Study, the results do not raise concern about bias in the youth interview estimates for the Wave 4 Cohort due to nonresponse at any primary wave since Wave 4.

6. Results of Wave 7 Cohort Nonresponse Bias Analyses

This chapter presents results of nonresponse bias analyses for the Wave 7 replenishment sample and the Wave 7 Cohort. Section 6.1.1 summarizes the Wave 7 household screener response rates for the entire Wave 7 replenishment sample, as well as for the Wave 7 replenishment AYS subsample where household members ages 9 and older were eligible for selection (see Section 2.9). Sections 6.2.1 and 6.3.1, respectively, examine adult and youth interview response rates for the Wave 7 replenishment sample, conditioning on Wave 7 household screener response. Sections 6.2.2 to 6.2.4 and 6.3.2 to 6.3.4 compare cross-sectional estimates of demographic and socio-economic characteristics, cigarette smoking, and ENDS use between the PATH Study Wave 7 Cohort and external sources. The results are presented separately for adults and youth.

As discussed in Section 2.9, the Wave 7 Cohort is comprised of participants recruited into the PATH Study at different times (Wave 1, Wave 4, or Wave 7). Members of the Wave 7 replenishment sample were asked to participate in the study for the first time at Wave 7; whereas those sampled at Wave 1 or Wave 4 had received multiple follow-up requests. As demonstrated by results for the Wave 4 Cohort presented in Tables C-1 and C-2, differences in response status prior to Wave 7 led to large differences in Wave 7 single-wave response rates. Furthermore, follow-up wave response rates are conditioning on Wave 4 response for those in the Wave 4 Cohort, whereas response rates for the adults and youth from the Wave 7 replenishment sample are conditioning on completion of the Wave 7 household screener. It is clear that the Wave 7 response rates for the different groups of study members comprising the Wave 7 Cohort are conceptually different, so no attempt was made to compute “blended” response rates for the Wave 7 Cohort at Wave 7. Instead, response rates were calculated separately for the Wave 4 Cohort²⁸ and the Wave 7 replenishment sample. The Wave 4 adult interview and youth interview single-wave response rates among Wave 4

²⁸Wave 4 Cohort members who at the time of Wave 7 were younger than age 15 and/or not in the CNP contribute to the Wave 4 Cohort single-wave response rates but do not belong to the Wave 7 Cohort. Because this group includes a relatively small number of cases (see Figure 2-1), their inclusion or exclusion has no meaningful impact on Wave 7 response rates.

respondents are presented in Appendix C, and Appendix D provides parent consent for shadow youth response rates.

6.1 Household Screener

6.1.1 Household Screener Response Rates for the Entire Wave 7 Replenishment Sample and the Adult, Youth, and Shadow Youth (AYS) Address Subsample

For the Wave 7 replenishment sample, the household screener combined typical screener functions (e.g., enumerating the household, collecting demographic information about each member and some household-level data, and selecting participants) with a special purpose for the PATH Study, collecting basic information on each adult's tobacco use for addresses in the AYS subsample. This allowed the study to classify the adult with sufficient validity for potential selection as a participant based on the PATH Study's sampling strata, even though some degree of misreporting of tobacco-use and demographic characteristics by the household screener respondent was expected.

The household screener response rates for the entire Wave 7 replenishment sample and the AYS address subsample were calculated using the following formulas:

$$RR_{HH} = C_{HH} / (C_{HH} + N_{HH} + e_{HH} \times U_{HH})$$

$$e_{HH} = (C_{HH} + N_{HH}) / (C_{HH} + N_{HH} + I_{HH})$$

where

RR_{HH} = Wave 7 replenishment sample household screener response rate;

C_{HH} = number of completed cases;

N_{HH} = number of nonresponding cases known to be eligible;

U_{HH} = number of nonresponding cases with unknown eligibility;

e_{HH} = estimated proportion of nonresponding cases with unknown eligibility that were eligible; and

I_{HH} = number of cases determined to be ineligible.

Unweighted counts and weighted counts based on the Wave 7 replenishment sample household IPS weight²⁹ were obtained for each response status category, for the unweighted response rates and weighted response rates, respectively. Table 6-1 provides overall household screener response rates for the entire Wave 7 replenishment sample and response rates by subgroups of sampled households. Subgroups were defined by the characteristics of the census block groups in which the sampled addresses were located, using information from the 5-year (2017 to 2021) ACS.³⁰ The “high” and “low” subgroup categories were defined relative to the nationwide percentage of persons having the characteristic: block groups with percentages below the national average for the characteristic were classified as low, and those with percentages above the national average were classified as high.

As shown in Table 6-1, the weighted household screener response rate for the entire Wave 7 replenishment sample was 55.6 percent. The weighted subgroup response rates differed from one another by 7.7 percentage points according to whether the percent of the neighborhood that identifies as White alone was above the national average, 7.2 percentage points by census region, 5.4 percentage points by whether the percent of the neighborhood that identifies as Hispanic was above the national average, and 4.8 percentage points by whether the percent of the neighborhood that identifies as Black alone was above the national average. The weighted subgroup response rates varied no more than 3.7 percentage points by education level or poverty status.

²⁹ For the Wave 7 replenishment AYS subsample, the corresponding AYS subsample household IPS weight was used.

³⁰ Information from the 5-year (2017 to 2021) rather than the 1-year (2021) ACS was used because 1-year ACS estimates are not provided for smaller geographies such as census tracts or block groups. These were the most recent 5-year ACS data available at the time response rates were first computed for Wave 7 of the PATH Study.

Table 6-1. Wave 7 household screener response rates for entire Wave 7 replenishment sample

Neighborhood characteristic ^a or census region	<i>CHH</i> Completed household screener (n)	<i>IHH</i> Ineligible (n)	<i>NHH</i> Nonresponse known to be eligible (n)	<i>UHH</i> Nonresponse with unknown eligibility status (n)	<i>RRHH</i> Unweighted response rate ^b (%)	<i>RRHH_w</i> Weighted response rate ^b (%)
Overall	125,706	17,676	84,953	16,497	55.7	55.6
Education (% with bachelor's degree)						
High (> 33.7%)	53,279	5,560	37,144	9,703	53.7	53.5
Low (<= 33.7%)	72,427	12,116	47,809	6,794	57.3	57.2
Race (% Black alone) ^c						
High (> 12.6%)	31,181	6,036	24,194	4,774	52.3	52.0
Low (<= 12.6%)	94,525	11,640	60,759	11,723	57.0	56.8
Race (% White alone)						
High (> 68.2%)	82,797	10,467	50,884	8,742	58.5	58.4
Low (<= 68.2%)	42,909	7,209	34,069	7,755	51.1	50.7
Ethnicity (% Hispanic) ^d						
High (> 18.4%)	33,645	4,615	26,220	5,491	51.9	51.7
Low (<= 18.4%)	92,061	13,061	58,733	11,006	57.3	57.1
Poverty status (% below poverty line)						
High (> 12.6%)	48,797	9,472	31,260	7,045	56.5	56.4
Low (<= 12.6%)	76,909	8,204	53,693	9,452	55.3	55.0
Census region						
Northeast	20,567	2,593	16,286	3,363	51.5	51.2
Midwest	29,402	3,880	18,134	3,157	58.4	58.4
South	47,706	8,413	32,674	6,013	55.6	55.4
West	28,031	2,790	17,859	3,964	56.6	56.4

^a The information used to define subgroups was from the 5-year (2017-2021) American Community Survey.

^b The unweighted household screener response rate formula is provided earlier in this section. The weighted response rates were calculated using a similar approach except that the Wave 7 replenishment sample household IPS weight was used to obtain weighted counts.

^c Black refers to Black or African American. Alone means that the participant reported only one race.

^d Hispanic refers to Hispanic or Latino.

The weighted household screener response rate for the Wave 7 replenishment AYS subsample was 30.3 percent (Table 6-2), as compared with 54.0 percent for households sampled at Wave 1 and 52.8 percent for the Wave 4 replenishment AYS subsample where all household members ages 10 and older and in the Wave 4 CNP were eligible for the PATH Study. The weighted subgroup response rates varied by 7.9 percentage points by census region, 6.4 percentage points by education level, 5.8 percentage points by poverty status, and 4.2 percentage points by whether the percent of the neighborhood that identifies as White alone was above the national average. The weighted subgroup response rates were less than 2.3 percentage points apart by whether the percent of the neighborhood that identifies as Black alone or the percent of neighborhood that identifies as Hispanic were above the respective national average.

During Wave 7, the PATH Study faced an unusual set of circumstances, including the ongoing COVID-19 pandemic and a national labor shortage. These challenges led to a higher-than-average field interviewer attrition rate, which was mitigated (but not eliminated) through the implementation of continuous recruiting and training cycles. With the available labor hours, and because of the need to completely replenish the pool of participants ages 14 and younger, the study chose to prioritize completing household screeners among the youth-targeted address subsamples. This was accomplished through the use of prioritization flags, a nonmonetary gamification program, and a monetary bonus program. As a result, the household screener response rates for the AYS subsample were notably lower than those for the entire Wave 7 replenishment sample. The other main reason for this pattern is that households in the youth-targeted address subsamples could be finalized as complete through verification that no youths resided in the household, whereas those in the AYS subsample could not (because of the need to also screen for adults). Finally, cases that refuse during the first contact typically receive a cooling-off period and a repeat visit where the field interviewer tries to convert the case into a complete. At Wave 7, again due to the labor shortage, the study was able to conduct refusal conversion efforts (which usually result in a 20 percent conversion rate) in only a very few PSUs.

Table 6-2. Wave 7 household screener response rates for Wave 7 replenishment AYS subsample

Neighborhood characteristic ^a or census region	C_{HH} Completed household screener (n)	I_{HH} Ineligible (n)	N_{HH} Nonresponse known to be eligible (n)	U_{HH} Nonresponse with unknown eligibility status (n)	RR_{HH} Unweighted response rate ^b (%)	$RR_{HH,w}$ Weighted response rate ^b (%)
Overall	10,013	3,058	20,028	3,141	30.5	30.3
Education (% with bachelor's degree)						
High (> 33.7%)	3,942	1,025	8,923	1,883	27.2	26.8
Low (<= 33.7%)	6,071	2,033	11,105	1,258	33.2	33.2
Race (% Black alone) ^c						
High (> 12.6%)	2,565	989	5,315	880	29.7	29.4
Low (<= 12.6%)	7,448	2,069	14,713	2,261	30.9	30.7
Race (% White alone)						
High (> 68.2%)	6,590	1,860	12,489	1,729	32.0	31.9
Low (<= 68.2%)	3,423	1,198	7,539	1,412	28.0	27.7
Ethnicity (% Hispanic) ^d						
High (> 18.4%)	2,727	776	5,848	972	28.9	28.7
Low (<= 18.4%)	7,286	2,282	14,180	2,169	31.2	31.0
Poverty status (% below poverty line)						
High (> 12.6%)	4,291	1,572	7,161	1,308	34.1	33.9
Low (<= 12.6%)	5,722	1,486	12,867	1,833	28.3	28.1
Census region						
Northeast	1,455	452	3,835	601	25.0	24.8
Midwest	2,385	683	4,414	601	32.7	32.7
South	3,999	1,380	7,461	1,192	32.0	31.6
West	2,174	543	4,318	747	30.4	30.2

^a The information used to define subgroups was from the 5-year (2017-2021) American Community Survey.

^b The unweighted household screener response rate formula is provided earlier in this section. The weighted response rates were calculated using a similar approach except that the Wave 7 replenishment AYS subsample household IPS weight was used to obtain weighted counts.

^c Black refers to Black or African American. Alone means that the participant reported only one race.

^d Hispanic refers to Hispanic or Latino.

6.2 Adult Interview

6.2.1 Response Rates for Wave 7 Replenishment Sample

For the Wave 7 replenishment sample, the adult interview response rate (conditioning on completion of the household screener) was calculated as the product of (1) the Phase 2 or individual screener response rate, and (2) the proportion of adults who completed the adult interview among those who completed the Phase 2 screener and were selected for the adult interview. The response rate was calculated as:

$$RR_A = (C_{P2}/(C_{P2} + N_{P2})) \times (C_A/(C_A + N_A))$$

where

RR_A = Wave 7 replenishment sample adult interview response rate;

C_{P2} = number of Phase 2 screener complete cases;

N_{P2} = number of Phase 2 screener nonrespondents;

C_A = number of Wave 7 replenishment sample adult interview complete cases; and

N_A = number of Wave 7 replenishment sample adult interview nonrespondents.

Unweighted response rates were calculated based on actual case counts, whereas weighted response rates were based on the sums of the Wave 7 replenishment AYS subsample IPS weights (calculated as the product of the AYS subsample household IPS weight and the inverse of the within-household probability of selection).

Table 6-3 provides the adult interview response rates. In addition to the overall row, the table includes rows on age, sex, race/ethnicity, and census region. Adults with missing values for a particular characteristic were excluded from the response rate calculation for that characteristic. Note that for the Wave 7 replenishment sample, $N_A = 0$ because all adults selected for interview after completing the Phase 2 screener completed the adult interview.

The adult interview weighted response rate (conditioning on household screener response) was 55.1 percent. The biggest subgroup difference occurred for census region, with adults from the West region having response rates more than 10 percentage points higher than those from the Northeast region. The second biggest response rate difference was by race/ethnicity; the weighted response

rate for non-Hispanic Black alone adults (58.6 percent) was more than 6 percentage points higher than for non-Hispanic adults of other or multiple races (52.2 percent). The variation in adult response rates was 5.4 percent by age and 5.0 percent by sex.

Table 6-3. Wave 7 adult interview response rates conditioning on household screener response (Wave 7 replenishment sample)

Characteristic reported by household screener respondent or census region	Phase 2 screener		Adult interview	RR _A : Unweighted response rate ^a (%)	RR _{A,W} : Weighted response rate ^a (%)
	CP2: Completed Phase 2 screener (n)	NP2: Phase 2 screener nonresponse (n)	CA: Completed adult interview (n)		
Overall	2,404	1,953	1,803	55.2	55.1
Age ^b					
18-24	757	709	678	51.6	51.3
25-44	660	518	461	56.0	55.3
45-64	584	433	398	57.4	56.7
65 or older	386	293	255	56.8	54.8
Sex					
Male	1,131	1,004	867	53.0	52.6
Female	1,273	944	936	57.4	57.6
Race/ethnicity ^{c, d}					
Hispanic	550	481	364	53.3	52.4
Non-Hispanic White alone	1,043	856	842	54.9	55.8
Non-Hispanic Black alone	618	441	446	58.4	58.6
Non-Hispanic other race or multiple races	174	157	145	52.6	52.2
Census region					
Northeast	298	302	216	49.7	48.8
Midwest	508	391	403	56.5	58.5
South	1,017	873	761	53.8	53.1
West	581	387	423	60.0	59.7

^a The adult interview unweighted response rate formula is provided earlier in this section. The weighted response rates were calculated using a similar approach except that the Wave 7 AYS replenishment subsample IPS weight was used to obtain weighted counts.

^b The sum of the counts across the categories does not equal the count in the overall row because a small number of participants were reported as under age 18.

^c The sum of the counts across the categories does not equal the count in the overall row due to missing values.

^d Hispanic refers to Hispanic or Latino; Black refers to Black or African American. Alone means that the participant reported only one race.

6.2.2 Comparison of Wave 7 Cohort Demographic and Socio-Economic Estimates to the American Community Survey

This nonresponse bias analysis investigates possible differences between weighted demographic and socio-economic estimates based on the Wave 7 Cohort of the PATH Study and corresponding estimates from the ACS. Although the response rates in Sections 6.1.1 and 6.2.1 pertain to the Wave 7 replenishment sample, the analyses in this section were conducted on the Wave 7 Cohort. The 1-year (2022) ACS estimates, calculated from the 2022 ACS PUMS,³¹ were used for comparison purposes because the majority of the Wave 7 data collection occurred in 2022. Estimates calculated from the ACS PUMS excluded institutional group quarters and persons in noninstitutional group quarters who were not college students. These exclusions correspond to the PATH Study's exclusions from the Wave 7 Cohort target population. By so doing, the PATH Study can assess the extent to which differential nonresponse among population subgroups may affect Wave 7 Cohort estimates. Estimates for the PATH Study were calculated using the Wave 7 Cohort cross-sectional weights, which were calibrated to match the 1-year 2021 ACS totals for sex, age group, race/ethnicity, education, and census region, because the 1-year 2022 ACS PUMS was not available at the time the weights were being created. In this section, results are presented for adults. Section 6.3.2 presents results from a similar analysis for youth.

Table 6-4 presents weighted estimates of adult demographic and socio-economic characteristics based on the Wave 7 Cohort adults. Adults with missing values for a particular characteristic were excluded from the Wave 7 Cohort estimates for that characteristic. The weighted distributions of sex, age, race/ethnicity, education, and census region are a close match to the 1-year 2022 ACS estimates. Because the Wave 7 cross-sectional weights were calibrated to 2021 ACS totals, a close, but not identical, match to the 2022 ACS estimates is the expected and desired result. Table 6-4 also shows that the estimated percentage of adults with health insurance based on the Wave 7 Cohort (88.4 percent) is 2.7 percentage points lower than the corresponding estimate from the 2022 ACS (91.1 percent).

³¹The ACS PUMS files provide a probability sample of individual records from the full ACS (United States Census Bureau, 2022), allowing data users to create custom tables that are not available through pre-tabulated ACS data products. Using the PUMS files allowed comparison estimates to be calculated that correspond to the target population for the PATH Study.

Table 6-4. Comparison of adult demographic and socio-economic estimates from PATH Study Wave 7 Cohort and 2022 ACS

Wave 7 characteristic	PATH Study Wave 7 Cohort		2022 ACS
	Unweighted count	Weighted percentage [95% confidence interval]	Weighted percentage [95% confidence interval]
Sex			
Male	14,062	48.5% [48.4%, 48.5%]	48.6% [48.6%, 48.6%]
Female	15,674	51.5% [51.5%, 51.6%]	51.4% [51.4%, 51.4%]
Age group			
18-24	10,310	11.6% [11.6%, 11.7%]	12.0% [12.0%, 12.0%]
25-44	10,496	34.2% [34.2%, 34.3%]	34.1% [34.1%, 34.2%]
45-64	5,832	32.5% [32.5%, 32.5%]	31.9% [31.9%, 31.9%]
65 or older	3,140	21.6% [21.6%, 21.6%]	22.0% [22.0%, 22.0%]
Race/ethnicity			
Hispanic	6,755	17.2% [17.1%, 17.2%]	17.2% [17.2%, 17.2%]
Non-Hispanic White alone	15,403	61.4% [61.3%, 61.5%]	60.6% [60.6%, 60.6%]
Non-Hispanic Black alone	4,586	11.0% [10.9%, 11.1%]	11.4% [11.4%, 11.4%]
Non-Hispanic other race or multiple races	2,454	10.5% [10.3%, 10.6%]	10.8% [10.8%, 10.8%]
Education			
Less than high school or GED	4,435	14.0% [14.0%, 14.1%]	13.8% [13.7%, 13.9%]
High school graduate	7,280	23.3% [23.2%, 23.4%]	23.3% [23.2%, 23.4%]
Some college but no degree, or associate's degree	10,307	29.7% [29.7%, 29.7%]	29.4% [29.4%, 29.5%]
Bachelor's degree or advanced degree	7,641	33.0% [32.9%, 33.0%]	33.5% [33.4%, 33.6%]
Census region			
Northeast	4,006	17.6% [17.6%, 17.6%]	17.5% [17.5%, 17.5%]
Midwest	6,759	20.7% [20.7%, 20.7%]	20.6% [20.6%, 20.6%]
South	11,472	38.0% [38.0%, 38.0%]	38.3% [38.3%, 38.3%]
West	7,543	23.6% [23.6%, 23.6%]	23.6% [23.6%, 23.6%]
Health insurance			
Yes	24,884	88.4% [87.9%, 88.9%]	91.1% [91.1%, 91.2%]
No	4,641	11.6% [11.1%, 12.1%]	8.9% [8.8%, 8.9%]

6.2.3 Comparison of Cigarette-Smoking Estimates to Other National Studies

This section compares adult cigarette-smoking estimates based on the Wave 7 Cohort respondents at Wave 7 of the PATH Study to similar estimates based on data from 2017-2020 NHANES, 2022 NHIS, 2022 NSDUH, and 2022 TUS-CPS.

Table 6-5 presents estimates of the prevalence of current cigarette smoking³² for the adult population as a whole and for subgroups. These estimates are accompanied by 95 percent confidence intervals. The point estimates from the PATH Study were calculated using the Wave 7 cross-sectional weight for the Wave 7 Cohort. The corresponding replicate weights were used to calculate variances and confidence intervals. Point estimates and 95 percent confidence intervals are reported for the other national studies as well.

Table 6-5 shows that the Wave 7 estimates of current cigarette smoking among adults tended to be lower than estimates from 2017-2020 NHANES and 2022 NSDUH and higher than estimates from 2022 NHIS and 2022 TUS-CPS. The estimates from Wave 7 of the PATH Study had non-overlapping 95 percent confidence intervals with about half the estimates from these external studies. The PATH Study confidence interval for the overall current cigarette-smoking estimate was non-overlapping with all four external studies. Wave 7 also had a non-overlapping confidence interval with two or three of the external studies for the estimate of current cigarette smoking among several subgroups of adults and for the overall estimate of every-day cigarette smoking among adults.

Table 6-5 does not show evidence of nonresponse bias in the Wave 7 estimates with respect to current cigarette smoking among adults in the sense that the Wave 7 Cohort estimates were within the range of estimates from comparable surveys.

³² For the PATH Study, following common practice for tobacco surveys, a person who currently smokes cigarettes is someone who (1) has smoked at least 100 cigarettes in their lifetime and (2) currently smokes cigarettes every day or some days. The questions used to define current cigarette smoking for each survey are provided in Appendix A.

Table 6-5. Comparison of cigarette-smoking estimates from Wave 7 adult interview (Wave 7 Cohort) and other national studies

Cigarette-smoking measure and subgroup ^a	Wave 7 (2022) PATH Study sample size	Percentage from Wave 7 (2022) PATH Study [95% confidence interval]	Percentage from 2017- 2020 NHANES [95% confidence interval]	Percentage from 2022 NHIS [95% confidence interval]	Percentage from 2022 NSDUH ^b [95% confidence interval]	Percentage from 2022 TUS-CPS [95% confidence interval]
Current, overall	29,764	13.1% [12.6%, 13.6%]	16.4% [14.4%, 18.5%]	11.6% [11.1%, 12.1%]	14.5% [13.9%, 15.2%]	9.4% [9.0%, 9.7%]
Current, male	14,051	14.7% [13.9%, 15.5%]	18.3% [16.0%, 20.8%]	13.2% [12.5%, 14.0%]	16.5% [15.6%, 17.5%]	10.4% [9.9%, 10.9%]
Current, female	15,669	11.6% [11.1%, 12.2%]	14.6% [12.6%, 16.8%]	10.0% [9.4%, 10.7%]	12.7% [11.8%, 13.5%]	8.4% [7.9%, 8.8%]
Current, age 18-24	10,301	4.7% [4.2%, 5.2%]	11.7% [9.0%, 15.1%]	4.8% [3.8%, 6.2%]	N/A	3.8% [3.0%, 4.9%]
Current, age 25-44	10,495	15.6% [14.7%, 16.5%]	21.6% [18.1%, 25.8%]	12.5% [11.7%, 13.4%]	N/A	9.7% [9.1%, 10.3%]
Current, age 45-64	5,828	16.8% [15.9%, 17.8%]	17.8% [15.8%, 20.0%]	15.1% [14.2%, 16.0%]	N/A	12.5% [11.8%, 13.2%]
Current, age 65 or older	3,138	8.1% [7.2%, 9.2%]	8.0% [6.6%, 9.6%]	8.7% [8.0%, 9.4%]	N/A	7.3% [6.7%, 7.9%]
Current, Hispanic	6,750	10.1% [9.2%, 11.1%]	13.0% [11.1%, 15.2%]	8.0% [7.0%, 9.0%]	9.7% [8.5%, 11.0%]	5.8% [5.1%, 6.6%]
Current, non-Hispanic White alone	15,399	13.9% [13.2%, 14.6%]	16.7% [14.2%, 19.5%]	12.7% [12.0%, 13.3%]	16.2% [15.5%, 17.0%]	10.4% [9.9%, 10.9%]
Current, non-Hispanic Black alone	4,582	16.0% [14.8%, 17.3%]	20.6% [17.3%, 24.3%]	14.2% [12.8%, 15.8%]	15.4% [13.8%, 17.2%]	10.5% [9.4%, 11.7%]
Current, non-Hispanic other race or multiple races	2,453	9.7% [8.5%, 11.2%]	15.0% [11.7%, 19.0%]	7.6% [6.4%, 9.0%]	11.2% [9.5%, 13.1%]	7.5% [6.4%, 8.9%]
Current every day, overall	29,768	10.2% [9.8%, 10.6%]	12.7% [10.9%, 14.8%]	8.9% [8.4%, 9.3%]	N/A	7.0% [6.7%, 7.3%]
Current some days, overall	29,765	2.9% [2.7%, 3.2%]	3.6% [3.1%, 4.3%]	2.7% [2.5%, 3.0%]	N/A	2.3% [2.2%, 2.5%]

^a For the race/ethnicity subgroups, Hispanic refers to Hispanic or Latino, Black refers to Black or African American, and alone means that the respondent reported only one race.

^b NSDUH's definition of a person who currently smokes cigarettes is someone who has smoked part or all of a cigarette in the past 30 days, which is more expansive than the definition used in the other surveys. However, NSDUH contains questions on lifetime smoking and current smoking. These questions were used to construct a modified measure of "current smoking" that is comparable to that of the other surveys (Ryan, Trosclair, & Gfroerer, 2012). The construction of this variable is described in Appendix A.

Aside from the possibility of nonresponse bias in one or more of the studies compared, the differences between estimates in Table 6-5 can be explained by several potential reasons. In addition to the varying degrees of sampling and measurement errors, the surveys differ in survey mode, question order, context, and timing of data collection.

Each of the five studies conducted at least a portion of their interviews in person. At Wave 7, most PATH Study participants completed the interview in person; however, some responded by telephone and a small percentage completed a Web interview. 2017-2020 NHANES was conducted entirely in person, NSDUH was conducted in person and on the Web, and NHIS and TUS-CPS were conducted in person and by telephone. For in-person data collection, the PATH Study and 2022 NSDUH used audio computer-assisted self-interview (ACASI) administration for the tobacco-use questions, so that the interviewer did not see or hear responses to the questions. In contrast, 2017-2020 NHANES, 2022 NHIS, 2022 TUS-CPS, and telephone data collection for Wave 7 of the PATH Study used direct questioning by an interviewer: Direct questions by an interviewer can produce underreports of socially undesirable behavior compared to self-administration (Tourangeau & Smith, 1996), and this may explain some of the differences between the cigarette-smoking estimates shown in Table 6-5.

The contexts and purposes of these surveys also differ: CPS is a general survey on unemployment, NHANES and NHIS are general health surveys, and NSDUH is a cross-sectional survey on substance use (including tobacco use) and health, including mental health. Unlike the cross-sectional prevalence surveys, the PATH Study uses a longitudinal cohort design to assess within-person changes and between-person differences in tobacco-use behaviors and health over time. Differences among the questions used in the instruments of these different studies are outlined in Appendix A.

The timing of data collection across the surveys compared is similar with respect to year except for 2017-2020 NHANES, which occurred prior to the COVID-19 pandemic. The differences between tobacco-use estimates from Wave 7 of the PATH Study and NHANES could be in part due to a material change in behavior in the population. Legislation that raised the Federal minimum age for sale of tobacco products from 18 to 21 years went into effect on December 20, 2019, near the close of 2017-2020 NHANES. In Table 6-5, the pattern of estimates for adults ages 18 to 24 might reflect lower tobacco use among the young adult population due to this regulation.

6.2.4 Comparison of ENDS-Use Estimates to Other National Studies

This section compares adult ENDS-use estimates based on the Wave 7 Cohort respondents at Wave 7 of the PATH Study to similar estimates based on data from 2022 NHIS and 2022 TUS-CPS. The primary measure of ENDS use is whether the adult currently uses ENDS every day or some days.³³ Note that this nonresponse bias analysis uses a different definition of adult current ENDS use for computing the PATH Study estimates than was used for the NRBA Reports for Waves 5 and 6. The reports for Waves 5 and 6 used an additional criterion—having ever used ENDS “fairly regularly”—when defining the adult ENDS-use outcome. However, using ENDS “fairly regularly” is sensitive to respondent interpretation, and NHIS and TUS-CPS do not ask respondents about ever using ENDS “fairly regularly.” For these reasons, the Wave 7 estimates of adult ENDS use are based only on whether the respondent uses ENDS every day, some days, or not at all.

Table 6-6 presents estimates of the prevalence of current ENDS use for the adult population as a whole and for subgroups. These estimates are accompanied by 95 percent confidence intervals. The point estimates from the PATH Study were calculated using the Wave 7 cross-sectional weight for the Wave 7 Cohort. The corresponding replicate weights were used to calculate variances and confidence intervals. Point estimates and 95 percent confidence intervals are reported for the other national studies as well.

³³ The questions used to define ENDS use for each survey are provided in Appendix B.

Table 6-6. Comparison of ENDS-use estimates from Wave 7 adult interview (Wave 7 Cohort) and other national studies

ENDS-use measure and subgroup^a	Wave 7 (2022) PATH Study sample size	Percentage from Wave 7 (2022) PATH Study [95% confidence interval]	Percentage from 2022 NHIS [95% confidence interval]	Percentage from 2022 TUS-CPS [95% confidence interval]
Current, overall	29,767	7.1% [6.8%, 7.4%]	6.0% [5.6%, 6.4%]	3.2% [2.9%, 3.4%]
Current, male	14,054	7.9% [7.3%, 8.5%]	6.9% [6.3%, 7.5%]	3.8% [3.4%, 4.2%]
Current, female	15,669	6.4% [6.0%, 6.8%]	5.2% [4.7%, 5.8%]	2.6% [2.3%, 2.9%]
Current, age 18-24	10,307	18.6% [17.7%, 19.5%]	15.3% [13.4%, 17.4%]	8.6% [7.2%, 10.3%]
Current, age 25-44	10,493	10.5% [9.8%, 11.3%]	9.0% [8.3%, 9.8%]	4.7% [4.2%, 5.2%]
Current, age 45-64	5,831	3.4% [3.0%, 3.9%]	3.0% [2.6%, 3.5%]	1.4% [1.2%, 1.6%]
Current, age 65 or older	3,134	1.0% [0.7%, 1.4%]	1.0% [0.7%, 1.2%]	0.5% [0.4%, 0.7%]
Current, Hispanic	6,750	6.8% [6.0%, 7.7%]	4.1% [3.4%, 4.9%]	2.6% [2.1%, 3.2%]
Current, non-Hispanic White alone	15,401	7.4% [7.0%, 7.8%]	6.9% [6.4%, 7.5%]	3.5% [3.2%, 3.9%]
Current, non-Hispanic Black alone	4,582	6.2% [5.4%, 7.2%]	4.7% [3.8%, 5.7%]	2.0% [1.5%, 2.7%]
Current, non-Hispanic other race or multiple races	2,454	7.1% [6.1%, 8.3%]	5.5% [4.4%, 6.8%]	3.2% [2.4%, 4.3%]
Current every day, overall	29,767	3.6% [3.4%, 3.8%]	3.2% [2.9%, 3.6%]	1.8% [1.6%, 2.0%]
Current some days, overall	29,767	3.5% [3.3%, 3.7%]	2.8% [2.5%, 3.1%]	1.3% [1.2%, 1.5%]

^a For the race/ethnicity subgroups, Hispanic refers to Hispanic or Latino, Black refers to Black or African American, and alone means that the respondent reported only one race.

The Wave 7 estimates of current ENDS use among adults were higher than the estimates from 2022 NHIS and 2022 TUS-CPS; however, about half of the thirteen 95 percent confidence intervals overlapped between the PATH Study and NHIS. The 95 percent confidence intervals for Wave 7 of the PATH Study were higher than those for 2022 TUS-CPS for the overall estimates and for all subgroups, with the exception of adults ages 65 and older (for whom the confidence intervals barely overlapped). As discussed in Section 6.2.3, some differences in the estimates may be attributable to differences in survey mode, context, question order and wording, and measurement error.

6.3 Youth Interview

6.3.1 Response Rates for Wave 7 Replenishment Sample

For the Wave 7 replenishment sample, the youth interview response rate was calculated conditioning on completion of the household screener. Household screener response rates for the entire Wave 7 replenishment sample are reported in Table 6-1 in Section 6.1.1. The youth interview response rate (conditioning on completion of the household screener) was calculated as:

$$RR_Y = C_Y / (C_Y + N_Y)$$

where

RR_Y = Wave 7 replenishment sample youth interview response rate;

C_Y = number of Wave 7 replenishment sample youth interview complete cases; and

N_Y = number of Wave 7 replenishment sample youth interview nonrespondents.

Unweighted response rates were calculated based on actual case counts, whereas weighted response rates were based on the sums of the Wave 7 replenishment sample IPS weights (calculated as the product of the Wave 7 replenishment sample household IPS weight and the inverse of the within-household probability of selection).

Table 6-7 provides the youth interview response rates. In addition to the overall row, the table includes rows on age, sex, and race/ethnicity based on information collected through the household screener, as well as census region. Youth with missing values for a particular characteristic were excluded from the response rate calculation for that characteristic.

Table 6-7. Wave 7 youth interview response rates conditioning on household screener response (Wave 7 replenishment sample)

Characteristic reported by household screener respondent or census region	<i>C_y</i> Completed youth interview (n)	<i>N_y</i> Youth interview nonresponse (n)	<i>RR_y</i> Unweighted response rate ^a (%)	<i>RR_{y w}</i> Weighted response rate ^a (%)
Overall	7,321	4,006	64.6	64.0
Age^b				
12-13	3,276	1,618	66.9	67.1
14-17	3,938	2,359	62.5	62.3
Sex^c				
Male	3,812	2,068	64.8	64.1
Female	3,509	1,935	64.5	64.0
Race/ethnicity^{c, d}				
Hispanic	2,164	1,152	65.3	64.9
Non-Hispanic White alone	3,511	2,003	63.7	63.1
Non-Hispanic Black alone	911	470	66.0	64.5
Non-Hispanic other race or multiple races	714	369	65.9	65.9
Census region				
Northeast	894	609	59.5	58.3
Midwest	1,652	897	64.8	64.3
South	2,884	1,651	63.6	63.0
West	1,891	849	69.0	68.6

^a The youth interview unweighted response rate formula is provided earlier in this section. The weighted response rates were calculated using a similar approach except that the Wave 7 replenishment sample IPS weight was used to obtain weighted counts.

^b The sum of the counts across the categories does not equal the count in the overall row because, for a small number of participants, either information needed for assignment to the 12-13 or 14-17 category was unavailable, or age was reported as outside the 12-17 range.

^c The sum of the counts across the categories does not equal the count in the overall row due to missing values.

^d Hispanic refers to Hispanic or Latino; Black refers to Black or African American. Alone means that the participant reported only one race.

The youth interview weighted response rate (conditioning on household screener response) was 64.0 percent. The biggest subgroup difference occurred for census region, with youth from the West region having response rates more than 10 percentage points higher than those from the Northeast region. Differences among subgroups on weighted response rates were 4.8 percentage points by age, 0.1 percentage points for sex and 2.8 percentage points for race/ethnicity.

6.3.2 Comparison of Wave 7 Cohort Demographic Estimates to the American Community Survey

Table 6-8 compares weighted estimates of youth demographic characteristics based on the Wave 7 Cohort youth with corresponding estimates based on the 2022 ACS PUMS. Estimates for the

PATH Study were calculated using the Wave 7 Cohort cross-sectional weight, and the ACS PUMS population exclusions were the same as those described in Section 6.2.2 for the adult analysis. Youth with missing values for a particular characteristic were excluded from the Wave 7 Cohort estimates for that characteristic.

Table 6-8. Comparison of youth demographic characteristics from PATH Study Wave 7 Cohort and 2022 ACS

Wave 7 characteristic	PATH Study Wave 7 Cohort		2022 ACS
	Unweighted count	Weighted percentage [95% confidence interval]	Weighted percentage [95% confidence interval]
Sex			
Male	5,560	51.2% [51.2%, 51.3%]	51.5% [51.3%, 51.6%]
Female	5,044	48.8% [48.7%, 48.8%]	48.5% [48.4%, 48.7%]
Age group			
12-13	3,273	33.8% [33.8%, 33.8%]	33.2% [32.9%, 33.4%]
14-17	7,377	66.2% [66.2%, 66.2%]	66.8% [66.6%, 67.1%]
Race/ethnicity			
Hispanic	3,099	25.9% [25.8%, 26.1%]	25.9% [25.7%, 26.1%]
Non-Hispanic White alone	4,920	48.5% [48.4%, 48.7%]	48.2% [48.0%, 48.4%]
Non-Hispanic Black alone	1,187	12.7% [12.6%, 12.9%]	12.9% [12.7%, 13.1%]
Non-Hispanic other race or multiple races	1,071	12.8% [12.6%, 13.0%]	13.0% [12.9%, 13.2%]
Census region			
Northeast	1,353	15.9% [15.9%, 15.9%]	15.8% [15.7%, 15.9%]
Midwest	2,389	20.9% [20.9%, 20.9%]	20.9% [20.8%, 21.1%]
South	4,084	39.1% [39.1%, 39.1%]	39.4% [39.2%, 39.5%]
West	2,824	24.1% [24.1%, 24.1%]	24.0% [23.8%, 24.1%]

The weighted distributions of sex, age, race/ethnicity, and census region are close to the 1-year 2022 ACS estimates. The Wave 7 Cohort cross-sectional weights were calibrated to 2021 ACS totals for these four characteristics, so a close, but not identical, match to the 2022 ACS estimates is the expected and desired result.

Both the PATH Study and 2022 ACS estimates show a gender imbalance among youth ages 12 to 17 in the United States, with 51.5 percent being male and 48.5 percent being female. Although this finding may not be intuitive, the U.S. Census Bureau has documented that the “sex ratio at birth in the United States has been around 105 males for every 100 females, however, since mortality at every age is generally higher for males, the sex ratio naturally declines with age” (<https://www2.census.gov/library/publications/decennial/2020/census-briefs/c2020br-06.pdf>).

6.3.3 Comparison of Cigarette-Smoking Estimates to Other National Studies

This section compares estimates for two common measures of youth cigarette smoking based on the Wave 7 Cohort respondents at Wave 7 of the PATH Study to similar estimates based on data from 2017-2020 NHANES, 2021 NYRBS, 2022 MTF, 2022 NSDUH, and 2022 NYTS. The primary measure of cigarette smoking among youth is whether the youth has ever tried smoking a cigarette, even one or two puffs. A second measure is having smoked cigarettes at all in the past 30 days. (See Appendix A for details.)

Tables 6-9 and 6-10 present estimates of the prevalence of cigarette smoking for youth ages 12 to 17, and for subgroups among these youth. In Table 6-9, comparisons of PATH Study estimates to NHANES and NSDUH are provided by age (12-13, 14-15, and 16-17) and comparisons to MTF are provided by grade (8, 10, and 12). In Table 6-10, estimates from the PATH Study are compared to NYRBS and NYTS estimates among middle and high school students only.³⁴ As mentioned in Section 3.2.4, the data from all five external sources were restricted to ages 12 to 17 to match the age range for PATH Study youth at Wave 7.

The point estimates from the PATH Study were calculated using the Wave 7 cross-sectional weight for the Wave 7 Cohort. The corresponding replicate weights were used to calculate variances and 95 percent confidence intervals. Point estimates and confidence intervals are reported for the other national studies as well.

³⁴ To be precise, the estimates were computed for youth in grades 6 to 12 who were ages 12 to 17.

Table 6-9. Comparison of cigarette-smoking estimates by age and grade from Wave 7 youth interview (Wave 7 Cohort) and other national studies

Cigarette use by age					Cigarette use by grade			
Cigarette-smoking measure and subgroup	Wave 7 (2022) PATH Study sample size	Percentage from Wave 7 (2022) PATH Study [95% confidence interval]	Percentage from 2017-2020 NHANES [95% confidence interval]	Percentage from 2022 NSDUH [95% confidence interval]	Cigarette-smoking measure and subgroup	Wave 7 (2022) PATH Study sample size	Percentage from Wave 7 (2022) PATH Study [95% confidence interval]	Percentage from 2022 MTF [95% confidence interval]
Ever tried cigarette smoking, even one or two puffs					Ever tried cigarette smoking, even one or two puffs			
Age 12-13, overall	3,270	2.7% [2.0%, 3.5%]	1.7% [0.9%, 3.4%]	2.6% [1.9%, 3.5%]	8 th grade	1,546	3.4% [2.6%, 4.5%]	6.5% [5.9%, 7.2%]
Age 12-13, male	1,684	2.7% [1.9%, 3.8%]	1.6% [0.6%, 4.3%]	2.2% [1.6%, 2.9%]	8 th grade, male	790	3.8% [2.6%, 5.5%]	6.0% [5.1%, 7.1%]
Age 12-13, female	1,566	2.7% [1.9%, 3.8%]	1.8% [0.7%, 4.4%]	3.0% [1.8%, 5.0%]	8 th grade, female	749	3.1% [2.0%, 4.8%]	5.8% [4.9%, 6.8%]
Age 14-15, overall	3,468	6.4% [5.6%, 7.4%]	11.0% [6.6%, 17.8%]	6.2% [5.0%, 7.7%]	10 th grade	1,660	5.9% [4.7%, 7.4%]	9.8% [9.1%, 10.7%]
Age 14-15, male	1,824	6.3% [5.1%, 7.6%]	13.6% [7.5%, 23.3%]	6.4% [4.8%, 8.5%]	10 th grade, male	854	5.7% [4.3%, 7.4%]	8.8% [7.6%, 10.2%]
Age 14-15, female	1,634	6.5% [5.2%, 8.1%]	8.2% [4.3%, 15.2%]	6.1% [4.6%, 8.0%]	10 th grade, female	803	6.2% [4.6%, 8.2%]	9.3% [8.4%, 10.3%]
Age 16-17, overall	3,872	10.4% [9.4%, 11.5%]	20.3% [14.7%, 27.3%]	11.4% [9.9%, 13.0%]	12 th grade	1,119	9.8% [8.2%, 11.8%]	14.7% [12.6%, 17.1%]
Age 16-17, male	2,028	10.5% [9.1%, 12.1%]	23.5% [15.0%, 34.8%]	11.4% [9.8%, 13.3%]	12 th grade, male	561	9.5% [7.1%, 12.7%]	16.1% [13.5%, 19.0%]
Age 16-17, female	1,829	10.2% [8.8%, 11.8%]	17.0% [11.0%, 25.2%]	11.3% [9.2%, 13.9%]	12 th grade, female	553	10.0% [7.8%, 12.8%]	12.4% [10.1%, 15.1%]
Has smoked cigarettes in past 30 days, age 12-17	10,635	1.1% [0.9%, 1.3%]	2.3% [1.1%, 4.5%]	1.3% [1.0%, 1.7%]	Has smoked cigarettes in past 30 days, 12 th grade	1,127	1.4% [0.8%, 2.4%]	3.5% [2.5%, 4.9%]

Table 6-10. Comparison of cigarette-smoking estimates by school level from Wave 7 youth interview (Wave 7 Cohort) and other national studies

Cigarette-smoking measure and subgroup ^a	Wave 7 (2022) PATH Study sample size	Percentage from Wave 7 (2022) PATH Study [95% confidence interval]	Percentage from 2021 NYRBS [95% confidence interval]	Percentage from 2022 NYTS [95% confidence interval]
Ever tried cigarette smoking, even one or two puffs				
Middle school, overall	3,518	3.0% [2.4%, 3.7%]	N/A	5.3% [4.5%, 6.3%]
Middle school, male	1,832	3.2% [2.5%, 4.2%]	N/A	5.0% [4.0%, 6.3%]
Middle school, female	1,666	2.8% [2.1%, 3.9%]	N/A	5.6% [4.5%, 7.1%]
Middle school, Hispanic	994	2.7% [1.8%, 4.0%]	N/A	5.5% [4.4%, 6.8%]
Middle school, non-Hispanic White alone	1,627	3.5% [2.7%, 4.6%]	N/A	5.4% [4.2%, 7.1%]
Middle school, non-Hispanic Black alone	421	1.3% [†] [0.6%, 2.9%]	N/A	6.0% [4.1%, 8.7%]
Middle school, non-Hispanic other race or multiple races	348	3.4% [†] [1.9%, 6.0%]	N/A	4.1% [2.6%, 6.4%]
High school, overall	6,096	7.5% [6.8%, 8.2%]	17.0% [15.1%, 19.1%]	9.9% [8.6%, 11.4%]
High school, male	3,160	7.5% [6.6%, 8.5%]	16.7% [14.9%, 18.7%]	10.0% [8.5%, 11.6%]
High school, female	2,917	7.5% [6.5%, 8.5%]	17.2% [14.7%, 20.0%]	9.8% [8.3%, 11.6%]
High school, Hispanic	1,816	6.3% [5.3%, 7.5%]	18.8% [15.7%, 22.3%]	9.0% [7.4%, 10.9%]
High school, non-Hispanic White alone	2,860	9.6% [8.4%, 10.8%]	18.4% [16.5%, 20.4%]	12.2% [10.4%, 14.4%]
High school, non-Hispanic Black alone	602	1.6% [†] [0.9%, 3.0%]	10.8% [8.3%, 14.0%]	4.3% [3.0%, 6.0%]
High school, non-Hispanic other race or multiple races	632	7.1% [5.3%, 9.5%]	13.2% [10.1%, 17.1%]	7.8% [5.7%, 10.7%]
Has smoked cigarettes in past 30 days, high school	6,120	1.2% [0.9%, 1.5%]	3.5% [3.0%, 4.1%]	1.7% [1.4%, 2.1%]

^a For the race/ethnicity subgroups, Hispanic refers to Hispanic or Latino, Black refers to Black or African American, and alone means the respondent reported only one race.

[†] Estimate should be interpreted with caution because it has low precision. It is based on a denominator sample size of less than 50, or the coefficient of variation of the estimate or its complement is larger than 30 percent.

When comparing by age, the estimates of ever cigarette smoking from Wave 7 of the PATH Study were uniformly lower than the estimates from 2017-2020 NHANES for both the age 14-and-15 and age 16-and-17 subgroups, but the 95 percent confidence intervals for these estimates were mostly overlapping—only age 16-and-17 overall and males ages 16 and 17 had non-overlapping confidence intervals (see Table 6-9). All other Wave 7 estimates were comparable with estimates both from NHANES and NSDUH with overlapping confidence intervals. Most of the confidence intervals did

not overlap between Wave 7 of the PATH Study and 2022 MTF for the grade-level estimates of ever cigarette smoking, with the MTF estimates being consistently higher.³⁵ The PATH Study estimate for cigarette smoking in the past 30 days among 12th graders was also lower than the MTF estimate, with no overlap in confidence intervals.

Table 6-10 shows that the PATH Study estimates of cigarette smoking among middle school students were generally lower than those from NYTS, with four out of seven estimates having non-overlapping 95 percent confidence intervals. The PATH Study estimates among high school students were well below those from NYRBS, with all non-overlapping confidence intervals. However, PATH Study estimates among high school students were more comparable to estimates from NYTS. Five out of eight of the NYTS estimates for high school students had 95 percent confidence intervals that overlapped with those from Wave 7 of the PATH Study.

It is unclear whether the PATH Study estimates of cigarette smoking among Wave 7 youth are generally lower due to nonresponse bias in the estimates from one or more of the studies compared, or for other reasons. However, a similar pattern has been observed in previous comparisons between estimates from the PATH Study and other national studies. The differences can be explained by a number of potential reasons, similar to those discussed in Section 6.2.3. In addition to the varying degrees of sampling and measurement errors, the surveys differ in terms of survey mode, context, question order and wording, and timing of data collection. The NYRBS, MTF, and NYTS are all self-administered school-based surveys. In 2021, the NYRBS was a multimode survey that was completed by students in the classroom using either pencil-and-paper instrument administration or electronic administration using tablets. In 2022, the MTF and NYTS were both fully online Web surveys that were administered in the classroom for schools conducting in-person classes and at home for schools conducting classes virtually. The NYTS is administered to students in grades 6 through 12, the MTF is administered to 8th, 10th, and 12th graders, and the NYRBS is administered to students in grades 9 through 12. The NHANES and NSDUH use ACASI for the in-person questions about tobacco use by youth and administer the survey individually in a mobile examination center or household setting, respectively. As discussed in Section 2.9, Wave 7 of the PATH Study was administered primarily in person using ACASI, but interviewer-administered

³⁵ In Table 6-9, the 2022 MTF estimates of ever cigarette smoking for 8th and 10th graders overall are lower than for both subgroup estimates by sex. This is because a relatively high percentage of respondents in these grades provided answers other than “Male” or “Female” or did not provide information about their sex (13.3 percent).

interviews were conducted over the telephone without ACASI as needed due to any public health considerations. Currivan, Nyman, Turner, and Biener (2004) found that even when telephone ACASI was used, estimates of youth smoking prevalence were lower for a telephone survey of youth smoking than for a school-based survey of the same population (see also Fowler & Stringfellow, 2001, for a discussion of higher smoking rates in school-based surveys). In addition, higher estimates of cigarette use in school-based studies as compared to home-based studies do not necessarily mean the school-based estimates are more accurate. As noted in Boyd, Veliz, Evans-Polce, Eisman, and McCabe (2020), school-based studies are often implemented in a group setting and the fact that peers could observe their answers may make respondents feel it more socially desirable to over-report risky behaviors.

As discussed in Section 6.2.3, FDA tobacco regulations went into effect on December 20, 2019, before the beginning of data collection for Wave 7 of the PATH Study, 2021 NYRBS, 2022 MTF, 2022 NSDUH, and 2022 NYTS, and near the close of data collection for 2017-2020 NHANES. In addition, the Wave 7, 2021 NYRBS, 2022 MTF, 2022 NSDUH, and 2022 NYTS responses were gathered after the start of the COVID-19 pandemic, whereas data collection for 2017-2020 NHANES occurred prior to the pandemic.³⁶ The differences between cigarette-smoking estimates from the PATH Study and NHANES could be in part due to a material change in behavior in the population caused by the new FDA regulations.

The contexts and purposes of these surveys also differ: NHANES and NYRBS are general health surveys; whereas MTF and NSDUH are cross-sectional surveys on substance use (including tobacco use) and health, and NYTS is a cross-sectional survey focusing on tobacco use. Unlike the cross-sectional prevalence surveys, the PATH Study uses a longitudinal cohort design to assess within-person changes and between-person differences in tobacco-use behaviors and health over time. Differences among the questions used in the instruments of these different studies are outlined in Appendix A.

³⁶ 2017-2020 NHANES stopped data collection in March of 2020, due to school closures from the COVID-19 pandemic.

6.3.4 Comparison of ENDS-Use Estimates to Other National Studies

This section compares estimates for youth ENDS use based on the Wave 7 Cohort respondents at Wave 7 of the PATH Study to similar estimates based on data from 2021 NYRBS, 2022 MTF, 2022 NSDUH, and 2022 NYTS. The primary measure of ENDS use among youth is whether the youth has ever tried using ENDS. A second measure is having used ENDS at all in the past 30 days. (See Appendix B for details.) The 2021 NYRBS asks about use of “electronic vapor products,” 2022 MTF asks respondents how often they have “vaped nicotine” in their lifetime, 2022 NSDUH asks whether respondents have ever “vaped nicotine with an e-cigarette or other vaping device,” and 2022 NYTS asks about “e-cigarette” use. However, the definitions provided to respondents immediately prior to asking these questions are virtually identical to the definition of “electronic nicotine products” provided to PATH Study respondents.

Tables 6-11 and 6-12 present estimates of the prevalence of ENDS use for youth ages 12 to 17, and for subgroups among these youth. Table 6-11 provides Wave 7 estimates by age (12-13, 14-15, and 16-17) to compare to 2022 NSDUH, and by grade (8, 10, and 12) to compare to 2022 MTF. Table 6-12 shows estimates only among middle and high school students³⁷ to compare Wave 7 of the PATH Study to 2021 NYRBS and 2022 NYTS. The data from all four external sources were restricted to ages 12 to 17 to match the age range for PATH Study youth at Wave 7.

The point estimates from the PATH Study were calculated using the Wave 7 cross-sectional weight for the Wave 7 Cohort. The corresponding replicate weights were used to calculate variances and 95 percent confidence intervals. Point estimates and confidence intervals are reported for the other national studies as well.

³⁷ To be precise, the estimates were computed for youth in grades 6 to 12 who were ages 12 to 17.

Table 6-11. Comparison of ENDS-use estimates by age and grade from Wave 7 youth interview (Wave 7 Cohort) and other national studies

ENDS use by age				ENDS use by grade			
ENDS-use measure and subgroup	Wave 7 (2022) PATH Study sample size	Percentage from Wave 7 (2022) PATH Study [95% confidence interval]	Percentage from 2022 NSDUH [95% confidence interval]	ENDS-use measure and subgroup	Wave 7 (2022) PATH Study sample size	Percentage from Wave 7 (2022) PATH Study [95% confidence interval]	Percentage from 2022 MTF [95% confidence interval]
Ever tried ENDS, even one or two times				Ever tried ENDS, even one or two times			
Age 12-13, overall	3,269	5.6% [4.9%, 6.5%]	7.2% [6.0%, 8.6%]	8 th grade	1,545	8.8% [7.4%, 10.5%]	17.4% [16.2%, 18.6%]
Age 12-13, male	1,685	4.8% [3.8%, 6.0%]	5.2% [3.9%, 6.8%]	8 th grade, male	791	7.1% [5.4%, 9.2%]	14.5% [12.7%, 16.4%]
Age 12-13, female	1,564	6.5% [5.4%, 7.9%]	9.2% [7.4%, 11.5%]	8 th grade, female	747	10.7% [8.5%, 13.5%]	19.8% [18.1%, 21.5%]
Age 14-15, overall	3,461	13.9% [12.8%, 15.1%]	19.1% [17.1%, 21.4%]	10 th grade	1,658	16.2% [14.4%, 18.2%]	27.5% [26.5%, 28.5%]
Age 14-15, male	1,822	12.1% [10.6%, 13.6%]	16.3% [14.1%, 18.8%]	10 th grade, male	851	15.1% [12.8%, 17.7%]	23.0% [21.6%, 24.4%]
Age 14-15, female	1,628	15.8% [14.1%, 17.6%]	22.3% [19.0%, 26.0%]	10 th grade, female	803	17.2% [14.4%, 20.3%]	31.1% [29.6%, 32.5%]
Age 16-17, overall	3,869	24.8% [23.3%, 26.4%]	31.6% [29.4%, 33.9%]	12 th grade	1,120	25.3% [22.3%, 28.5%]	38.1% [34.5%, 41.8%]
Age 16-17, male	2,026	23.7% [21.8%, 25.8%]	29.1% [25.6%, 32.9%]	12 th grade, male	563	24.1% [20.4%, 28.2%]	34.5% [30.2%, 39.0%]
Age 16-17, female	1,828	26.0% [23.9%, 28.2%]	34.1% [31.0%, 37.3%]	12 th grade, female	552	26.4% [22.3%, 31.0%]	41.1% [36.8%, 45.5%]
Has used ENDS in past 30 days, age 12–17	10,612	5.4% [5.0%, 5.9%]	7.0% [6.4%, 7.8%]	Has used ENDS in past 30 days, 12 th grade	1,124	9.3% [7.7%, 11.2%]	19.2% [16.3%, 22.5%]

Table 6-12. Comparison of ENDS-use estimates by school level from Wave 7 youth interview (Wave 7 Cohort) and other national studies

ENDS-use measure and subgroup^a	Wave 7 (2022) PATH Study sample size	Percentage from Wave 7 (2022) PATH Study [95% confidence interval]	Percentage from 2021 NYRBS [95% confidence interval]	Percentage from 2022 NYTS [95% confidence interval]
Ever tried ENDS, even one or two puffs				
Middle school, overall	3,516	6.3% [5.4%, 7.3%]	N/A	9.3% [8.0%, 10.8%]
Middle school, male	1,833	5.4% [4.2%, 6.7%]	N/A	7.8% [6.5%, 9.4%]
Middle school, female	1,633	7.3% [6.2%, 8.7%]	N/A	11.0% [9.4%, 12.8%]
Middle school, Hispanic	995	6.0% [4.5%, 8.0%]	N/A	11.2% [9.7%, 12.8%]
Middle school, non-Hispanic White alone	1,626	6.8% [5.6%, 8.4%]	N/A	8.9% [7.3%, 10.8%]
Middle school, non-Hispanic Black alone	419	4.5% [2.9%, 6.9%]	N/A	9.1% [6.9%, 11.9%]
Middle school, non-Hispanic other race or multiple races	347	6.7% [4.4%, 10.1%]	N/A	8.4% [4.7%, 14.4%]
High school, overall	6,089	18.8% [17.8%, 19.9%]	35.7% [33.1%, 38.3%]	27.2% [25.0%, 29.6%]
High school, male	3,157	17.8% [16.5%, 19.2%]	31.2% [28.7%, 33.7%]	24.7% [22.2%, 27.3%]
High school, female	2,912	19.8% [18.4%, 21.3%]	40.5% [37.2%, 43.9%]	29.7% [27.2%, 32.3%]
High school, Hispanic	1,812	16.8% [15.1%, 18.6%]	40.5% [36.7%, 44.4%]	25.9% [23.3%, 28.7%]
High school, non-Hispanic White alone	2,857	22.4% [20.6%, 24.2%]	35.9% [33.3%, 38.5%]	30.7% [27.9%, 33.6%]
High school, non-Hispanic Black alone	601	11.1% [8.5%, 14.3%]	33.1% [30.0%, 36.4%]	21.1% [17.8%, 24.8%]
High school, non-Hispanic other race or multiple races	636	17.4% [14.4%, 20.8%]	28.7% [23.2%, 35.0%]	22.8% [16.6%, 30.4%]
Has used ENDS in past 30 days, high school	6,110	6.8% [6.1%, 7.4%]	17.5% [15.8%, 19.3%]	13.1% [11.6%, 14.8%]

^a For the race/ethnicity subgroups, Hispanic refers to Hispanic or Latino, Black refers to Black or African American, and alone means the respondent reported only one race.

The estimates of ever use of ENDS from Wave 7 of the PATH Study were uniformly lower than the estimates from 2022 NSDUH, with non-overlapping 95 percent confidence intervals for all age 14-and-15 subgroups, the age 16-and-17 group overall, and for females ages 16 and 17 (see Table 6-11). The PATH Study estimates were comparable with the NSDUH estimates for youth ages 12 and 13, with overlapping confidence intervals. The PATH Study estimate of past 30-day ENDS use among youth ages 12 to 17 was lower than the NSDUH estimate with a non-overlapping confidence interval. The estimates of youth ever use of ENDS from Wave 7 of the PATH Study were lower than those from 2022 MTF for all grade and sex subgroups, as was the estimate of past 30-day ENDS use among 12th graders, with no overlap in confidence intervals.

As shown in Table 6-12, the estimates of ever use of ENDS from Wave 7 of the PATH Study among middle and high school students ages 12 to 17 were uniformly much lower than those from 2021 NYRBS and 2022 NYTS. None of the NYRBS 95 percent confidence intervals overlapped with those from the PATH Study, and only five of the fifteen PATH Study confidence intervals overlapped with those from NYTS, four of those being among middle school students. The 2022 NYTS estimates of ever use of ENDS were also lower than the 2021 NYRBS estimates among high school students, with only one overlap of confidence intervals for non-Hispanic youth of other or multiple races. The PATH Study estimate of ENDS use in the past 30 days among high-school students ages 12 to 17 was lower than the NYTS estimate, which in turn was lower than the NYRBS estimate—none of the corresponding confidence intervals overlapped.

As discussed in Section 6.3.3, some differences in the estimates may be attributable to differences in survey mode, context, question order and wording, measurement error, and timing of data collection. In particular, 2021 NYRBS was administered in the classroom, and 2022 MTF and 2022 NYTS were administered in the classroom for schools conducting in-person classes and at home for schools conducting classes virtually, whereas Wave 7 of the PATH Study was administered over the telephone or at home (with the exception of a small percentage of participants who completed the interview via the Web). It is uncertain which factors lead to the observed differences in ENDS-use estimates for youth ages 12 to 17 between the PATH Study and other national studies.

7. Summary of Findings

This chapter summarizes the PATH Study's Wave 7 nonresponse bias analysis findings separately for the Wave 1 Cohort, Wave 4 Cohort, and Wave 7 Cohort. Each of the following components of the analysis is reviewed:

- Wave 7 interview response rates.
- The representativeness of Wave 7 interview respondents and statistical weighting adjustments to reduce potential nonresponse bias.
- The degree to which Wave 7 estimates of cigarette-smoking and ENDS use align with similar estimates from other national studies.

7.1 Wave 1 Cohort

7.1.1 Adult Interview

The Wave 1 Cohort unweighted all-waves response rate was slightly lower than the weighted all-waves response rate for the Wave 7 adult interview (conditioning on Wave 1 response) overall and for many subgroups. There were also differences in weighted all-waves response rates across subgroups.

- The adult weighted all-waves response rate was 48.6 percent.
- Wave 7 adults who were younger than age 18 and ages 18 to 24 at Wave 1 had lower weighted all-waves response rates (42.2 percent and 40.8 percent, respectively) than those in the older age groups (who had response rates ranging from 47.3 percent to 55.3 percent).
- Adults with a bachelor's degree or advanced degree at Wave 1 had a higher propensity to be all-waves respondents (56.7 percent) than adults with less education (who had response rates between 45.5 percent and 48.5 percent).
- Adults with health insurance at Wave 1 were more likely to be all-waves respondents (51.3 percent) than adults without health insurance (43.2 percent).
- The weighted all-waves response rates were 8.3 percentage points lower for Wave 1 adults with current established tobacco use at Wave 1 than for Wave 1 adults with no current established tobacco use at Wave 1, and 9.3 percentage points lower for Wave 1

youth who had ever used tobacco at Wave 1 than for Wave 1 youth who had never used tobacco at Wave 1.

However, these differences in response rates do not necessarily indicate nonresponse bias in the Wave 7 adult estimates for the Wave 1 Cohort.

As a result of comparing select demographic and socio-economic characteristics between those eligible to be Wave 7 adult all-waves respondents and the Wave 7 adult all-waves respondents, the largest differences prior to Wave 7 weighting adjustments to address nonresponse since Wave 1 were: an underrepresentation of the male population among the all-waves respondents, and overrepresentations of adults ages 45 to 64 at Wave 1 and adults with a bachelor's or advanced degree at Wave 1. When similar comparisons were made with respect to tobacco-use characteristics, the differences consistently indicated an underrepresentation of Wave 1 tobacco use among the Wave 7 adult all-waves respondents.

After Wave 7 weighting adjustments were applied, the differences between those eligible to be Wave 7 adult all-waves respondents and the Wave 7 adult all-waves respondents were greatly reduced. However, among the Wave 7 adult all-waves respondents, there were slight overrepresentations of adults ages 18 to 64 at Wave 1 and adults with a bachelor's degree or advanced degree at Wave 1, and a slight underrepresentation of adults ages 65 and older at Wave 1. Underrepresentations of Wave 1 adults with current established tobacco use at Wave 1 persisted overall and for a number of subgroups, with the largest underrepresentations occurring for males and adults residing in the South at Wave 1. Additionally, the Wave 7 adult all-waves respondents tended to overrepresent Wave 1 current established tobacco use for non-Hispanic Black alone adults at Wave 1 and Wave 1 ever use of tobacco for non-Hispanic Black alone youth at Wave 1. (See Section 4.1.2 for a full description of findings.) Despite the use of both demographic and tobacco-use variables from Wave 1 for calibrating the Wave 7 all-waves weight, which improved the representativeness of the Wave 7 adult all-waves respondents, these results demonstrate the challenges faced by panel studies as waves of nonresponse accrue.

7.1.2 Youth Interview

The Wave 1 Cohort unweighted all-waves response rate was generally similar to the weighted all-waves response rate for the Wave 7 youth interview (conditioning on Wave 1 response) overall and

for subgroups, although there were differences in weighted all-waves response rates across subgroups.

- The youth weighted all-waves response rate was 49.3 percent.
- Moderate differences were found by Wave 1 race/ethnicity and census region, with the largest difference indicating that Hispanic youth (50.0 percent) had a higher propensity to be all-waves respondents than non-Hispanic youth of other race or multiple races (42.7 percent).

These differences in response rates do not necessarily indicate nonresponse bias in the Wave 7 youth estimates for the Wave 1 Cohort.

Comparing select demographic characteristics between those eligible to be Wave 7 youth all-waves respondents and the Wave 7 youth all-waves respondents reflected positively on the calibration of the Wave 7 all-waves weights to demographic variables from Wave 1. However, the assessment of nonresponse bias is moderated by the fact that nearly all Wave 7 youth in the Wave 1 Cohort were age 9 at Wave 1 and limited data are available for this group because they were shadow youth who did not complete an interview at Wave 1. After applying the Wave 7 all-waves weight for the Wave 1 Cohort, the Wave 7 youth all-waves respondents slightly underrepresented Wave 7 youth who resided in the Midwest census region at Wave 1.

7.2 Wave 4 Cohort

7.2.1 Adult Interview

The Wave 4 Cohort unweighted all-waves response rate was lower than the weighted all-waves response rate for the Wave 7 adult interview (conditioning on Wave 4 response) overall and for most subgroups. There were also differences in weighted all-waves response rates across subgroups.

- The adult weighted all-waves response rate was 63.1 percent.
- The subgroups with the largest differences in the weighted all-waves response rates for Wave 1 Cohort adults also had large differences in the weighted all-waves response rates for Wave 4 Cohort adults.
- The weighted all-waves response rates for Wave 4 Cohort adults were lower for those younger than age 18 or ages 18 to 24 at Wave 4 (52.8 percent and 52.3 percent,

respectively) than for those in the older age groups (who had response rates ranging from 62.0 percent to 69.4 percent).

- Adults with a bachelor's degree or advanced degree at Wave 4 had a higher propensity to respond at Waves 5, 6, and 7 (71.9 percent) than adults with less education (who had response rates ranging from 58.6 percent to 63.0 percent).
- Adults with health insurance at Wave 4 were more likely to respond at Wave 5, 6, and 7 (65.3 percent) than adults without health insurance (55.5 percent).
- The weighted all-waves response rates were 9.7 percentage points lower for Wave 4 adults with current established tobacco use at Wave 4 than for Wave 4 adults with no current established tobacco use at Wave 4, and 9.3 percentage points lower for Wave 4 youth who had ever used tobacco at Wave 4 than for Wave 4 youth who had never used tobacco at Wave 4.
- There was also a large difference in the weighted all-waves response rate for those recruited at Wave 4 (50.7 percent) compared to those recruited at Wave 1 (65.3 percent). The Wave 1 sample members of the Wave 4 Cohort have persisted with the PATH Study over a number of years, and all completed an interview at Wave 4.

However, these differences in response rates do not necessarily indicate nonresponse bias in the Wave 7 adult estimates for the Wave 4 Cohort.

As a result of comparing select demographic and socio-economic characteristics between those eligible to be Wave 7 adult all-waves respondents and the Wave 7 adult all-waves respondents, prior to Wave 7 weighting adjustments, the most noticeable underrepresentations among the Wave 4 Cohort were for males, adults ages 24 and younger at Wave 4, Wave 4 adults with at most a high school diploma, and adults without health insurance at Wave 4. There were corresponding overrepresentations of adults ages 45 to 64 at Wave 4 and Wave 4 adults with a bachelor's degree or advanced degree. When similar comparisons were made with respect to tobacco-use characteristics, the differences consistently indicated underrepresentations of Wave 4 adult current established tobacco use and Wave 4 youth ever use of tobacco among the Wave 7 adult all-waves respondents. (See Section 5.1.2 for a full description of findings.)

After Wave 7 weighting adjustments were applied, the differences between those eligible to be Wave 7 adult all-waves respondents and the Wave 7 adult all-waves respondents from the Wave 4 Cohort were mostly negligible with some residual underrepresentation of those younger than age 18 at Wave 4 and adults without health insurance at Wave 4. Among the Wave 7 adult all-waves respondents some underrepresentation of Wave 4 youth ever use of tobacco persisted overall, for

males, non-Hispanic White alone persons, and for those who resided in the South census region at Wave 4.

7.2.2 Youth Interview

The Wave 4 Cohort unweighted all-waves response rate was slightly higher than the weighted all-waves response rate for the Wave 7 youth interview (conditioning on Wave 4 response) overall and for most subgroups, although there were differences in weighted response rates across subgroups.

- The youth weighted all-waves response rate was 51.8 percent.
- The weighted all-waves response rates varied by 15.1 percentage points for Wave 4 age and 16.6 percentage points for recruitment wave. Response rates were lower for participants younger than age 12 at Wave 4 (47.6 percent) and for those recruited at Wave 4 (48.6 percent)—these are highly correlated characteristics because all Wave 4 shadow youth in the Wave 4 Cohort are from the Wave 4 replenishment sample. Shadow youth were not required to complete a PATH Study interview at the time of their enrollment to become a study participant.
- Youth residing in the West census region at Wave 4 had a higher propensity to respond at Waves 5, 6, and 7 (57.9 percent) than youth in other census regions (who had response rates ranging from 48.8 percent to 50.3 percent).
- The weighted all-waves response rate was higher for Wave 4 youth who had never used tobacco at Wave 4 (63.0 percent) than for those who had ever used tobacco at Wave 4 (52.3 percent).

These differences in response rates do not necessarily indicate nonresponse bias in the Wave 7 youth estimates for the Wave 4 Cohort.

As a result of comparing select demographic and tobacco-use characteristics between those eligible to be Wave 7 youth all-waves respondents and the Wave 7 youth all-waves respondents, the noticeable underrepresentation prior to Wave 7 weighting adjustments to address nonresponse among the Wave 4 Cohort was for youth younger than age 12 at Wave 4, i.e., Wave 4 shadow youth. There was also a slight overrepresentation of people residing in the West among the Wave 7 youth all-waves respondents.

After Wave 7 weighting adjustments, the underrepresentation of Wave 4 shadow youth was reduced but remained to a small degree. There was also a slight underrepresentation of people residing in the

Midwest at Wave 4. Wave 4 youth who had ever used tobacco at that time were appropriately represented among the Wave 7 youth all-waves respondents, both before and after applying the Wave 7 weighting adjustments.

7.3 Wave 7 Cohort

7.3.1 Household Screener

Among the households screened for the entire Wave 7 replenishment sample, as well as those screened for the Wave 7 replenishment AYS subsample, the unweighted response rates were similar to the weighted response rates overall and by subgroups. However, there were differences in weighted response rates across subgroups, for both the entire Wave 7 replenishment sample and the Wave 7 replenishment AYS subsample.

For the entire Wave 7 replenishment sample:

- The weighted household screener response rate was 55.6 percent.
- The weighted subgroup response rates varied by 7.7 percentage points by whether the percent of the neighborhood that identifies as White alone was above the national average, 7.2 percentage points by census region, 5.4 percentage points by whether the percent of the neighborhood that identifies as Hispanic was above the national average, and 4.8 percent by whether the percent of the neighborhood that identifies as Black alone was above the national average. There was less variation in response rates by whether the percent of the neighborhood with a bachelor's degree was above the national average or whether the percent of the neighborhood below the poverty line was above the national average.

For the Wave 7 replenishment AYS subsample:

- The weighted household screener response rate was 30.3 percent, compared to 54.0 percent for households sampled at Wave 1 and 52.8 percent for the Wave 4 replenishment AYS subsample (for which all household members ages 10 and older and in the Wave 4 CNP were eligible). The lower response rate can be partly attributed to higher-than-average field interviewer attrition rates (amid the ongoing COVID-19 pandemic and a national labor shortage) which led the PATH Study to prioritize completing household screeners among the youth-targeted address subsamples at Wave 7. In addition, households targeted only for youth screening were allowed to be finalized as complete through verification that no youths resided in the household; whereas the AYS subsample could not use this verification process because of the need to also screen for adults.

- The weighted subgroup response rates differed by 7.9 percentage points by census region, 6.4 percentage points by whether the percent of the neighborhood with a bachelor's degree was above the national average, 5.8 percentage points by whether the percent of the neighborhood below the poverty level was above the national average, and 4.2 percentage points by whether the percent of the neighborhood that identifies as White alone was above the national average. There was less variation in response rates by whether the percent of the neighborhood that identifies as Black alone was above the national average or whether the percent of the neighborhood that identifies as Hispanic was above the national average.

7.3.2 Adult Interview

This response rate summary applies only to the Wave 7 Cohort adults selected as part of the Wave 7 replenishment AYS subsample. For these adults, the unweighted response rate was similar to the weighted response rate for the Wave 7 adult interview (conditioning on completion of the household screener) overall and by subgroups, although there were differences in weighted response rates across subgroups.

- The adult weighted response rate was 55.1 percent.
- The weighted response rate was noticeably lower among adults residing in the Northeast census region (48.8 percent) compared to adults residing in the Midwest (58.5 percent) or West (59.7 percent) census regions.
- The weighted response rate by race/ethnicity was highest for non-Hispanic Black alone adults (58.6 percent) and lowest for non-Hispanic adults of other or multiple races (52.2 percent).
- Younger adults (ages 18 to 24) had a lower weighted response rate (51.3 percent) than older adults (who had response rates ranging from 54.8 percent to 56.7 percent), and males had a lower weighted response rate (52.6 percent) than females (57.6 percent).

Weighted demographic estimates based on all Wave 7 Cohort adults were a close match to corresponding estimates based on the 2022 ACS, consistent with the calibration of the Wave 7 cross sectional weights to 2021 ACS totals. However, the PATH Study estimate of the percentage of adults with health insurance was the furthest from the 2022 ACS estimate for the subgroups examined.

There is no evidence of nonresponse bias in the Wave 7 Cohort with respect to current cigarette-smoking among adults in the sense that the PATH Study's estimates were all within the range of estimates from other national studies. In general, the Wave 7 Cohort adult estimates tended to be

lower than estimates from 2017-2020 NHANES and 2022 NSDUH and higher than estimates from 2022 NHIS and 2022 TUS-CPS.

The Wave 7 Cohort estimates of adult current ENDS use at Wave 7 were higher than the estimates from 2022 NHIS and 2022 TUS-CPS; however, the 95 percent confidence intervals for the PATH Study estimates overlapped with those from NHIS for about half of the thirteen estimates.

When interpreting the results of these cross-study comparisons, caution should be taken due to varying degrees of sampling and measurement errors as well as differences in survey mode, question order, and context. Another possible reason for differences in the estimates between studies is the timing of data collection, especially with Wave 7 of the PATH Study, 2022 NHIS, 2022 NSDUH, and 2022 TUS-CPS collecting responses after the start of the COVID-19 pandemic, whereas 2017-2020 NHANES was completed prior to the pandemic.

7.3.3 Youth Interview

This response rate summary applies to the Wave 7 Cohort youth selected as part of the Wave 7 replenishment sample. For these youth, the unweighted response rate was similar to the weighted response rate for the Wave 7 youth interview (conditioning on completion of the household screener) overall and by subgroups, although there were differences in weighted response rates across subgroups.

- The youth weighted response rate was 64.0 percent.
- The weighted response rates varied by 4.8 percentage points for age and 10.3 percentage points for census region. Response rates were higher among youth ages 12 and 13 (67.1 percent) compared to youth ages 14 to 17 (62.3 percent). Youth residing in the West census region had the highest response rate (68.6 percent) and those residing in the Northeast had the lowest (58.3 percent).

Weighted demographic estimates based on all Wave 7 Cohort youth were close to corresponding estimates based on the 2022 ACS, consistent with the calibration of the Wave 7 cross sectional weights to 2021 ACS totals.

When comparing by age and grade, the 95 percent confidence intervals for the PATH Study's youth cigarette-smoking estimates based on the Wave 7 Cohort overlapped with most of those from 2017-2020 NHANES and all from 2022 NSDUH. Only two of the confidence intervals overlapped

between Wave 7 of the PATH Study and 2022 MTF; with the MTF estimates being consistently higher. For middle school students, PATH Study estimates of cigarette smoking were generally lower than those from 2022 NYTS, with a majority of the estimates having non-overlapping confidence intervals. For high school students, the Wave 7 estimates of cigarette smoking were lower than the estimates from 2021 NYRBS, with none of the confidence intervals overlapping. However, the estimates of cigarette smoking among high school students for the PATH Study and 2022 NYTS were much more closely aligned, with 95 percent confidence intervals overlapping for five out of the eight estimates.

The PATH Study's Wave 7 Cohort estimates of ever ENDS use were consistently lower than the estimates from 2022 NSDUH for youth ages 14 and 15, as well as ages 16 and 17 with mostly non-overlapping confidence intervals, whereas the PATH Study estimates were generally comparable to the NSDUH estimates for youth ages 12 and 13. However, the Wave 7 ever ENDS-use estimates by grade and school level were noticeably lower than the estimates from 2021 NYRBS, 2022 MTF, and 2022 NYTS, and the differences were bigger than those observed for youth cigarette smoking.

It is uncertain whether the differences in estimates of youth cigarette smoking and ENDS use are due to nonresponse bias in one or more of the studies compared or a variety of other differences between the studies, including the survey mode and the timing of data collection.

7.4 General Conclusions

If the demographic, socio-economic, and tobacco-use characteristics examined in this report are correlated with key tobacco- and health-related outcome measures, these results raise no serious concerns about potential nonresponse bias in youth estimates from Wave 7 of the PATH Study, for the Wave 1 Cohort, Wave 4 Cohort, or Wave 7 Cohort. However, almost all the Wave 7 youth in the Wave 1 Cohort were age 9 at Wave 1 and limited data are available for assessing nonresponse bias for this group.

Due to its age, the Wave 1 Cohort has an increased potential for nonresponse bias, and the analyses presented in this report suggest that the attrition of participants who use tobacco has had a small effect on estimates for adults based on all-waves respondents. There is limited evidence of potential nonresponse bias in Wave 7 adult estimates based on all-waves respondents for the Wave 4 Cohort,

although researchers should consider the results presented in Section 5.1.2 when analyzing Wave 7 adults who were youth at Wave 4. The cross-sectional weights developed for the newly established Wave 7 Cohort produce estimates that closely align with those from external data sources. These results confirm the timeliness of the introduction of replenishment samples at Wave 4 and Wave 7 of the PATH Study.

There is little evidence of nonresponse bias with respect to Wave 7 estimates of current cigarette smoking and ENDS use among adults in the sense that the Wave 7 Cohort estimates were generally within the range of estimates from other national studies, after accounting for sampling error. The Wave 7 estimates of youth cigarette smoking were similar to the 2022 NSDUH estimates, but generally lower than the estimates from other national studies. The Wave 7 Cohort produces adult estimates of current ENDS use that are higher than those from 2022 NHIS and 2022 TUS-CPS, but the 95 percent confidence intervals for the PATH Study and NHIS overlapped for about half of the estimates. The PATH Study's estimates of youth ever ENDS use are lower at Wave 7 than other external studies, consistent with findings from previous nonresponse bias analyses. It is difficult to determine the primary reason(s) for these results due to the many differences between the PATH Study and the surveys compared.

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Appendix A

Cigarette-Smoking Questions in the PATH Study and Other Surveys

Table A-1 lists the questions used to ask about the current cigarette-smoking status of adults in the PATH Study, as well as in the surveys used for comparison, and describes the populations included in the estimates from those surveys. Table A-2 provides similar information for youth ever and past 30-day cigarette smoking.

Note that although the questions used to define current cigarette smoking are similar among the surveys, small differences could have an effect on the answers given. For example, in the PATH Study, the question used to establish whether a respondent has smoked at least 100 cigarettes in their lifetime has closed response categories:

“How many cigarettes have you smoked in your entire life? A pack usually has 20 cigarettes in it.”

For adults:

1. 1 or more puffs but never a whole cigarette;
2. 1 to 10 cigarettes (about ½ pack total);
3. 11 to 20 cigarettes (about ½ pack to 1 pack);
4. 21 to 50 cigarettes (more than 1 pack but less than 3 packs);
5. 51 to 99 (more than 2½ packs but less than 5 packs); and
6. 100 or more cigarettes (5 packs or more).

For youth:

1. 1 or more puffs but never a whole cigarette;
2. 1 cigarette;
3. 2 to 10 cigarettes (about ½ pack total);
4. 11 to 20 cigarettes (about ½ pack to 1 pack);
5. 21 to 50 cigarettes (more than 1 pack but less than 3 packs);
6. 51 to 99 (more than 2½ packs but less than 5 packs); and
7. 100 or more cigarettes (5 packs or more).

In NHANES, NHIS, NSDUH, and TUS-CPS, however, the question “Have you smoked at least 100 cigarettes in your entire life?” calls for a yes/no response.

The positioning of the questions and context also differ among the surveys. For example, the cigarette-smoking questions are near the beginning of the PATH Study’s adult and youth questionnaires, and respondents know that the questionnaires are about tobacco-use behaviors. In contrast to the PATH Study, NHANES and NHIS have cigarette-smoking questions in the middle of the survey, and respondents know that these surveys cover a range of health topics. Another example is the TUS-CPS, where the cigarette-smoking questions are near the beginning of the adult questionnaire on tobacco, but the survey is administered as part of the CPS. These differences among the surveys may be associated with differences in responses to the cigarette-smoking questions.

Table A-1. Questions and algorithms used to define adult current cigarette smoking in the PATH Study, NHANES, NHIS, NSDUH, and TUS-CPS

Wave 7 PATH Study	2017-2020 NHANES	2022 NHIS	2022 NSDUH	2022 TUS-CPS*
Questions to define current cigarette use (answers defining current cigarette use given in parentheses)				
“How many cigarettes have you smoked in your entire life? A pack usually has 20 cigarettes in it.” (Wave 1 – Wave 7, 100 or more cigarettes (5 packs or more)) “Have you ever smoked a cigarette, even one or two puffs?” (Wave 1, Wave 4, Wave 5, Wave 5.5, Wave 6, Wave 7 adult, yes) “Have you ever tried cigarette smoking, even one or two puffs?” (Wave 1 – Wave 6 youth, yes) “In the past 30 days, have you smoked a cigarette, even one or two puffs?” (Wave 2 – Wave 7 adult, yes) “In the past 12 months, have you smoked a cigarette, even one or two puffs?” (Wave 2 – Wave 7, yes) “Do you now smoke cigarettes every day, some days, or not at all?” (Wave 7, every day, some days)	“{Have you/Has SP} smoked at least 100 cigarettes in {your/his/her} entire life?” (yes) “{Do you/Does SP} now smoke cigarettes every day, some days, or not at all?” (every day, some days)	“Have you smoked at least 100 cigarettes in your ENTIRE LIFE?” (yes) “Do you NOW smoke cigarettes every day, some days or not at all?” (every day, some days)	“Have you smoked at least 100 cigarettes in your entire life?” (yes) “Have you ever smoked part or all of a cigarette?” (yes) “During the past 30 days, have you smoked part or all of a cigarette?” (yes)	“(Have/has) (you/name) smoked at least 100 cigarettes in (your/his/her) entire life?” (yes) “(Do/does) (you/name) now smoke cigarettes every day, some days, or not at all?” (every day, some days)

Table A-1. Questions and algorithms used to define adult current cigarette smoking in the PATH Study, NHANES, NHIS, NSDUH, and TUS-CPS (continued)

Wave 7 PATH Study	2017-2020 NHANES	2022 NHIS	2022 NSDUH	2022 TUS-CPS*
Coding of “Yes” and “No” answers				
If R07R_A_CUR_ESTD_CIGS = 1 then “Yes” Else if R07R_A_CUR_ESTD_CIGS = 2 then “No”	If SMQ020 = 1 and SMQ040 in (1 2) then “Yes” Else if SMQ020 = 2 or SMQ040 = 3 then “No”	If SMKEV_A = 1 and SMKNOW_A in (1 2) then “Yes” Else if SMKEV_A = 2 or SMKNOW_A = 3 then “No”	If CIGEVER = 1 and CIGREC in (1 11) and CIG100LF in (1 3 5) then “Yes” Else if CIGEVER = 2 or CIGREC in (2 3 4 14 19 29) or CIG100LF = 2 then “No”	If PEA1 = 1 and PEA3 in (1 2) then “Yes” Else if PEA1 = 2 or PEA3 = 3 then “No”
Additional question to define every-day cigarette smoking (answer defining every-day cigarette smoking given in parentheses)				
“Do you now smoke cigarettes every day, some days, or not at all?” (Wave 7, every day)	“(Do you/Does SP) now smoke cigarettes every day, some days, or not at all?” (every day)	“Do you NOW smoke cigarettes every day, some days or not at all?” (every day)	N/A	“(Do/does) (you/name) now smoke cigarettes every day, some days, or not at all?” (every day)
Coding of “Yes” and “No” answers				
If R07R_A_EDY_CIGS = 1 then “Yes” Else if R07R_A_EDY_CIGS = 2 then “No”	If SMQ020 = 1 and SMQ040 = 1 then “Yes” Else if SMQ020 = 2 or SMQ040 in (2 3) then “No”	If SMKEV_A = 1 and SMKNOW_A = 1 then “Yes” Else if SMKEV_A = 2 or SMKNOW_A in (2 3) then “No”	N/A	If PEA1 = 1 and PEA3 = 1 then “Yes” Else if PEA1 = 2 or PEA3 in (2 3) then “No”
Additional question to define some-days cigarette smoking (answer defining some-days cigarette smoking given in parentheses)				
“Do you now smoke cigarettes every day, some days, or not at all?” (Wave 7, some days)	“(Do you/Does SP) now smoke cigarettes every day, some days, or not at all?” (some days)	“Do you NOW smoke cigarettes every day, some days or not at all?” (some days)	N/A	“(Do/does) (you/name) now smoke cigarettes every day, some days, or not at all?” (some days)
Coding of “Yes” and “No” answers				
If R07R_A_SDY_CIGS = 1 then “Yes” Else if R07R_A_SDY_CIGS = 2 then “No”	If SMQ020 = 1 and SMQ040 = 2 then “Yes” Else if SMQ020 = 2 or SMQ040 in (1 3) then “No”	If SMKEV_A = 1 and SMKNOW_A = 2 then “Yes” Else if SMKEV_A = 2 or SMKNOW_A in (1 3) then “No”	N/A	If PEA1 = 1 and PEA3 = 2 then “Yes” Else if PEA1 = 2 or PEA3 in (1 3) then “No”
Subsetting used for estimates				
Ages 18+	Ages 18+	Ages 18+	Ages 18+	Ages 18+ Excludes proxy respondents

Table A-1. Questions and algorithms used to define adult current cigarette smoking in the PATH Study, NHANES, NHIS, NSDUH, and TUS-CPS (continued)

Wave 7 PATH Study	2017-2020 NHANES	2022 NHIS	2022 NSDUH	2022 TUS-CPS*
Exclusions from population				
The Wave 7 adult target population for the Wave 7 Cohort is the resident (not incarcerated) U.S. population ages 18 and older at Wave 7 who were in the U.S. civilian, noninstitutionalized population at the time of Wave 7.	Includes only the U.S. civilian, noninstitutionalized population.	Includes only the civilian, noninstitutionalized population residing in the U.S. at the time of the interview. Several segments of the population are excluded, such as: persons in long-term care institutions; persons on active duty with the Armed Forces; persons in correctional facilities; and U.S. nationals living in foreign countries.	Includes only the U.S. civilian, noninstitutionalized population. Excludes homeless persons who do not use shelters, military personnel on active duty, and residents of institutional group quarters, such as jails and hospitals.	Includes only the U.S. civilian, noninstitutionalized population age 18 or older at the time of the interview.
Proxy responses allowed				
No	Yes, when there is a serious physical or mental condition, a proxy respondent may be used to conduct the interview.	Yes, for individuals physically or mentally incapable of responding.	No	Yes
Other comments				
	Suspended field operations in March 2020, due to the COVID-19 pandemic. Data collected from 2019 to March 2020 were combined with data from the NHANES 2017-2018 cycle to form a nationally representative sample of NHANES 2017-March 2020 pre-pandemic data.			

* A proxy response is allowed only after four failed callback attempts to the self-respondent or if the self-respondent will not return before closeout. See https://cancercontrol.cancer.gov/sites/default/files/2024-02/22-23_TUS-CPSTechDoc_CPS22-508-02132024.pdf, p1-3.

Table A-2. Questions and algorithms used to define youth cigarette smoking in the PATH Study, NHANES, NYRBS, MTF, NSDUH, and NYTS

Wave 7 PATH Study	2017-2020 NHANES	2021 NYRBS	2022 MTF	2022 NSDUH	2022 NYTS
Questions to define ever tried cigarette smoking (answers defining ever tried cigarette smoking given in parentheses)					
“Have you ever tried cigarette smoking, even one or two puffs?” (Wave 1 – Wave 7 youth, yes) “In the past 12 months, have you smoked a cigarette, even one or two puffs?” (Wave 2 – Wave 7 youth, yes)	“About how many cigarettes have you smoked in your entire life?” (1 or more puffs but never a whole cigarette, 1 cigarette, 2 to 5 cigarettes, 6 to 15 cigarettes, 16 to 25 cigarettes, 26 to 99 cigarettes, 100 or more cigarettes)	“Have you ever tried cigarette smoking, even one or two puffs?” (yes)	“Have you ever smoked cigarettes?” (yes)	“Have you ever smoked part or all of a cigarette?” (yes)	“Have you ever smoked a cigarette, even one or two puffs?” (yes)
Coding of “Yes” and “No” answers					
If R07R_Y_EVR_CIGS = 1 then “Yes” Else if R07R_Y_EVR_CIGS = 2 then “No”	If SMQ621 in (2 3 4 5 6 7 8) then “Yes” Else if SMQ621 = 1 then “No”	If QN30 = 1 then “Yes” Else if QN30 = 2 then “No”	For 8 th and 10 th grades: If V7101D = 1 then “Yes” Else if V7101D = 0 then “No” For 12 th grade: If V2101D = 1 then “Yes” Else if V2101D = 0 then “No”	If CIGEVER = 1 then “Yes” Else if CIGEVER = 2 then “No”	If QN35 = 1 then “Yes” Else if QN35 = 2 then “No”

Table A-2. Questions and algorithms used to define youth cigarette smoking in the PATH Study, NHANES, NYRBS, MTF, NSDUH, and NYTS (continued)

Wave 7 PATH Study	2017-2020 NHANES	2021 NYRBS	2022 MTF	2022 NSDUH	2022 NYTS
Additional questions to define past 30-day cigarette use (answers defining past 30-day cigarette use in parentheses)*					
“When was the last time you smoked a cigarette, even one or two puffs?” (Wave 7; earlier today, not today but sometime in the past 7 days, not in the past 7 days but sometime in the past 30 days)	“On how many of the past 30 days did {you/SP} smoke a cigarette?” (1-30)	“During the past 30 days, on how many days did you smoke cigarettes?” (1 or 2 days, 3 to 5 days, 6 to 9 days, 10 to 19 days, 20 to 29 days, all 30 days)	“How frequently have you smoked cigarettes during the past 30 days?” (less than 1 cigarette per day, 1 to 5 cigarettes per day, about one-half pack per day, about one pack per day, about one and one-half packs per day, two packs or more per day)	“During the past 30 days, have you smoked part or all of a cigarette?” (yes)	“During the past 30 days, on how many days did you smoke cigarettes?” (1-30)
Coding of “Yes” and “No” answers					
If R07R_Y_CUR_CIGS = 1 then “Yes” Else if R07R_Y_CUR_CIGS = 2 then “No”	If 1 SMD641 in (1 2 ... 30) then “Yes” Else if SMQ621 = 1 or SMD641 = 0 then “No”	If QN32 = 1 then “Yes” Else if QN32 = 2 then “No”	For 8 th and 10 th grades: If V7102D = 1 then “Yes” Else if V7102D = 0 then “No” For 12 th grade: If V2102D = 1 then “Yes” Else if V2102D = 0 then “No”	If CIGEVER = 1 and CIGREC in (1 11) then “Yes” Else if CIGEVER = 2 or CIGREC in (2 3 4 14 19 29) then “No”	If QN38 in (1 2 ... 30) then “Yes” Else if QN35 = 2 or QN38 = 0 then “No”
Subsetting used for estimates					
Ages 12-17 Excludes respondents who report being “ungraded” or “home-schooled” from estimates of cigarette smoking and ENDS use by grade or for high school students.	Ages 12-17	Ages 12-17 Excludes respondents who report being “ungraded” from estimates of cigarette smoking and ENDS use for high school students.	Ages 12-17	Ages 12-17	Ages 12-17 Excludes respondents who report being “ungraded” from estimates of cigarette smoking and ENDS use for high school students.

Table A-2. Questions and algorithms used to define youth cigarette smoking in the PATH Study, NHANES, NYRBS, MTF, NSDUH, and NYTS (continued)

Wave 7 PATH Study	2017-2020 NHANES	2021 NYRBS	2022 MTF	2022 NSDUH	2022 NYTS
Exclusions from population					
The Wave 7 youth target population for the Wave 7 Cohort is the resident (not incarcerated) U.S. population ages 12 to 17 at Wave 7 who were in the U.S. civilian, noninstitutionalized population at the time of Wave 7.	Includes only the U.S. civilian, noninstitutionalized population.	Includes only students in grades 9 to 12 who are attending public and private schools in the 50 States and the District of Columbia. U.S. territories and the Virgin Islands are excluded from the sampling frame.	Includes only youth in grades 8, 10, and 12 who are attending public and private schools within the 48 contiguous states.	Includes only the U.S. civilian, noninstitutionalized population. Excludes homeless persons who do not use shelters, military personnel on active duty, and residents of institutional group quarters, such as jails and hospitals.	Includes only youth who are public and private school students enrolled in regular middle schools and high schools in grades 6 through 12 in the 50 U.S. States and the District of Columbia. Alternative schools, special education schools, Department of Defense operated schools, Bureau of Indian Affairs schools, vocational schools that serve only pull-out populations, and students enrolled in regular schools unable to complete the questionnaire without special assistance are excluded.

Table A-2. Questions and algorithms used to define youth cigarette smoking in the PATH Study, NHANES, NYRBS, MTF, NSDUH, and NYTS (continued)

Wave 7 PATH Study	2017-2020 NHANES	2021 NYRBS	2022 MTF	2022 NSDUH	2022 NYTS
Other comments					
		Self-administered survey in classroom.	Separate online surveys for 8 th and 10 th grades vs. 12 th grade, self-administered in the classroom or at home if schooling remotely.		

* To reduce respondent burden, the PATH Study, NHANES, NSDUH, NYTS, and MTF use computer-assisted interviews (CAIs) whereby skip patterns based on answers to questions about having ever tried cigarettes are used to determine whether to present the past 30-day cigarette-smoking question. For the PATH Study, NHANES, and NSDUH, youth who indicated that they had never tried cigarettes or who did not answer the question were not asked about cigarette smoking in the past 30 days. Thus, when creating estimates for these three studies, youth who answered that they had never tried cigarettes were logically coded a “No” value for past 30-day cigarette smoking. For NYTS and MTF, the skip pattern rules for these two questions were slightly different: youth who had never tried cigarettes were not asked the past 30-day question and were logically coded a “No” value when computing estimates of past 30-cigarette smoking; but youth who did not answer the question about having ever tried cigarettes were asked about cigarette smoking in the past 30 days. NYTS and MTF responses to the past 30-day question were used when creating estimates of past 30-day cigarette smoking, even if the youth had not responded about having ever tried cigarettes; however, their missing value was retained when computing estimates of ever cigarette smoking, regardless of the response to the past 30-day question.

For NYRBS youth were able to answer the question about past 30-day cigarette smoking regardless of their answer about having ever tried cigarettes. When computing cigarette-smoking estimates for NYRBS, no attempt was made to resolve inconsistencies between the responses to these two questions; the responses to the ever question were used to compute estimates of ever cigarette smoking, and the responses to the past 30-day question were used to compute estimates of past 30-day cigarette smoking.

Appendix B

ENDS-Use Questions in the PATH Study and Other Surveys

Table B-1 lists the questions used to ask about current ENDS use of adults in the PATH Study, NHIS, and TUS-CPS. The observations in Appendix A regarding the differences in positioning of the cigarette-smoking questions among the surveys also apply to the questions about use of ENDS.

Table B-2 lists the questions used to ask about ever and past 30-day ENDS use of youth in the PATH Study as well as in the surveys used for comparison. Administration of the questions differs between the PATH Study, NYRBS, MTF, NSDUH, and NYTS (see Sections 6.2.3 and 6.3.3 for details).

The information provided in Tables A-1 (for adults) and A-2 (for youth) about age restrictions, population exclusions, proxy responses, and “other comments” applies to both the cigarette-smoking and ENDS-use estimates and so is not repeated in Tables B-1 and B-2.

Table B-1. Questions and algorithms used to define adult current ENDS use in the PATH Study, NHIS, and TUS-CPS

Wave 7 PATH Study	2022 NHIS	2022 TUS-CPS
Questions to define current ENDS use (answers defining current ENDS use given in parentheses)		
“Have you ever used an e-cigarette, such as NJOY, Blu, or Smoking Everywhere, even one or two times?” (Wave 1, yes) “Have you ever used an electronic nicotine product, even one or two times?” (Wave 2, Wave 3 youth, Wave 4, Wave 5, Wave 5.5, Wave 6, Wave 7, yes) “In the past 30 days, have you used an electronic nicotine product, even one or two times?” (Wave 3 – Wave 7 adult, yes) “In the past 12 months, have you used an electronic nicotine product, even one or two times?” (Wave 3 – Wave 7, yes) “Do you now use electronic nicotine products every day, some days, or not at all?” (Wave 7, every day, some days)	“Have you ever used an e-cigarette or other electronic vaping product, even just one time, in your entire life?” (yes) “Do you NOW use e-cigarettes or other electronic vaping products every day, some days, or not at all?” (every day, some days)	“(Have/Has) (you/name) EVER used E-cigarettes EVEN ONE TIME?” (yes) “(Do you/Does [name]) NOW use an e-cigarette every day, some days or not at all?” (every day, some days)
Coding of “Yes” and “No” answers		
If R07R_A_CUR_EDSD_EPRODS = 1 then “Yes” Else if R07R_A_CUR_EDSD_EPRODS = 2 then “No”	If ECIGEV_A = 1 and ECIGNOW_A in (1 2) then “Yes” Else if ECIGEV_A = 2 or ECIGNOW_A = 3 then “No”	If PEJ1A3_5 = 1 and PEJ2A3_5 in (1 2) then “Yes” Else if PEJ1A3_5 = 2 or PEJ2A3_5 = 3 then “No”
Additional question to define every-day ENDS use (answer defining every-day ENDS use given in parentheses)		
“Do you now use electronic nicotine products every day, some days, or not at all?” (Wave 7, every day)	“Do you NOW use e-cigarettes or other electronic vaping products every day, some days, or not at all?” (every day)	“(Do you/Does [name]) NOW use an e-cigarette every day, some days or not at all?” (every day)

Table B-1. Questions and algorithms used to define adult current ENDS use in the PATH Study, NHIS, and TUS-CPS (continued)

Wave 7 PATH Study	2022 NHIS	2022 TUS-CPS
Coding of “Yes” and “No” answers		
If R07R_A_CUR_EDSD_EPRODS = 1 and R07_AV1003 = 1 then “Yes” Else if R07R_A_CUR_EDSD_EPRODS in (1 2) then “No”	If ECIGEV_A = 1 and ECIGNOW_A = 1 then “Yes” Else if ECIGEV_A = 2 or ECIGNOW_A in (2 3) then “No”	If PEJ1A3_5 = 1 and PEJ2A3_5 = 1 then “Yes” Else if PEJ1A3_5 = 2 or PEJ2A3_5 in (2 3) then “No”
Additional question to define some-days ENDS use (answer defining some-days ENDS use given in parentheses)		
“Do you now use electronic nicotine products every day, some days, or not at all?” (Wave 7, some days)	“Do you NOW use e-cigarettes or other electronic vaping products every day, some days, or not at all?” (some days)	“(Do you/Does [name]) NOW use an e-cigarette every day, some days, or not at all?” (some days)
Coding of “Yes” and “No” answers		
If R07R_A_CUR_EDSD_EPRODS = 1 and R07_AV1003 = 2 then “Yes” Else if R07R_A_CUR_EDSD_EPRODS in (1 2) then “No”	If ECIGEV_A = 1 and ECIGNOW_A = 2 then “Yes” Else if ECIGEV_A = 2 or ECIGNOW_A in (1 3) then “No”	If PEJ1A3_5 = 1 and PEJ2A3_5 = 2 then “Yes” Else if PEJ1A3_5 = 2 or PEJ2A3_5 in (1 3) then “No”

Table B-2. Questions and algorithms used to define youth ENDS use in the PATH Study, NYRBS, MTF, NSDUH, and NYTS

Wave 7 PATH Study	2021 NYRBS	2022 MTF	2022 NSDUH	2022 NYTS
Questions to define ever ENDS use (answers defining ever ENDS use given in parentheses)				
“Have you ever used an e-cigarette, such as NJOY, Blu, or Smoking Everywhere, even one or two times?” (Wave 1 youth, yes) “Have you ever used an electronic nicotine product, even one or two times?” (Wave 2 – Wave 7 youth, yes) “In the past 12 months, have you used an electronic nicotine product, even one or two times?” (Wave 3 – Wave 7 youth, yes)	“Have you ever used an electronic vapor product?” (yes)	“In your LIFETIME how often have you vaped nicotine?” (once or twice, occasionally but not regularly, regularly in the past, regularly now)	“Have you ever, even once, vaped nicotine with an e-cigarette or other vaping device?” (yes)	“Have you ever used an e-cigarette, even once or twice?” (yes)
Coding of “Yes” and “No” answers				
If R07R_Y_EVR_EPRODS = 1 then “Yes” Else if R07R_Y_EVR_EPRODS = 2 then “No”	If QN34 = 1 then “Yes” Else if QN34 = 2 then “No”	If V7780 in (2 3 4 5) then “Yes” Else if V7780 = 1 then “No”	If NICVAPEVER = 1 then “Yes” Else if NICVAPEVER = 2 then “No”	If QN6 = 1 then “Yes” Else if QN6 = 2 then “No”

Table B-2. Questions and algorithms used to define youth ENDS use in the PATH Study, NYRBS, MTF, NSDUH, and NYTS (continued)

PATH Study Wave 7	NYRBS 2021	MTF 2022	NSDUH 2022	NYTS 2022
Additional question to define past 30-day ENDS use (answers defining past 30-day ENDS use given in parentheses)				
“When was the last time you used an electronic nicotine product, even one or two times?” (Wave 7, earlier today, not today but sometime in the past 7 days, not in the past 7 days but sometime in the past 30 days)	“During the past 30 days, on how many days did you use an electronic vapor product?” (1 or 2 days, 3 to 5 days, 6 to 9 days, 10 to 19 days, 20 to 29 days, all 30 days)	“On how many DAYS (if any) during the LAST 30 DAYS have you vaped nicotine?” (1-2 days, 3-5 days, 6-9 days, 10-19 days, 20-29 days, 30 days)	“How long has it been since you last vaped nicotine with an e-cigarette or other vaping device?” (within the past 30 days)	“During the past 30 days, on how many days did you use e-cigarettes?” (1-30)
Coding of “Yes” and “No” answers				
If R07R_Y_CUR_EPRODS = 1 then “Yes” Else if R07R_Y_CUR_EPRODS = 2 then “No”	If QN35 = 1 then “Yes” Else if QN35 = 2 then “No”	If V7782 in (2 3 4 5 6 7) then “Yes” Else if V7782 = 1 then “No”	If NICVAPEVER = 1 and NICVAPREC in (1 11) then “Yes” Else if NICVAPEVER = 2 or NICVAPREC in (2 3 4 14 19 29) then “No”	If QN9 in (1 2 ... 30) then “Yes” Else if QN6 = 2 or QN9 = 0 then “No”

Appendix C

Wave 7 Single-Wave Response Rates for the Wave 4 Cohort

Appendix C presents Wave 7 adult and youth interview single-wave response rates for the Wave 4 Cohort, conditioning on Wave 4 response. Single-wave response rates reflect a perspective that considers the eligibility and response status of each participant only at the wave to which the response rate pertains. As described in Chapter 2, the Wave 7 Cohort was created by pooling respondents from the Wave 7 replenishment sample with Wave 7 respondents from the Wave 4 Cohort who were in the CNP and ages 15 and older at the time of Wave 7. Although the response rates presented in this appendix include a relatively small number of Wave 4 Cohort members who at the time of Wave 7 were younger than age 15 and/or not in the CNP (see Figure 2-1), their inclusion has no meaningful impact on the resulting Wave 7 response rates. Therefore, the response rates presented below complement the Wave 7 replenishment sample response rates in Chapter 6.

The unweighted and weighted single-wave response rates were calculated for the Wave 7 adult interview among those in the Wave 4 Cohort using the following formulas:

$$RR_{A_SW} = C_{A_SW} / (C_{A_SW} + N_{A_SW} + e_{A_SW} \times U_{A_SW})$$

$$e_{A_SW} = (C_{A_SW} + N_{A_SW}) / (C_{A_SW} + N_{A_SW} + I_{A_SW})$$

where

RR_{A_SW} = Wave 7 adult interview single-wave response rate;

C_{A_SW} = number of Wave 7 adult interview respondents;

N_{A_SW} = number of Wave 7 adult nonrespondents known to be eligible at Wave 7;

U_{A_SW} = number of Wave 7 adult nonrespondents whose eligibility status could not be determined;

e_{A_SW} = estimated proportion of eligible adults among the Wave 7 adult nonrespondents with unknown eligibility status; and

I_{A_SW} = number of Wave 7 adults determined to be ineligible at Wave 7.³⁸

The Wave 7 youth interview unweighted response rate for the Wave 4 Cohort (RR_{Y_SW}) was calculated using the same approach (substituting “Y” in place of “A” in the formulas above).

Weighted response rates ($RR_{A_W_SW}$ and $RR_{Y_W_SW}$) were calculated by replacing each count in the response rate formulas with the corresponding sum of the Wave 4 cross-sectional weights. The reason for using the Wave 4 cross-sectional weight to calculate weighted single-wave response rates is the same as was provided for calculating the Wave 4 all-waves response rates in Section 5.1.1.

Tables C-1 and C-2 show the Wave 7 adult and youth single-wave response rates, respectively, for the Wave 4 Cohort conditioning on Wave 4 response. In addition to the overall row, the tables include rows for subgroups based on: response status since Wave 4; Wave 4 sex, age, race/ethnicity, census region, education level (adults only), health insurance status (adults only), and tobacco-use status; and recruitment wave. Participants for whom the Wave 4 interview or Wave 4 household screener (in the case of shadow youth) had missing information for the tobacco-use status, socio-economic, or demographic characteristic were excluded from the response rate calculation for that characteristic. In Table C-1, the education level and health insurance status rows exclude the Wave 7 adults who were youth at Wave 4 because they were not asked about highest level of education attained or health insurance status at that time. In Table C-2, the tobacco-use rows exclude the Wave 7 youth who were age 10 or 11 at Wave 4 (shadow youth) because they were not asked about tobacco use at that time. Also, in Table C-2, note that most youth in the Wave 4 age group “12-17” were age 12 at Wave 4.

The Wave 7 adult interview weighted response rate (conditioning on Wave 4 response) was 66.9 percent (Table C-1). For the youth interview, the Wave 7 weighted response rate was 55.9 percent (Table C-2). In part, these response rates reflect decisions made by the PATH Study to stop asking cohort members to participate if they had a very low propensity to respond (see Section 2.9). For both adults and youth, the biggest variation in response rates was associated with response history since Wave 4. Participants who responded at both Waves 5 and 6 had a substantially higher propensity to respond at Wave 7 than

³⁸ Excludes participants determined to be permanently ineligible (i.e., deceased, moved out of the United States permanently, or incarcerated long term) prior to Wave 7.

participants who did not respond at Waves 5 and 6. The second biggest variation in response rates for adults and youth was by recruitment wave. The weighted response rate for Wave 7 adults who were recruited at Wave 1 (69.2 percent) was more than 15 percentage points higher than for those recruited at Wave 4 (54.1 percent). Wave 7 youth who were recruited at Wave 1 also were more likely to respond (69.4 percent) than those recruited at Wave 4 (52.7 percent). A possible explanation for the large differences in subgroup response rates by recruitment wave is that Wave 4 Cohort members who have responded to multiple previous waves may be more cooperative than other cohort members. The Wave 1 sample members of the Wave 4 Cohort have persisted with the PATH Study over a number of years and all completed an interview at Wave 4.

Table C-1. Wave 7 adult interview single-wave response rates conditioning on Wave 4 response (Wave 4 Cohort)

Characteristic ^a	<i>C_{A,SW}</i> Completed Wave 7 adult interview (n)	<i>I_{A,SW}</i> Ineligible at Wave 7 ^b (n)	<i>N_{A,SW}</i> Nonresponse known to be eligible at Wave 7 (n)	<i>U_{A,SW}</i> Nonresponse with unknown eligibility status at Wave 7 (n)	<i>RR_{A,SW}</i> Unweighted single-wave response rate ^c (%)	<i>RR_{A,W,SW}</i> Weighted single-wave response rate ^c (%)
Overall	28,165	463	6,680	10,165	62.8	66.9
Response history since Wave 4						
Responded at each previous primary wave	25,366	250	2,508	2,236	84.3	86.8
Did not respond in at least one previous primary wave	2,799	213	4,172	7,929	19.1	17.5
Wave 4 sex						
Male	13,341	282	3,549	5,366	60.2	64.7
Female	14,784	181	3,115	4,782	65.3	69.0
Wave 4 age						
Under 18	7,569	70	1,987	3,304	59.0	58.4
18-24	6,392	73	1,830	2,803	58.1	58.4
25-44	7,217	74	1,526	2,334	65.3	67.3
45-64	5,276	132	940	1,256	70.9	72.4
65 or older	1,709	114	397	466	67.1	66.2
Wave 4 race/ethnicity ^d						
Hispanic	6,440	88	1,469	2,296	63.3	66.9
Non-Hispanic White alone	14,652	236	3,715	5,413	61.8	67.2
Non-Hispanic Black alone	4,207	93	802	1,379	66.1	68.0
Non-Hispanic other race or multiple races	2,309	34	543	861	62.4	64.8
Wave 4 census region						
Northeast	3,895	46	1,073	1,583	59.6	64.7
Midwest	6,528	105	1,569	2,360	62.6	67.3
South	10,604	216	2,441	3,992	62.5	65.9
West	7,138	96	1,597	2,228	65.3	69.8
Wave 4 education (Wave 4 adults only)						
Less than high school or GED	3,591	113	837	1,451	61.5	63.3
High school graduate	4,674	104	1,246	1,811	60.7	62.8
Some college but no degree, or associate's degree	7,341	113	1,763	2,387	64.0	67.5
Bachelor's degree or advanced degree	4,926	59	818	1,166	71.4	74.3

Table C-1. Wave 7 adult interview single-wave response rates conditioning on Wave 4 response (Wave 4 Cohort) (continued)

Characteristic ^a	<i>C_{A,SW}</i> Completed Wave 7 adult interview (n)	<i>I_{A,SW}</i> Ineligible at Wave 7 ^b (n)	<i>N_{A,SW}</i> Nonresponse known to be eligible at Wave 7 (n)	<i>U_{A,SW}</i> Nonresponse with unknown eligibility status at Wave 7 (n)	<i>RR_{A,SW}</i> Unweighted single-wave response rate ^c (%)	<i>RR_{A,W,SW}</i> Weighted single-wave response rate ^c (%)
Wave 4 health insurance (Wave 4 adults only)						
Yes	17,281	306	3,826	5,330	65.6	68.6
No	3,124	81	809	1,441	58.4	62.1
Wave 4 current established tobacco use ^e (Wave 4 adults only)						
Yes	7,217	203	1,787	2,810	61.4	61.9
No	13,317	189	2,885	4,026	66.0	69.4
Wave 4 ever tobacco use ^f (Wave 4 youth only)						
Yes	1,566	17	507	828	54.1	53.1
No	5,618	50	1,346	2,248	61.1	60.6
Recruitment wave						
Wave 1	23,575	390	5,138	7,363	65.5	69.2
Wave 4	4,590	73	1,542	2,802	51.6	54.1

^a For each characteristic, the sum of the counts across the categories may not be equal to the count in the overall row due to missing values.

^b Excludes participants determined to be permanently ineligible (i.e., deceased, moved out of the United States permanently, or incarcerated long term) prior to Wave 7.

^c The adult interview unweighted single-wave response rate formula is provided earlier in this section. The weighted response rates were calculated using a similar approach except that the Wave 4 cross-sectional weight was used to obtain weighted counts.

^d Hispanic refers to Hispanic or Latino; Black refers to Black or African American. Alone means that the participant reported only one race.

^e Current established use of a tobacco product means the person currently uses the product every day or some days and—for cigarettes—has smoked at least 100 cigarettes in their lifetime and—for any other tobacco product—has reported they ever used that product fairly regularly. This characteristic considers the following products covered by the Wave 4 adult interview: cigarettes, traditional cigars, cigarillos, filtered cigars, pipes, smokeless tobacco, snus, hookah, and electronic nicotine delivery systems or ENDS (such as e-cigarettes, vape pens, personal vaporizers and mods, e-cigars, e-pipes, e-hookahs, and hookah pens).

^f Ever use of a tobacco product means the person has ever used the product, even one or two times. This characteristic considers the following products covered by the Wave 4 youth interview: cigarettes, traditional cigars, cigarillos, filtered cigars, pipes, smokeless tobacco, snus, hookah, electronic nicotine delivery systems or ENDS (such as e-cigarettes, vape pens, personal vaporizers and mods, e-cigars, e-pipes, e-hookahs, and hookah pens), dissolvable tobacco, bidis, and kreteks.

Table C-2. Wave 7 youth interview single-wave response rates conditioning on Wave 4 response (Wave 4 Cohort)

Characteristic ^a	<i>C_{Y,SW}</i> Completed Wave 7 youth interview (n)	<i>I_{Y,SW}</i> Ineligible at Wave 7 ^b (n)	<i>N_{Y,SW}</i> Nonresponse known to be eligible at Wave 7 (n)	<i>U_{Y,SW}</i> Nonresponse with unknown eligibility status at Wave 7 (n)	<i>RR_{Y,SW}</i> Unweighted single-wave response rate ^c (%)	<i>RR_{Y,W,SW}</i> Weighted single-wave response rate ^c (%)
Overall	3,452	8	778	1,796	57.3	55.9
Response history since Wave 4						
Responded at each previous primary wave	3,198	1	288	148	88.0	87.7
Did not respond in at least one previous primary wave	254	7	490	1,648	10.7	10.5
Wave 4 sex						
Male	1,816	5	408	966	57.0	55.6
Female	1,629	3	369	826	57.7	56.2
Wave 4 age						
Under 12	2,266	5	549	1,441	53.3	51.8
12-17	1,186	3	229	354	67.1	66.4
Wave 4 race/ethnicity ^d						
Hispanic	1,001	3	234	495	57.9	56.8
Non-Hispanic White alone	1,638	1	393	869	56.5	55.1
Non-Hispanic Black alone	382	2	77	208	57.3	54.9
Non-Hispanic other race or multiple races	341	1	61	192	57.5	56.7
Wave 4 census region						
Northeast	480	1	109	287	54.8	54.2
Midwest	780	3	200	447	54.7	53.2
South	1,210	2	296	651	56.1	54.7
West	982	2	173	411	62.7	61.1
Wave 4 ever tobacco use ^e (Wave 4 youth only)						
Yes	48	0	13	19	60.0	57.5
No	1,115	2	213	330	67.3	66.6

Table C-2. Wave 7 youth interview single-wave response rates conditioning on Wave 4 response (Wave 4 Cohort) (continued)

Characteristic ^a	<i>C_{Y,SW}</i> Completed Wave 7 youth interview (n)	<i>I_{Y,SW}</i> Ineligible at Wave 7 ^b (n)	<i>N_{Y,SW}</i> Nonresponse known to be eligible at Wave 7 (n)	<i>U_{Y,SW}</i> Nonresponse with unknown eligibility status at Wave 7 (n)	<i>RR_{Y,SW}</i> Unweighted single-wave response rate ^c (%)	<i>RR_{Y,W,SW}</i> Weighted single-wave response rate ^c (%)
Recruitment wave						
Wave 1	838	3	160	201	69.9	69.4
Wave 4	2,614	5	618	1,595	54.2	52.7

^a For each characteristic, the sum of the counts across the categories may not be equal to the count in the overall row due to missing values.

^b Excludes participants determined to be permanently ineligible (i.e., deceased, moved out of the United States permanently, or incarcerated long term) prior to Wave 7.

^c The youth interview unweighted single-wave response rate formula is provided earlier in this section. The weighted response rates were calculated using a similar approach except that the Wave 4 cross-sectional weight was used to obtain weighted counts.

^d Hispanic refers to Hispanic or Latino; Black refers to Black or African American. Alone means that the participant reported only one race.

^e Ever use of a tobacco product means the person has ever used the product, even one or two times. This characteristic considers the following products covered by the Wave 4 youth interview: cigarettes, traditional cigars, cigarillos, filtered cigars, pipes, smokeless tobacco, snus, hookah, electronic nicotine delivery systems or ENDS (such as e-cigarettes, vape pens, personal vaporizers and mods, e-cigars, e-pipes, e-hookahs, and hookah pens), dissolvable tobacco, bidis, and kreteks.

Appendix D

Wave 7 Replenishment Sample Shadow Youth Response Rates

If a child age 9 to 11 was selected for the PATH Study at the conclusion of the household screener, the parent (or guardian) was asked to consent to their child joining the study as a shadow youth to be interviewed after turning 12. If the parent provided consent, as well as the child's date of birth, the child was considered a shadow youth respondent. If the parent did not provide consent, or did not provide the date of birth, the sampled child was treated as a shadow youth nonrespondent.

Table D-1 presents the weighted and unweighted shadow youth response rates for the Wave 7 replenishment sample. The shadow youth response rates (conditioning on completion of the household screener) were calculated as:

$$RR_{SY} = C_{SY} / (C_{SY} + N_{SY})$$

where

RR_{SY} = Wave 7 replenishment sample shadow youth response rate;

C_{SY} = number of Wave 7 shadow youth respondents; and

N_{SY} = number of Wave 7 shadow youth nonrespondents.

Unweighted response rates were calculated based on actual case counts, whereas weighted response rates were based on the sums of the Wave 7 replenishment sample IPS weights (calculated as the product of the Wave 7 replenishment sample household IPS weight and the inverse of the within-household probability of selection).

In addition to the overall row, the table includes rows on age, sex, and race/ethnicity based on information collected through the household screener, as well as census region. Shadow youth with missing values for a particular characteristic were excluded from the response rate calculation for that characteristic.

The decision to exclude shadow youth with missing values for a particular characteristic from the corresponding response rate calculation leads to results in Table D-1 that may seem inconsistent.

This is because response rates (not shown) were lower for shadow youth for whom sex, race, and/or ethnicity were not provided by the household screener respondent. For example, the shadow youth weighted response rates for males and females were both greater than 79 percent, and yet the overall response rate (which includes shadow youth for whom sex information was missing) was only 77.9 percent. The lower overall response rate also reflects the study procedures applied to household members for whom the household screener respondent did not provide age information at all. These household members were treated as shadow youth for sampling purposes and the 253 among them that were selected for the PATH Study were immediately designated as nonrespondents.

Table D-1. Wave 7 shadow youth response rates conditioning on household screener response (Wave 7 replenishment sample)

Characteristic reported by household screener respondent or census region	<i>C_{SY}</i> Shadow youth response (n)	<i>N_{SY}</i> Shadow youth nonresponse (n)	<i>RR_{SY}</i> Unweighted response rate ^a (%)	<i>RR_{SY.W}</i> Weighted response rate ^a (%)
Overall	5,739	1,538	78.9	77.9
Age ^b				
9	1,895	414	82.1	82.2
10	1,870	431	81.3	81.1
11	1,891	429	81.5	81.5
Age not in expected range or not reported ^c	76	253	23.1	18.7
Sex ^b				
Male	3,035	695	81.4	81.0
Female	2,700	684	79.8	79.6
Race/ethnicity ^{b, d}				
Hispanic	1,590	404	79.7	79.4
Non-Hispanic White alone	2,872	623	82.2	82.1
Non-Hispanic Black alone	644	187	77.5	76.1
Non-Hispanic other race or multiple races	612	159	79.4	78.8
Census region				
Northeast	731	243	75.1	74.1
Midwest	1,340	342	79.7	78.7
South	2,227	624	78.1	76.8
West	1,441	329	81.4	81.1

^a The unweighted response rate formula is provided earlier in this section. The weighted response rates were calculated using a similar approach except that the Wave 7 replenishment sample IPS weight was used to obtain weighted counts.

^b The sum of counts across categories does not equal the overall count due to missing values. For example, for the age characteristic, 18 cases were excluded where the household screener respondent said a household member was age 9 to 11 but did not provide an exact age.

^c "Age not in expected range" refers to cases where the age reported by the household screener respondent was older than age 11, but prior to starting an interview the age of the household member was corrected to be within the 9 to 11 age range. Household members for whom no age information was provided by the household screener respondent were treated as shadow youth for sampling purposes; those selected were immediately designated as nonrespondents.

^d Hispanic refers to Hispanic or Latino; Black refers to Black or African American. Alone means that the participant reported only one race.

The weighted shadow youth response rate for the Wave 7 replenishment sample was 77.9 percent. The biggest subgroup difference occurred for census region, with weighted response rates of 81.1 percent for shadow youth in the West region and 74.1 percent for those in the North region. Differences among subgroups on weighted response rates were 1.4 percentage points for sex and 6.0 percentage points for race/ethnicity.